A MATHEMATICAL COMPANION · OCTOBER 2026

The Mathematics of General Relativity

From Calculus to Einstein's Field Equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$



three geodesics on a sphere: a triangle whose angles add up to more than 180°. That excess is curvature.

Companion to "General Relativity: A Beginner's Guide"

From derivatives and matrices through index notation, the metric, Christoffel symbols, curvature and the field equations, ending with a symbol-by-symbol anatomy of the great equations of general relativity.

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A mathematical companion to "General Relativity: A Beginner's Guide"

In the Beginner's Guide you met general relativity (GR) as a set of ideas: gravity is the curvature of spacetime, free fall is the natural state of motion, and clocks lower down run slow. This companion teaches the mathematics those ideas are written in, one prerequisite at a time, from the derivatives and Taylor series you know to Einstein's field equations. The aim is literacy rather than mastery: by the end you should be able to look at $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$, the Schwarzschild metric or the Riemann tensor and say what every symbol is, what operation each piece asks for, and why the equation has the shape it has. You will not yet solve the field equations for merging black holes (almost nobody can by hand), but you will know what you are looking at, which mathematics you still need, and what to ask a professor of GR. Every symbol is introduced at first use and briefly re-explained when it returns much later, so you can dip into any section.

How to read this document

The main body is organised by prerequisite, not by history: Part I builds the calculus and linear algebra beneath tensors, Part II flat spacetime and tensor algebra, and Part III curved spacetime, up to the field equations and the action principle. Every teaching section has the same six parts: what the mathematics is, why GR needs it, how it works step by step, worked examples (from plain numbers to GR), where it appears in Einstein's equations, and what to remember.

Part IV, "Anatomy of the Great Equations", dissects twenty equations symbol by symbol; Part V holds an equation dependency map, a notation reference, a study plan and a bibliography.

Three tags say how far to trust a result: **EXACT** (it follows from the definitions or from GR with no approximation), **APPROXIMATE** (it holds only under stated conditions, such as weak gravity or slow motion) and **ANALOGY** (a picture for intuition, not the real mathematics). Asides marked "Needs further study" flag topics that need a future course before you can use them on your own. Eq. (16.4) is the fourth numbered equation of Section 16.

Conventions used throughout

Metric signature $(-,+,+,+)$. The flat-spacetime interval is $\Delta s^2 = -(c\,\Delta t)^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$, so the Minkowski metric is $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$, where $\mathrm{diag}(\dots)$ lists the entries of a diagonal matrix (one whose off-diagonal entries are all zero). This is the convention of Misner, Thorne and Wheeler (MTW), Carroll, Wald, Tong and the Beginner's Guide (Carroll 1997, eq. 1.8; Tong GR notes). Timelike separations (ones a clock can travel) have $\Delta s^2 < 0$.

Units. SI units are the default, with $c = 299\,792\,458$ m/s and $G = 6.674\,30 \times 10^{-11}$ m³ kg⁻¹ s⁻² (NIST CODATA). The time coordinate is $x^0 = ct$, in metres, so all four coordinates share units. Some books set $c = 1$ (as Carroll does) or $G = c = 1$ ("geometric units", where masses become lengths: one solar mass is 1.477 km or 4.925 µs); this document says so wherever it uses either shortcut.

Indices. Greek indices $\mu, \nu, \rho, \sigma, \lambda, \alpha, \beta$ run over $0, 1, 2, 3$ (time and three space directions); Latin indices i, j, k run over $1, 2, 3$ (space only). A superscript on a coordinate is a label, not a power: x^2 is the coordinate y. The sign $\equiv$ means "is defined as", $\approx$ "is approximately" and $\propto$ "is proportional to". An index repeated once up and once down in a term is summed over (the Einstein summation convention, Section 11).

Curvature signs (Carroll/MTW). The Riemann tensor is $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$, the Ricci tensor is the contraction $R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$, and the field equations carry $+8\pi G/c^4$ (Carroll 1997, eqs. 3.67, 3.90, 4.52). With these choices a sphere has positive curvature. A dot over a symbol means a derivative

with respect to proper time τ or a path parameter λ, except in Newtonian formulas (such as $m\ddot{x} = -kx$ or $\ddot{\mathbf{x}} = -\nabla\Phi$) and in cosmology ($\dot{a}$, $\dot{\rho}$), where it means d/dt as in Physics I.

Textbooks differ in up to three independent signs, which MTW label [S1] (metric signature), [S2] (sign of the Riemann tensor) and [S3] (sign in the field equation). The physics is identical; only bookkeeping changes (Wikipedia: Einstein field equations, reporting MTW's classification).

Text	Signature	[S1] [S2] [S3]	What you will notice
MTW (1973); Carroll (1997 notes, 2004 book); Tong (2019 notes)	$(-,+,+,+)$	+ + +	Same as this document
Weinberg, *Gravitation and Cosmology* (1972)	$(-,+,+,+)$	+ − −	A sphere's scalar curvature comes out negative; $-8\pi G$ in the field equation
Peebles (1980)	$(+,-,-,-)$	− + +	Timelike intervals positive
Rindler (1977); Peacock (1999)	$(+,-,-,-)$	− + −	Timelike intervals positive; field-equation sign flipped
Landau & Lifshitz; most particle-physics books	$(+,-,-,-)$	− (others vary)	$u_\mu u^\mu = +c^2$ instead of $-c^2$

When you open a new book, check four things: the sign of the time term in flat space, whether a sphere gets positive curvature, the sign in front of $8\pi G$, and whether c and G are visible.

Prerequisite roadmap

The mathematics of GR is one long chain, from the derivative through matrices, coordinate changes, index notation and flat spacetime to the metric, the Christoffel symbols, curvature and the field equations, with two side branches: differential equations, which solve what the chain produces (orbits obey ordinary differential equations, the field equations are partial ones), and the calculus of variations, a second route to the geodesic equation and the field equations.

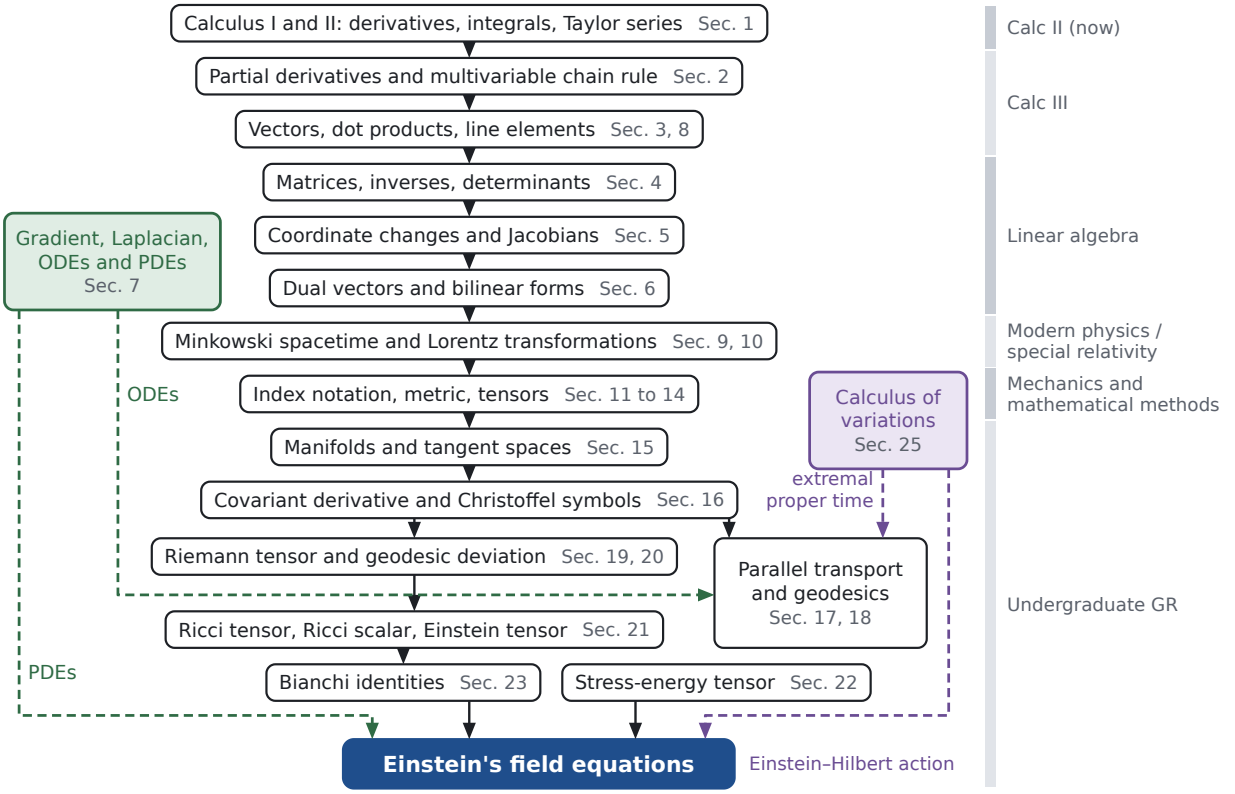


Figure 0.1. The prerequisite roadmap. Each arrow means "is needed for"; section numbers show where each topic is taught, and the right-hand bar shows the university course where it is usually met.

Difficulty ratings at a glance

Ratings are informal, relative to your current knowledge: 1 means you could do it today, 5 about as demanding as the hardest parts of Calculus II, 10 graduate-level work you should not expect to master here. They lean on instructors' reports that index manipulation, covariant derivatives and curvature are where students struggle most (Hartle 2006; Moore 2006).

Section	Topic	Rating	What makes it hard
1	Derivatives, parameters, Taylor approximations	2	A clock reading as the variable
2	Partial derivatives, multivariable chain rule	4	Tracking what is held fixed
3	Vectors, coordinates, dot products, line elements	4	Basis vectors that turn
4	Matrices, inverses, determinants	4	Order matters; new algebra rules
5	Coordinate transformations, Jacobians	5	Matrices of derivatives at a point
6	Dual vectors, bilinear forms	6	Components that change oppositely
7	Gradient, Laplacian, ODEs and PDEs	5	3D derivatives; boundary conditions
8	Curved surfaces, line elements	4	Measuring from inside a surface
9	Four-vectors, Minkowski interval	4	A minus sign that changes geometry
10	Lorentz transformations	5	Hyperbolic functions; simultaneity
11	Index notation, upper and lower indices	6	Implied sums; free vs dummy indices
12	Metric tensor, raising and lowering	6	Which object raises, which lowers
13	Tensor algebra, transformation laws	6	One Jacobian per index

Section	Topic	Rating	What makes it hard
14	Proper time, four-velocity, four-acceleration	5	Derivatives by the traveller's clock
15	Manifolds, tangent spaces	6	No surrounding space
16	Covariant derivative, Christoffel symbols	8	Correction terms; index bookkeeping
17	Parallel transport	7	A vector differential equation
18	Geodesics	7	Coupled nonlinear ODEs
19	Curvature, Riemann tensor	9	Four indices; long computations
20	Geodesic deviation	8	Second covariant derivatives
21	Ricci tensor, Ricci scalar, Einstein tensor	8	Contractions and their meaning
22	Stress-energy tensor, conservation	7	Momentum flux as a tensor
23	Bianchi identities, field equations as PDEs	9	Identities among curvature derivatives
24	Killing vectors, conserved quantities	7	Symmetry as a differential equation
25	Variational calculus, Einstein–Hilbert action	9	Varying a whole function; the metric as unknown

PART I

Foundations

Part I collects the calculus and linear algebra that tensors are built from: first- and second-year university mathematics (Calculus III, linear algebra, differential equations), taught only as far as GR needs.

The running example is the flat plane in polar coordinates (r, θ), the gentlest place to meet every tool of GR, because each result can be checked against Cartesian geometry you already trust.

SECTION 1

Functions, derivatives and Taylor approximations

Difficulty: 2/10. Calculus I and II, with a moving clock's own reading as the variable.

1. What Is This Mathematics?

A function turns one input into one output, such as position as a function of time, $x(t)$. Its derivative is the instantaneous rate of change, the slope of the tangent line: velocity is the derivative of position, acceleration that of velocity.

A path can be labelled by any counter, a **parameter**, often written λ (lambda), as a film's frames can be numbered by the studio clock or by the actor's wristwatch. For a moving object relativity prefers its own wristwatch reading, the **proper time** τ (tau).

A **Taylor series** approximates a smooth function near a point by its tangent line plus curvature terms; its most-used case is the **binomial approximation**: for small x, $(1 + x)^n \approx 1 + nx$.

2. Why Does General Relativity Need It?

In GR a particle's history is a curve labelled by proper time, so velocities and accelerations are derivatives with respect to τ. Gravity near Earth is weak ($|\Phi|/c^2$, the Newtonian potential divided by c^2, is about 7×10^{-10} at the surface), so the first Taylor term almost always suffices: that is how GR reduces to Newton's theory and yields the GPS clock corrections.

3. How Does the Mathematics Work?

The derivative of f at t is a limit of slopes over a small step h:

$$f'(t) = \lim_{h \to 0} \frac{f(t+h) - f(t)}{h}. \tag{1.1}$$

Three notations mean the same thing: $\frac{dx}{dt}$ (Leibniz), $x'(t)$ (Lagrange) and $\dot{x}$ (Newton). In GR books the dot almost always means $d/d\tau$ or $d/d\lambda$ (Hartle 2006).

If x and t are both functions of τ, the chain rule converts rates between the two counters:

$$\frac{dx}{dt} = \frac{dx/d\tau}{dt/d\tau}. \tag{1.2}$$

The Taylor series of f about a is (OpenStax Calculus Vol. 2, §6.3)

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots \tag{1.3}$$

and for any real power n the binomial series converges for $-1 < x < 1$ (OpenStax Vol. 2, §6.4):

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots \tag{1.4}$$

GR uses two cases constantly. For $f(x) = (1-x)^{1/2}$: $f(0) = 1$; $f'(x) = -\frac{1}{2}(1-x)^{-1/2}$, so $f'(0) = -\frac{1}{2}$; $f''(x) = -\frac{1}{4}(1-x)^{-3/2}$, so $f''(0) = -\frac{1}{4}$ and the x^2 coefficient is $-\frac{1}{4}/2 = -\frac{1}{8}$. The same steps for $(1-x)^{-1/2}$ give coefficients $\frac{1}{2}$ and $\frac{3}{8}$:

$$\sqrt{1-x} \approx 1 - \frac{x}{2} - \frac{x^2}{8}, \qquad \frac{1}{\sqrt{1-x}} \approx 1 + \frac{x}{2} + \frac{3x^2}{8}. \tag{1.5}$$

4. Worked Examples

Worked Example 1.1 — How good is "first order"?

Take $x = 0.1$. Exactly, $\sqrt{0.9} = 0.948683$; first order gives $1 - 0.05 = 0.95$, an error of 1.3×10^{-3}; adding the second-order term gives $0.95 - 0.00125 = 0.94875$, an error of 6.7×10^{-5}. At $x = 0.01$ the first-order error is 1.3×10^{-5}: ten times smaller x, about a hundred times smaller error, the signature of a remainder shrinking like x^2. But at $v = 0.6c$ the Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ is exactly $1/\sqrt{0.64} = 1.25$, while $1 + \frac{1}{2}(0.36) = 1.18$ is 5.6% low: the shortcut needs $x \ll 1$.

Worked Example 1.2 — A clock reading as the parameter

A particle moves along x at $v = 0.6c$, so $\gamma = 1.25$. By its own clock, $t(\tau) = 1.25\,\tau$ and $x(\tau) = 0.75\,c\,\tau$, so $dt/d\tau = 1.25$ and $dx/d\tau = 0.75c$, and by Eq. (1.2), $dx/dt = 0.75c/1.25 = 0.6c$. Each tick of the particle's clock is 1.25 ticks of the laboratory clock: time dilation, read off as a derivative.

Worked Example 1.3 — GPS: the speed term

A GPS satellite orbits at radius $a = 2.656175 \times 10^7$ m around Earth, whose $GM = 3.986004418 \times 10^{14}$ m³/s² (Ashby 2003). For a circular orbit, Physics I gives $v^2/a = GM/a^2$, so

$$\begin{aligned} v^2 &= \frac{GM}{a} = \frac{3.986004 \times 10^{14}}{2.656175 \times 10^7} = 1.500656 \times 10^7 \ \text{m}^2/\text{s}^2, \\ \frac{v^2}{c^2} &= \frac{1.500656 \times 10^7}{8.987552 \times 10^{16}} = 1.66970 \times 10^{-10}. \end{aligned} \tag{1.6}$$

So $v = 3873.8$ m/s, and by Eq. (1.5) the clock ticks at the rate $\sqrt{1 - v^2/c^2} \approx 1 - 8.3485 \times 10^{-11}$ (the second-order term is about 10^{-21}). Over a day of 86 400 s it loses 7.21×10^{-6} s, **7.2 µs per day**, from its speed; the larger, gravitational half of the GPS budget needs the metric (Worked Example 14.4).

5. Connection to Einstein's Equations

The geodesic equation (Section 18), GR's law of motion, begins with a second derivative with respect to proper time, $d^2x^\mu/d\tau^2$. Weak-gravity clock rates, $d\tau/dt \approx 1 + \Phi/c^2 - v^2/2c^2$, are a first-order Taylor expansion APPROXIMATE (valid when $|\Phi|/c^2 \ll 1$ and $v \ll c$), and so is the whole Newtonian limit (Equation 11 in Part IV), in Φ/c^2.

6. What I Should Remember

- A derivative is a rate and a slope; in GR the natural parameter for a path is proper time τ, and a dot usually means $d/d\tau$, not d/dt.
- The chain rule, $dx/dt = (dx/d\tau)/(dt/d\tau)$, converts between parameters.
- $\sqrt{1-x} \approx 1 - x/2$ and $1/\sqrt{1-x} \approx 1 + x/2$ for small x, with an error that shrinks like x^2.

SECTION 2

Partial derivatives and the multivariable chain rule

Difficulty: 4/10. Calculus I derivatives with the other variables frozen; normally Calculus III.

1. What Is This Mathematics?

For a quantity depending on several inputs, such as a hillside's height $h(x, y)$, a **partial derivative** asks how fast it changes when you nudge one input and freeze the others: $\partial h/\partial x$ is the slope stepping due east (holding y fixed), $\partial h/\partial y$ the slope stepping due north. The curly ∂ ("partial") warns that other variables are held constant.

The **multivariable chain rule** gives your rate of climb along a path across the hill: east-slope times eastward speed plus north-slope times northward speed, one term per route of influence.

2. Why Does General Relativity Need It?

The metric of GR is a set of functions of four coordinates, $g_{\mu\nu}(t, x, y, z)$. Its first partial derivatives build the Christoffel symbols, the "gravitational force" terms of the equation of motion; its second build curvature, whose 20 independent components are counted using $\partial_x\partial_y f = \partial_y\partial_x f$ (Carroll 1997, §2). The chain rule for changing coordinates seeds every tensor transformation law.

3. How Does the Mathematics Work?

The partial derivative with respect to x is the ordinary limit with y frozen (OpenStax Calculus Vol. 3, §4.3):

$$\frac{\partial f}{\partial x} = \lim_{h\to 0} \frac{f(x+h, y) - f(x, y)}{h}. \tag{2.1}$$

Treat every other variable as a constant and differentiate as in Calculus I. The notations $\partial f/\partial x$, $\partial_x f$, f_x and $f_{,x}$ are equivalent; GR writes $\partial_\mu f \equiv \partial f/\partial x^\mu$, where μ labels one of the four coordinates. **Clairaut's theorem**: if the mixed second partials are continuous, the order does not matter,

$$\frac{\partial}{\partial y}\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\frac{\partial f}{\partial y}. \tag{2.2}$$

GR assumes smooth (infinitely differentiable) metrics, so Eq. (2.2) always holds.

If $z = f(x, y)$ and both x and y depend on t, then (OpenStax Vol. 3, §4.5)

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}. \tag{2.3}$$

Draw a tree with a branch per intermediate variable, multiply along each path, and add. A thermometer on a post measures the **partial** time derivative $\partial T/\partial t$; one carried by a cyclist measures the **total** derivative $dT/dt = \partial T/\partial t + (\partial T/\partial x)(dx/dt)$. If the air cools at 0.1 °C/s and warms by 0.5 °C per metre along the road, the post reads -0.1 °C/s while a cyclist riding at 2 m/s reads $-0.1 + 0.5 \times 2 = +0.9$ °C/s.

The same rule changes coordinates. With $x = r\cos\theta$, $y = r\sin\theta$,

$$\frac{\partial f}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial f}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial f}{\partial y}. \tag{2.4}$$

Common confusion. Partial derivatives are not fractions: $\partial x/\partial r$ (with θ fixed) is $\cos\theta$, and $\partial r/\partial x$ (with y fixed) is $x/r = \cos\theta$ too, so their product is $\cos^2\theta$, not 1, because different variables are held fixed.

4. Worked Examples

Worked Example 2.1 — Partials and mixed partials

Let $f(x,y) = x^2 y + 3y^3$ at $(1,2)$. Freezing y: $f_x = 2xy = 4$. Freezing x: $f_y = x^2 + 9y^2 = 1 + 36 = 37$. Mixed: $f_{xy} = \partial_y(2xy) = 2x$ and $f_{yx} = \partial_x(x^2 + 9y^2) = 2x$, both equal to 2 at this point, as Clairaut promises. A common error is to assume a mixed partial vanishes because the variable "was held fixed".

Worked Example 2.2 — Gravity from a potential

Newton's potential of a point mass is $\Phi = -GM/r$ with $r = \sqrt{x^2+y^2+z^2}$. Freeze y and z and use the one-variable chain rule:

$$\begin{aligned}\frac{\partial \Phi}{\partial x} &= -GM \cdot \left(-\tfrac{1}{2}\right)(x^2+y^2+z^2)^{-3/2} \cdot 2x \\ &= \frac{GMx}{(x^2+y^2+z^2)^{3/2}}.\end{aligned} \tag{2.5}$$

At $(R,0,0)$ on Earth's surface this is GM/R^2; with $GM = 3.9860 \times 10^{14}$ m³/s² and mean radius $R = 6.371 \times 10^6$ m (NASA Earth Fact Sheet), $GM/R^2 = 9.820$ m/s², NASA's mean surface gravity. The field $g_x = -\partial\Phi/\partial x$ points toward the centre.

Worked Example 2.3 — Cartesian to polar derivatives

Let $f = xy$. Then $\partial f/\partial x = y$, $\partial f/\partial y = x$, $\partial x/\partial r = \cos\theta$, $\partial y/\partial r = \sin\theta$, $\partial x/\partial\theta = -r\sin\theta$ and $\partial y/\partial\theta = r\cos\theta$. Eq. (2.4) and its θ partner give

$$\begin{aligned}\partial_r f &= y\cos\theta + x\sin\theta = 2r\sin\theta\cos\theta = r\sin 2\theta, \\ \partial_\theta f &= -yr\sin\theta + xr\cos\theta = r^2\cos 2\theta.\end{aligned} \tag{2.6}$$

Substituting first, $f = r^2\cos\theta\sin\theta = \tfrac{1}{2}r^2\sin 2\theta$, gives the same partials.

5. Connection to Einstein's Equations

The Christoffel symbols (Section 16) are built entirely from first partial derivatives of the metric:

$$\Gamma^{\sigma}_{\mu\nu} = \tfrac{1}{2} g^{\sigma\rho}\left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right). \tag{2.7}$$

(Here $g^{\sigma\rho}$ is the inverse metric, Section 4, and an index repeated in a term, such as ρ, is summed over its four values, Section 11; for now, notice only that every term holds one first derivative of the metric.)

Curvature takes one more derivative, so the field equations are second-order in $g_{\mu\nu}$. With indices, Eq. (2.4) becomes $\partial f/\partial x'^{\mu} = (\partial x^{\nu}/\partial x'^{\mu})\,\partial f/\partial x^{\nu}$, the transformation law of a gradient (Carroll 1997, eq. 1.41), which Section 13 extends to every tensor.

6. What I Should Remember

- A partial derivative is an ordinary derivative with every other variable frozen (always know which); $\partial f/\partial x$, $\partial_x f$, f_x and $f_{,x}$ are the same, and $\partial_\mu f$ is the four-coordinate version.
- Mixed partials of smooth functions commute; curvature will measure what does *not* commute.
- The chain rule adds one term per route of influence and converts derivatives between coordinate systems.

SECTION 3

Vectors, coordinates, dot products and the first line element

Difficulty: 4/10. Physics I vectors, but polar reference arrows turn from place to place.

1. What Is This Mathematics?

A **vector** is an arrow, with a size and a direction. Its **components** are instructions relative to reference arrows, the **basis vectors** ("go 3 along the first and 4 along the second"); a **coordinate system** labels points. Changing coordinates changes the arrow's description, not the arrow, like one sentence in English and in French (Tong, Vector Calculus, §6).

Cartesian basis vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$ point the same way everywhere. Polar ones do not: "outward" ($\hat{\mathbf{e}}_r$) and "around" ($\hat{\mathbf{e}}_\theta$) rotate as you walk around the origin (Figure 3.1).

The **dot product** measures how much two arrows point the same way, scaled by their lengths; an arrow dotted with itself gives its squared length. For a tiny step, that squared length is the **line element** ds^2: Pythagoras for infinitesimal displacements, and the first appearance of the metric.

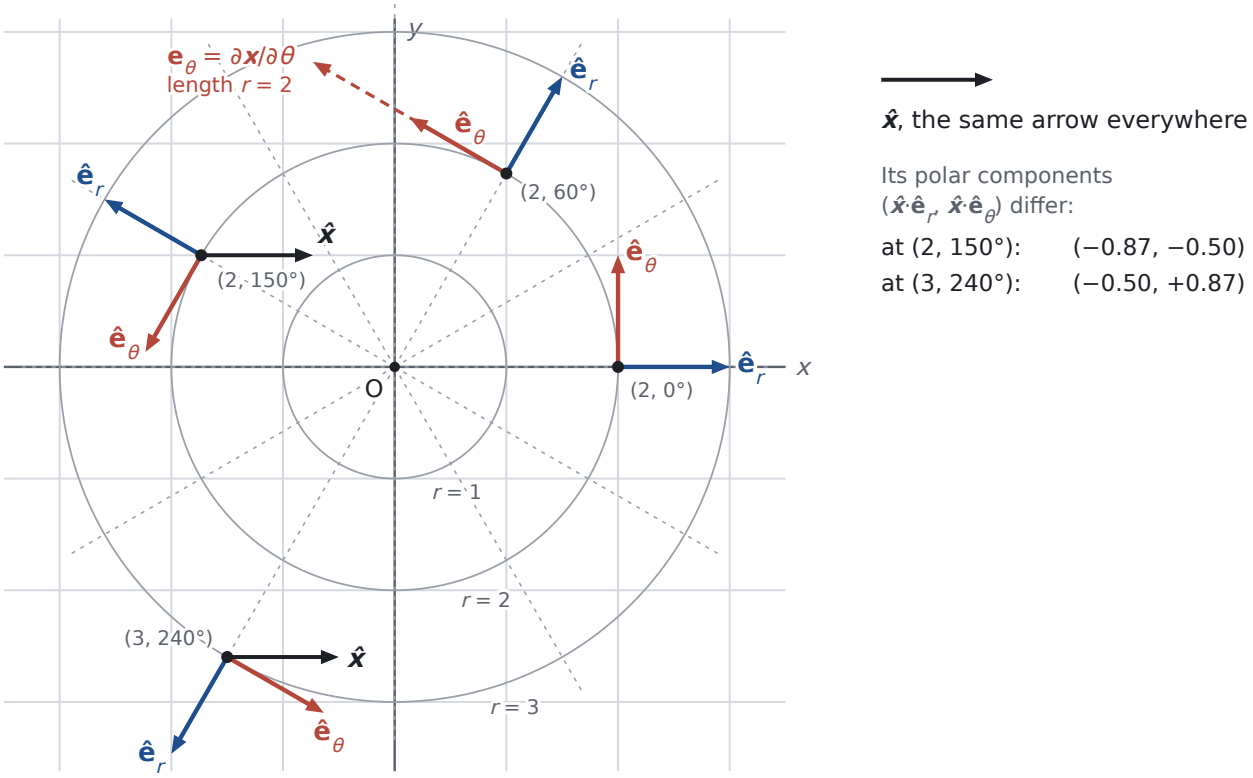


Figure 3.1. Polar basis vectors turn from point to point. The unit arrows $\hat{e}_r$ and $\hat{e}_\theta$ (blue and red) point outward and around at each location, so a fixed arrow such as $\hat{x}$ (black) has different polar components at different points. The dashed arrow is the coordinate basis vector e_θ, whose length is r.

2. Why Does General Relativity Need It?

GR's metric is a generalised dot product, and in curved spacetime no basis vectors stay the same from place to place, so every GR calculation has the polar-coordinate problem built in (Tong GR notes, ch. 1).

3. How Does the Mathematics Work?

In Cartesian components, $\mathbf{V} = V^x\hat{\mathbf{x}} + V^y\hat{\mathbf{y}}$; the superscripts x, y are labels, not powers. The dot product and length are

$$\mathbf{A}\cdot\mathbf{B} = A^xB^x + A^yB^y = |\mathbf{A}|\,|\mathbf{B}|\cos\alpha, \qquad |\mathbf{A}| = \sqrt{\mathbf{A}\cdot\mathbf{A}}, \tag{3.1}$$

where α is the angle between the arrows. With polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, the unit basis vectors, in Cartesian form, depend on θ:

$$\hat{\mathbf{e}}_r = (\cos\theta,\ \sin\theta), \qquad \hat{\mathbf{e}}_\theta = (-\sin\theta,\ \cos\theta). \tag{3.2}$$

Differentiating shows how they turn: $\partial\hat{\mathbf{e}}_r/\partial\theta = \hat{\mathbf{e}}_\theta$ and $\partial\hat{\mathbf{e}}_\theta/\partial\theta = -\hat{\mathbf{e}}_r$. GR prefers the **coordinate basis**, obtained by nudging one coordinate: $\mathbf{e}_r = \partial\mathbf{x}/\partial r = (\cos\theta, \sin\theta)$ and $\mathbf{e}_\theta = \partial\mathbf{x}/\partial\theta = (-r\sin\theta,\ r\cos\theta)$, of length r.

For the line element, start from Pythagoras, with the differentials from the chain rule, Eq. (2.3):

$$\begin{aligned} dx &= \cos\theta\, dr - r\sin\theta\, d\theta, \\ dy &= \sin\theta\, dr + r\cos\theta\, d\theta. \end{aligned} \tag{3.3}$$

Square and add. The cross terms $\mp 2r\sin\theta\cos\theta\, dr\, d\theta$ cancel, and $\sin^2\theta + \cos^2\theta = 1$ twice:

$$ds^2 = dx^2 + dy^2 = dr^2 + r^2\,d\theta^2. \tag{3.4}$$

The coefficients are the **metric components**, $g_{rr} = 1$, $g_{\theta\theta} = r^2$, $g_{r\theta} = 0$: the coordinates' built-in ruler, by which a step $d\theta$ is worth $r\,d\theta$ of distance. (ds^2 means $(ds)^2$, not "d of s^2".)

4. Worked Examples

Worked Example 3.1 — One arrow, changing components

Take the fixed arrow $\mathbf{V} = \hat{\mathbf{x}} = (1, 0)$. At the point $(1, 1)$, $r = \sqrt{2}$ and $\theta = 45^\circ$, so $\hat{\mathbf{e}}_r = (0.7071, 0.7071)$, $\hat{\mathbf{e}}_\theta = (-0.7071, 0.7071)$, and the components are $\mathbf{V}\cdot\hat{\mathbf{e}}_r = 0.7071$ and $\mathbf{V}\cdot\hat{\mathbf{e}}_\theta = -0.7071$. At $(0, 2)$, $\theta = 90^\circ$, $\hat{\mathbf{e}}_r = (0, 1)$, $\hat{\mathbf{e}}_\theta = (-1, 0)$, and the components are 0 and -1. The arrow never changed; the basis turned.

Worked Example 3.2 — Centripetal acceleration from turning bases

Write position as $\mathbf{x} = r\,\hat{\mathbf{e}}_r$ and differentiate with the product rule, using $d\hat{\mathbf{e}}_r/dt = \dot\theta\,\hat{\mathbf{e}}_\theta$ and $d\hat{\mathbf{e}}_\theta/dt = -\dot\theta\,\hat{\mathbf{e}}_r$ (here the dot is d/dt):

$$\begin{aligned} \dot{\mathbf{x}} &= \dot r\,\hat{\mathbf{e}}_r + r\dot\theta\,\hat{\mathbf{e}}_\theta, \\ \ddot{\mathbf{x}} &= (\ddot r - r\dot\theta^2)\,\hat{\mathbf{e}}_r + (r\ddot\theta + 2\dot r\dot\theta)\,\hat{\mathbf{e}}_\theta. \end{aligned} \tag{3.5}$$

For uniform circular motion with $r = 2$ m and $\dot\theta = 3$ rad/s, $a_r = -2 \times 9 = -18$ m/s^2, a term that exists only because $\hat{\mathbf{e}}_r$ rotates. Setting $\ddot{\mathbf{x}} = 0$ (no force) gives $\ddot r - r\dot\theta^2 = 0$ and $\ddot\theta + 2\dot r\dot\theta/r = 0$: a straight line in polar coordinates, and a preview of the geodesic equation.

Worked Example 3.3 — Tiny steps

At $r = 2$ take $dr = 0.01$ and $d\theta = 0.005$: Eq. (3.4) gives $ds^2 = 0.0001 + 4 \times 0.000025 = 0.0002$, so $ds = 0.014142$, against a straight chord from $(2, 0)$ to $(2.01\cos 0.005,\ 2.01\sin 0.005)$ of 0.014160. The 0.12% difference vanishes as the steps shrink.

5. Connection to Einstein's Equations

Every GR metric is a line element $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$ (Section 12), of which Eq. (3.4) is the simplest curvilinear case. The basis derivatives $\partial\mathbf{e}_r/\partial\theta = \frac{1}{r}\mathbf{e}_\theta$ and $\partial\mathbf{e}_\theta/\partial\theta = -r\,\mathbf{e}_r$ are the Christoffel symbols $\Gamma^\theta_{r\theta} = 1/r$ and $\Gamma^r_{\theta\theta} = -r$ of Section 16, nonzero even in flat space: nonzero Christoffel symbols do not by themselves mean curvature.

6. What I Should Remember

- A vector is not its components: components depend on the basis, and polar (like all curved-space) basis vectors change from point to point.
- The dot product turns two vectors into a number; the line element ds^2 is Pythagoras for tiny steps.
- In polar coordinates $ds^2 = dr^2 + r^2 d\theta^2$: the coefficients are a position-dependent ruler, the first metric.

SECTION 4

Matrices: multiplication, inverses and determinants

Difficulty: 4/10. New but mechanical arithmetic in which order matters.

1. What Is This Mathematics?

A **matrix** is a table of numbers acting as a machine that turns one list (a vector) into another, each output entry being one row dotted with the input. Multiplying matrices chains machines, and order matters, as with socks and shoes. The **determinant** of a square matrix is its area (2D) or volume (3D) scale factor, negative if it flips orientation and zero if it squashes space flat irreversibly; the **inverse** undoes the matrix.

2. Why Does General Relativity Need It?

The metric $g_{\mu\nu}$ is a symmetric 4×4 matrix at every point; its inverse $g^{\mu\nu}$ appears in nearly every formula, and its determinant g appears as $\sqrt{-g}$ in every invariant volume. Lorentz transformations are 4×4 matrices, and two boosts in different directions do not commute (Carroll 1997, §1).

3. How Does the Mathematics Work?

With A_{ij} the entry in row i, column j, matrix–vector and matrix–matrix products are

$$(A\mathbf{v})_i = \sum_j A_{ij} v_j, \qquad (AB)_{ij} = \sum_k A_{ik} B_{kj}. \tag{4.1}$$

The **identity** I has 1s on the diagonal and 0s elsewhere, so $AI = IA = A$. The **transpose** A^T swaps rows and columns, $(A^T)_{ij} = A_{ji}$, and $(AB)^T = B^T A^T$. A matrix is **symmetric** if $A^T = A$ and **diagonal** if every off-diagonal entry is zero. "Usually AB is different from BA" (Strang, *Introduction to Linear Algebra*, §1.4). For a 2×2 matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \tag{4.2}$$

and the inverse exists only when $ad - bc \neq 0$. A 3×3 determinant expands along the first row: $\det = a(ei - fh) - b(di - fg) + c(dh - eg)$ for rows (a, b, c), (d, e, f), (g, h, i). A diagonal matrix inverts entry by entry (take each reciprocal); a non-diagonal one does not.

4. Worked Examples

Worked Example 4.1 — Order matters

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Row by column, AB has entries $(1)(0) + (2)(1) = 2$, $(1)(1) + (2)(0) = 1$, $(3)(0) + (4)(1) = 4$, $(3)(1) + (4)(0) = 3$, so $AB = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$: the columns of A swapped. The same steps give $BA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$: the rows swapped. $AB \neq BA$.

Worked Example 4.2 — Determinant and inverse

$\det A = 1 \cdot 4 - 2 \cdot 3 = -2$: A doubles areas and flips orientation. By Eq. (4.2), $A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$. Check AA^{-1}: $1(-2) + 2(1.5) = 1$, $1(1) + 2(-0.5) = 0$, $3(-2) + 4(1.5) = 0$, $3(1) + 4(-0.5) = 1$. That is I.

Worked Example 4.3 — Metric matrices from GR

The polar metric is $g = \mathrm{diag}(1, r^2)$, with inverse $\mathrm{diag}(1, 1/r^2)$ and determinant r^2. The Minkowski metric $\eta = \mathrm{diag}(-1, 1, 1, 1)$ is its own inverse, with determinant -1. The Schwarzschild metric (Equation 12 of Part IV) in coordinates (ct, r, θ, ϕ), writing $f = 1 - r_s/r$, where $r_s = 2GM/c^2$ (the Schwarzschild radius) is a fixed length set by the mass M, is diagonal:

$$g = \operatorname{diag}\left(-f,\ f^{-1},\ r^2,\ r^2\sin^2\theta\right),$$
$$\det g = -f \cdot f^{-1} \cdot r^2 \cdot r^2\sin^2\theta = -r^4\sin^2\theta. \tag{4.3}$$

So $\sqrt{-g} = r^2\sin\theta$, as for flat space in spherical coordinates: the f and $1/f$ cancel.

Worked Example 4.4 — Inverting a non-diagonal metric

The 60° grid of Section 11 has metric $g = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$. Then $\det g = 1 - \frac{1}{4} = 0.75$ and $g^{-1} = \frac{1}{0.75}\begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix} = \begin{pmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{pmatrix}$ (check: $\frac{4}{3} - \frac{2}{3}\cdot\frac{1}{2} = 1$ and $\frac{4}{3}\cdot\frac{1}{2} - \frac{2}{3} = 0$), not the reciprocals of the entries, 1 and 2.

5. Connection to Einstein's Equations

The statement "$g^{\mu\nu}$ is the inverse of $g_{\mu\nu}$" is written $g^{\mu\nu}g_{\nu\sigma} = \delta^\mu_\sigma$, where the Kronecker delta δ^μ_σ is 1 when $\mu = \sigma$ and 0 otherwise: the identity matrix in index form (Carroll 1997, eq. 2.28). Lorentz transformations are the matrices with $\Lambda^T\eta\Lambda = \eta$ (Section 10), and $\sqrt{-g}$ makes the Einstein–Hilbert action's volume element $\sqrt{-g}\,d^4x$ coordinate-independent (Section 25).

6. What I Should Remember

- Row times column is the only multiplication rule, and $AB \neq BA$ in general.
- The determinant is an area or volume scale factor; an inverse exists only when it is nonzero, and only a diagonal matrix inverts entry by entry.
- The metric is a symmetric matrix, its inverse raises indices, and $\det g$ supplies $\sqrt{-g}$.

SECTION 5

Coordinate transformations and Jacobians

Difficulty: 5/10. Partial derivatives arranged as matrices: Calculus III's change of variables.

1. What Is This Mathematics?

A change of coordinates, such as Cartesian to polar, is a set of functions; zoomed in, it looks like a matrix, the **Jacobian matrix** of first partial derivatives of one set of coordinates with respect to the other. Its determinant, the **Jacobian**, is the area stretch of a tiny coordinate rectangle. ANALOGY An exchange rate that differs from city to city: the Jacobian at a point is the local rate between "coordinate units".

2. Why Does General Relativity Need It?

GR allows any coordinates and constantly translates between them, one Jacobian factor per tensor index (Section 13); and since coordinate volumes are not physical volumes, every spacetime integral carries the Jacobian-like factor $\sqrt{-g}$ (Carroll 1997, eq. 2.48).

3. How Does the Mathematics Work?

For polar coordinates the Jacobian matrix and its determinant are (OpenStax Vol. 3, §5.7)

$$J = \begin{pmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det J = r. \tag{5.1}$$

Eq. (3.3) is J acting on $(dr, d\theta)$. A polar "rectangle" dr by $d\theta$ has real sides dr and $r\,d\theta$, so its area is $dA = r\,dr\,d\theta$ (Figure 5.1). The reverse change uses the inverse matrix:

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta/r & \cos\theta/r \end{pmatrix} = J^{-1}. \tag{5.2}$$

The metric changes by a Jacobian on each side. Starting from the Cartesian metric I:

$$g_{\text{polar}} = J^T I J = J^T J, \qquad \det g_{\text{polar}} = (\det J)^2, \tag{5.3}$$

so $\sqrt{\det g} = |\det J|$: the metric "knows" the Jacobian, which is why GR writes invariant volumes as $\sqrt{|g|}\,d^n x$.

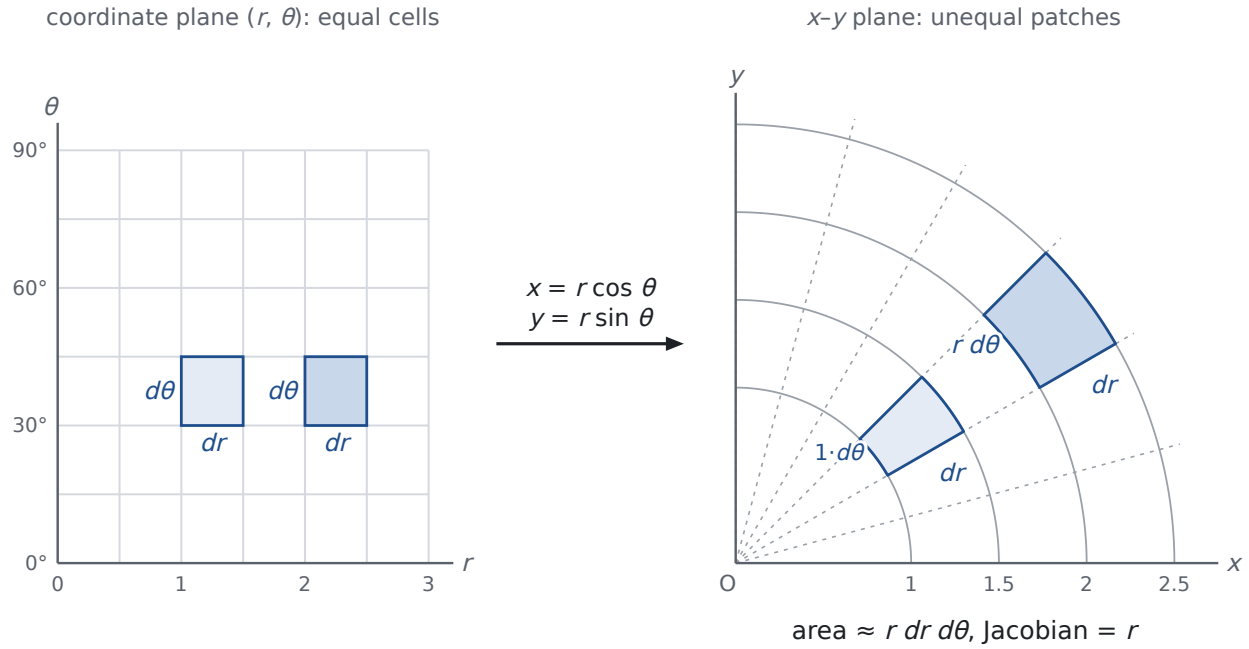


Figure 5.1. The Jacobian as an area stretch. A coordinate rectangle dr by dθ becomes a patch of real size dr by r dθ, so its area is r dr dθ; the Jacobian determinant r is the stretch factor, and it grows with distance from the origin.

4. Worked Examples

Worked Example 5.1 — A Jacobian at one point

At $r = 2$, $\theta = \pi/6$ ($\cos\theta = 0.86603$, $\sin\theta = 0.5$), Eq. (5.1) gives $J = \begin{pmatrix} 0.86603 & -1 \\ 0.5 & 1.73205 \end{pmatrix}$ and $\det J = 1.5 + 0.5 = 2 = r$. Eq. (5.2) gives $J^{-1} = \begin{pmatrix} 0.86603 & 0.5 \\ -0.25 & 0.43301 \end{pmatrix}$. Check the four entries of JJ^{-1}: $0.75 + 0.25 = 1$, $0.43301 - 0.43301 = 0$, $0.43301 - 0.43301 = 0$, $0.25 + 0.75 = 1$.

Worked Example 5.2 — Areas and volumes

The quarter-annulus $1 \le r \le 2$, $0 \le \theta \le \pi/2$ has area $\int_0^{\pi/2}\int_1^2 r\,dr\,d\theta = \frac{3}{2}\cdot\frac{\pi}{2} = 3\pi/4$, a quarter of $\pi(2^2 - 1^2)$. In 3D spherical coordinates the Jacobian is $r^2\sin\theta$, and a ball of radius R has volume $\int_0^{2\pi}\int_0^{\pi}\int_0^{R} r^2\sin\theta\,dr\,d\theta\,d\phi = \frac{R^3}{3}\cdot 2\cdot 2\pi = \frac{4}{3}\pi R^3$.

Worked Example 5.3 — The metric from the Jacobian

Multiply out Eq. (5.3) with the matrix of Eq. (5.1):

$$J^T J = \begin{pmatrix} \cos^2\theta + \sin^2\theta & -r\sin\theta\cos\theta + r\sin\theta\cos\theta \\ \cdot & r^2\sin^2\theta + r^2\cos^2\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}. \tag{5.4}$$

(The dot marks the entry equal to its mirror image.) This is $ds^2 = dr^2 + r^2 d\theta^2$ again, and $\sqrt{\det g} = r = \det J$.

5. Connection to Einstein's Equations

The Einstein–Hilbert action integrates curvature with the invariant volume element $\sqrt{-g}\,d^4x$, the curved four-dimensional version of $r\,dr\,d\theta$ (Section 25), and the tensor transformation law puts one Jacobian entry, $\partial x'^{\mu}/\partial x^{\nu}$ or $\partial x^{\nu}/\partial x'^{\mu}$, on each index (Carroll 1997, eq. 2.19).

6. What I Should Remember

- The Jacobian matrix holds the partial derivatives of one set of coordinates with respect to the other; the reverse change uses its inverse.
- Its determinant is the local area or volume stretch: $dA = r\,dr\,d\theta$ in polar coordinates.
- The metric transforms as $g' = J^T g J$, so $\sqrt{|\det g|}$ supplies the Jacobian automatically; a Lorentz boost is a constant Jacobian.

SECTION 6

Linear algebra for tensors: dual vectors and bilinear forms

Difficulty: 6/10. Short but abstract ideas from linear algebra.

1. What Is This Mathematics?

A **vector space** is any collection you can add and scale (arrows, lists of numbers, functions); a **basis** is a minimal set of reference vectors, components are the recipe for building any vector from them, and a **linear map** respects adding and scaling.

A **dual vector** (covector, one-form) is a linear machine that eats a vector and returns a number: if vectors are columns, dual vectors are rows (Carroll 1997, §1). ANALOGY A price list eats a shopping basket and returns the bill; count eggs in dozens and the quantities shrink by 12 while the unit prices grow by 12, leaving the bill unchanged.

A **bilinear form**, such as the dot product, eats two vectors and returns a number, linear in each slot. A **tensor** is the same idea with any number of slots (Tong, Vector Calculus, §6).

2. Why Does General Relativity Need It?

The gradient is the prime dual vector: fed a direction, it returns the rate of change, no ruler needed (Carroll 1997, eq. 1.40). The metric and the stress-energy tensor $T_{\mu\nu}$ are symmetric bilinear forms. Vector and dual-vector components coincide in Cartesian space, which is why school physics never separated them; in spacetime they differ by a sign, and in curved coordinates by more (Carroll 1997, after eq. 1.62).

3. How Does the Mathematics Work?

Label vector components with upper indices, V^1, V^2, and dual-vector components with lower ones, ω_1, ω_2. The dual vector acting on the vector is

$$\omega(\mathbf{V}) = \omega_1 V^1 + \omega_2 V^2, \tag{6.1}$$

and linearity means $\omega(a\mathbf{V}+b\mathbf{W}) = a\,\omega(\mathbf{V}) + b\,\omega(\mathbf{W})$. Rescaling the basis scales vector components oppositely and dual components alike, so $\omega(\mathbf{V})$ never changes: hence the different index positions. A bilinear form with matrix G acts as

$$g(\mathbf{U},\mathbf{W}) = \sum_{i,j} G_{ij} U^i W^j = \mathbf{U}^T G\,\mathbf{W}, \tag{6.2}$$

and it is symmetric when $G^T = G$. The gradient of f is the dual vector $df = (\partial_1 f,\ \partial_2 f)$; acting on a velocity it gives the rate of change along the path, by the chain rule (2.3).

4. Worked Examples

Worked Example 6.1 — Opposite scaling

Let $\omega = (2,-1,3)$ and $\mathbf{V} = (1,4,2)$. Then $\omega(\mathbf{V}) = 2-4+6 = 4$. Double every basis vector: the vector's components halve to $(0.5,2,1)$, the dual vector's double to $(4,-2,6)$, and $4(0.5)-2(2)+6(1) = 4$ again.

Worked Example 6.2 — The gradient eats a velocity

Let $f = x^2 + 3y$ at $(1,2)$. Its gradient components are $(\partial_x f, \partial_y f) = (2x, 3) = (2,3)$. Acting on the velocity $\mathbf{V} = (0.5,1)$: $2(0.5)+3(1) = 4$. Check directly: along the path $(1+s/2,\ 2+s)$, $\frac{d}{ds}[(1+s/2)^2 + 3(2+s)]$ at $s=0$ is $2(1)(\frac{1}{2}) + 3 = 4$.

Worked Example 6.3 — The polar metric as a machine

At $r = 2$ the polar metric has matrix $G = \mathrm{diag}(1,4)$, acting on components $(dr, d\theta)$. For $\mathbf{U} = (1,\ 0.5)$ and $\mathbf{W} = (2,\ -0.25)$, Eq. (6.2) gives $1\cdot 1\cdot 2 + 4\cdot 0.5\cdot(-0.25) = 2 - 0.5 = 1.5$. Feeding the same vector twice gives its squared length: $g(\mathbf{U},\mathbf{U}) = 1 + 4(0.25) = 2$.

5. Connection to Einstein's Equations

Both sides of Einstein's equations are symmetric bilinear machines: $T_{\mu\nu}$ fed an observer's four-velocity u^μ twice, divided by c^2, gives the energy density that observer measures, and $G_{\mu\nu}$ fed the same gives a measure of curvature. "There are ten independent equations (since both sides are symmetric two-index tensors)" (Carroll 1997, after eq. 4.52).

> **Needs further study.** A linear algebra course (vector spaces, bases and change of basis, dual spaces, determinants, eigenvalues) is the single most useful mathematics course for GR after Calculus III; graduate courses treat "finite-dimensional vector spaces" as essential background (Cambridge Part III GR guide).

6. What I Should Remember

- A dual vector is a row that eats a column and returns a number; the gradient is the standard example.
- Vector components (upper index) and dual components (lower index) change in opposite ways, so the number they produce is basis-independent.
- A bilinear form eats two vectors (the metric is a symmetric one, and $g(\mathbf{U},\mathbf{U})$ is a squared length); a tensor is a multilinear machine, defined by how it acts, not as a table of numbers.

SECTION 7

Gradients, the Laplacian, and ODEs versus PDEs

Difficulty: 5/10. Three-dimensional derivatives and differential-equation language from Calculus III and a differential-equations course.

1. What Is This Mathematics?

The **gradient** ∇f ("del f") points steepest uphill on a field f, with length equal to the steepness; the **divergence** $\nabla \cdot \mathbf{F}$ is a vector field's net outflow per unit volume (a faucet is a source, a drain a sink); the **Laplacian** $\nabla^2 f$ compares f at a point with its average on a tiny surrounding sphere of radius h, which is $f + \frac{h^2}{6}\nabla^2 f$ plus smaller terms.

A **differential equation** relates an unknown function to its derivatives: an **ordinary differential equation** (ODE) if the unknown depends on one variable, a **partial differential equation** (PDE) if on several. ANALOGY An ODE is one aeroplane's flight plan; a PDE is the weather over a whole country.

2. Why Does General Relativity Need It?

Newton's gravity is a PDE, Poisson's equation $\nabla^2\Phi = 4\pi G\rho$, to which Einstein's equations were built to reduce for weak, slow gravity (Carroll 1997, eq. 4.35). GR's law of motion is a system of ODEs and its field equations a system of PDEs, which symmetry can turn back into ODEs.

3. How Does the Mathematics Work?

In Cartesian coordinates,

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right), \qquad \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}, \tag{7.1}$$

and $\nabla^2 f = \nabla \cdot \nabla f = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f$. Newton's field is $\mathbf{g} = -\nabla\Phi$. A point mass M sends a flux $-(GM/r^2)(4\pi r^2) = -4\pi GM$ through every sphere around it; the divergence theorem turns that into Gauss's law, $\nabla \cdot \mathbf{g} = -4\pi G\rho$, and so (Tong, Vector Calculus, §5)

$$\nabla^2\Phi = 4\pi G\rho. \tag{7.2}$$

The 4π is the area of a unit sphere. For a function of r alone the Laplacian is not d^2f/dr^2 but

$$\nabla^2 f = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right). \tag{7.3}$$

The **order** of a differential equation is its highest derivative. It is **linear** if the unknown and its derivatives appear only to the first power and are not multiplied together; then solutions can be added (superposed). An nth-order ODE needs n initial conditions; a PDE on a region needs boundary conditions (Paul's Online Notes).

4. Worked Examples

Worked Example 7.1 — Two Laplacians

For $f = x^2 + y^2 + z^2$: $\nabla^2 f = 2 + 2 + 2 = 6$. For $f = 1/r$ with $r \neq 0$, use Eq. (7.3): $df/dr = -1/r^2$, so $r^2\, df/dr = -1$, whose derivative is 0. Hence $\nabla^2(1/r) = 0$ away from the origin, so $\Phi = -GM/r$ solves Poisson's equation outside a mass, where $\rho = 0$.

Worked Example 7.2 — A uniform ball

A ball of radius R has uniform density ρ_0 and mass $M = \frac{4}{3}\pi R^3 \rho_0$. Inside, try $\Phi_{\text{in}} = A + Br^2$. Using Eq. (7.3), $\nabla^2(r^2) = \frac{1}{r^2}\frac{d}{dr}(2r^3) = 6$, so $6B = 4\pi G\rho_0$ and $B = GM/(2R^3)$. Outside, $\nabla^2\Phi = 0$ gives $\Phi_{\text{out}} = C/r$ (a constant term is dropped so that $\Phi \to 0$ far away). Matching slopes at R: $2BR = -C/R^2$, so $C = -GM$. Matching values: $A + BR^2 = -GM/R$, so $A = -3GM/(2R)$. The result, an **EXACT** solution of Eq. (7.2) for this density, is

$$\Phi_{\text{in}} = -\frac{GM\,(3R^2 - r^2)}{2R^3}, \qquad \Phi_{\text{out}} = -\frac{GM}{r}. \tag{7.4}$$

For an idealised uniform Earth ($GM = 3.9860 \times 10^{14}$ m³/s², $R = 6.371 \times 10^6$ m), surface gravity is 9.820 m/s², halfway down 4.910 m/s², and the implied density 5513 kg/m³, NASA's mean density (NASA Earth Fact Sheet); the real Earth has a dense core, so only the outside solution fits it.

Worked Example 7.3 — Nonlinearity kills superposition

The ODE $y' = y^2$ is solved by $y_1 = 1/(1-t)$, since $y_1' = 1/(1-t)^2 = y_1^2$. Try $2y_1$: the left side is $2/(1-t)^2$ but the right side is $4/(1-t)^2$. Doubling a solution does not give a solution. Carroll's GR version: "two known solutions cannot be superposed to find a third" (Carroll 1997, §4).

Equation	Type	Order	Linear?	Needs
$m\ddot{x} = -kx$ (spring)	ODE	2	yes	2 initial conditions
$\nabla^2\Phi = 4\pi G\rho$ (Poisson)	PDE	2	yes	boundary conditions
Geodesic equation (Section 18)	4 coupled ODEs	2	no	initial position and velocity
Einstein's equations (Section 23)	10 coupled PDEs	2	no	initial data, boundary and coordinate choices

5. Connection to Einstein's Equations

In the weak-field limit the time-time component of Einstein's equations becomes Eq. (7.2) **APPROXIMATE** (weak, static field; slow motion; pressure much less than ρc^2). The full equations, "ten, coupled partial differential equations", have known exact solutions that exploit symmetry (Tong GR notes, ch. 5): spherical symmetry turns them into ODEs in r, as in Worked Example 7.2.

6. What I Should Remember

- Gradient: scalar to vector (strictly, a dual vector); divergence: vector to scalar; Laplacian: "average minus centre".
- Newtonian gravity is $\nabla^2\Phi = 4\pi G\rho$; the 4π is the area of a unit sphere.
- ODEs have one independent variable, PDEs several, and nonlinear ones cannot be superposed: the geodesic equation is a nonlinear ODE system, Einstein's equations a nonlinear PDE system that symmetry can reduce to ODEs.

PART II

Spacetime and Tensor Algebra

Part II moves from the plane to curved surfaces and the flat spacetime of special relativity, and introduces the index notation in which all of GR is written. By its end you will be able to read $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$, raise and lower indices, and say how any tensor changes when coordinates change. Curvature of spacetime waits for Part III.

SECTION 8

Curved surfaces and line elements

Difficulty: 4/10. The chain rule and $\sin^2 + \cos^2 = 1$, plus the new idea of measuring a surface from inside.

1. What Is This Mathematics?

Two numbers, such as latitude and longitude, label every point on a curved surface, and a line element turns small changes of those labels into true distances along it. ANALOGY A map whose scale bar changes from place to place, as on a Mercator map; the metric is that scale bar written as a formula.

2. Why Does General Relativity Need It?

Spacetime has no "outside" from which to measure it, so everything GR says comes from coordinates plus a metric; the sphere, described by its own two angles, is the training case (Moore 2006).

3. How Does the Mathematics Work?

Let θ be the **colatitude**, the angle down from the north pole ($0 \le \theta \le \pi$; latitude is $90^\circ - \theta$), and ϕ the longitude. A sphere of radius a sits in space as $x = a\sin\theta\cos\phi$, $y = a\sin\theta\sin\phi$, $z = a\cos\theta$. By the chain rule (2.3),

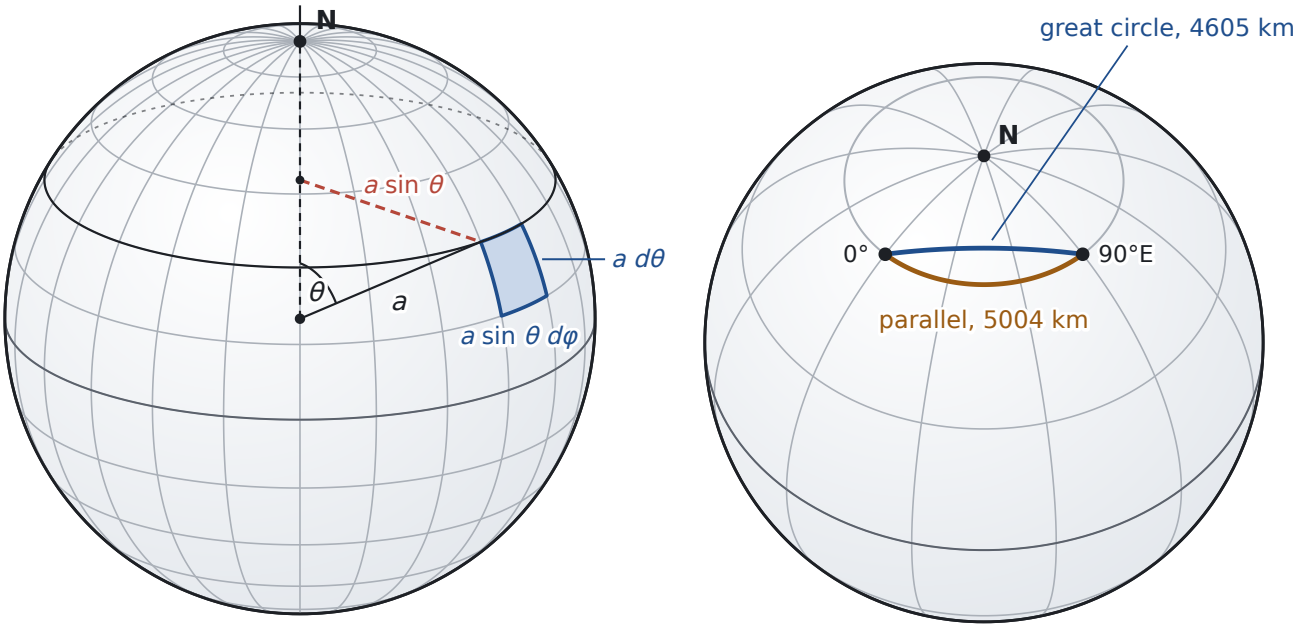
$$\begin{aligned} dx &= a\cos\theta\cos\phi\, d\theta - a\sin\theta\sin\phi\, d\phi, \\ dy &= a\cos\theta\sin\phi\, d\theta + a\sin\theta\cos\phi\, d\phi, \\ dz &= -a\sin\theta\, d\theta. \end{aligned} \tag{8.1}$$

Square each line and add. The $d\theta\, d\phi$ cross terms, $\mp 2a^2\sin\theta\cos\theta\sin\phi\cos\phi$, cancel. The $d\theta^2$ terms give $a^2\cos^2\theta + a^2\sin^2\theta = a^2$ and the $d\phi^2$ terms give $a^2\sin^2\theta$:

$$ds^2 = a^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right). \tag{8.2}$$

Read off $g_{\theta\theta} = a^2$, $g_{\phi\phi} = a^2\sin^2\theta$, $g_{\theta\phi} = 0$ (Carroll 1997, eqs. 2.31–2.33). Along a meridian, $ds = a\, d\theta$; along a circle of latitude, $ds = a\sin\theta\, d\phi$, which shrinks to zero at the poles (angles in radians). The symbol $d\Omega^2 \equiv$

$d\theta^2 + \sin^2\theta\, d\phi^2$ is shorthand for the unit sphere.



colatitude θ measured from N; cell at $\theta = 60°$–$75°$

two routes on the 60°N parallel, from 0° to 90°E

Figure 8.1. Line elements on a sphere. A grid cell has north–south side a dθ but east–west side a sin θ dφ, which shrinks toward the poles. Between two points on the 60°N parallel, the great circle (blue) is shorter than the parallel (orange).

4. Worked Examples

Worked Example 8.1 — Degrees on the Earth

With Earth's mean radius $a = 6371.0$ km (NASA Earth Fact Sheet) and $1° = 0.0174533$ rad, one degree of latitude is $a\,d\theta = 111.19$ km everywhere. One degree of longitude is $a\sin\theta\,d\phi$: 111.19 km at the equator ($\sin\theta = 1$), 78.63 km at 45°N ($\sin\theta = 0.70711$) and 55.60 km at 60°N ($\sin\theta = 0.5$).

Worked Example 8.2 — Parallels are not straight

From longitude 0° to 90°E along the 60°N parallel, $L = a\sin\theta\,\Delta\phi = 6371 \times 0.5 \times 1.5708 = 5003.8$ km. The great circle between the same points has angular length $\Delta\sigma$, the angle between the two points' position vectors, so by the dot product (3.1) and the coordinates above, $\cos\Delta\sigma = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2\cos\Delta\phi = \cos^2 30° + \sin^2 30°\cos 90° = 0.75$ (colatitude $30°$ is latitude $60°$N), so $\Delta\sigma = 0.72273$ rad and $L = 4604.5$ km, 399 km shorter.

Worked Example 8.3 — Curvature from inside

Points at surface distance r from the north pole have $\theta = r/a$, so the circle through them has circumference $C = 2\pi a\sin(r/a)$. Taylor-expanding $\sin x \approx x - x^3/6$:

$$C \approx 2\pi r - \frac{\pi r^3}{3a^2}, \qquad K \equiv \lim_{r\to 0}\frac{3\,(2\pi r - C)}{\pi r^3} = \frac{1}{a^2}. \tag{8.3}$$

K is the **Gaussian curvature**, found entirely from measurements within the surface. On Earth a circle of radius 10 km falls short of $2\pi r$ by 2.6 cm and one of 1000 km by 25.8 km: an r^3 effect, which is why small regions look flat.

5. Connection to Einstein's Equations

The Schwarzschild metric ends with $r^2(d\theta^2 + \sin^2\theta\, d\phi^2)$, Eq. (8.2) with a replaced by r, and the cosmological metric (FLRW, after Friedmann, Lemaître, Robertson and Walker) contains a curved three-dimensional version. As Carroll puts it, "the metric tensor contains all the information we need to describe the curvature of the manifold" (Carroll 1997, §2).

6. What I Should Remember

- The line element converts coordinate steps into true distances: on a sphere $ds^2 = a^2(d\theta^2 + \sin^2\theta\, d\phi^2)$, with θ measured from the pole.
- Curvature can be measured from inside, for example from the circumference shortfall $\pi r^3/3a^2$.
- Coordinate lines need not be the shortest paths.

SECTION 9

Four-vectors, Minkowski spacetime and the interval

Difficulty: 4/10. Pythagoras with one sign changed, applied to the Beginner's Guide's ideas.

1. What Is This Mathematics?

An **event** is a point in spacetime, a place at an instant, labelled by four numbers $x^\mu = (x^0, x^1, x^2, x^3) = (ct, x, y, z)$, where the index μ runs over 0, 1, 2, 3 and $\mu = 0$ is time measured in metres. A particle's history is a curve, its **worldline**. A **four-vector** is any set of four components that changes between observers the way the displacement Δx^μ between two events does. Observers in relative motion disagree about time and space separations but agree on one combination, the **interval**, whose minus sign makes time different from a fourth direction of space.

2. Why Does General Relativity Need It?

Every small freely falling laboratory obeys special relativity, so the Minkowski metric is the local standard against which curvature is measured, and "the metric determines causality, by defining the speed of light faster than which no signal can travel" (Carroll 1997, §2).

3. How Does the Mathematics Work?

The interval between two events is

$$\Delta s^2 = -(c\,\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2. \tag{9.1}$$

Here the superscript 2 is a genuine square of a bracket. Packaging the coefficients $(-1, 1, 1, 1)$ in the Minkowski metric,

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \Delta s^2 = \sum_{\mu=0}^{3}\sum_{\nu=0}^{3} \eta_{\mu\nu}\,\Delta x^\mu \Delta x^\nu. \tag{9.2}$$

Of the sixteen terms in the double sum only the four diagonal ones survive. The sign classifies how two events are related (Carroll 1997, §1):

Sign of Δs^2	Name	Meaning	What you can compute
negative	timelike	a clock can be present at both events; one can cause the other	proper time $\Delta\tau = \sqrt{-\Delta s^2}/c$
zero	null (lightlike)	only light connects them	nothing ticks along light
positive	spacelike	no signal can connect them	proper distance $\sqrt{\Delta s^2}$

One metre of time is $1/c = 3.33564$ ns.

> **Common confusion.** Δs^2 can be negative. It is the name of a quantity, not the square of a real number, and it is never square-rooted directly.

4. Worked Examples

Worked Example 9.1 — Three kinds of separation

With components in metres: $(c\Delta t, \Delta x) = (5, 3)$ gives $\Delta s^2 = -25 + 9 = -16$ m^2, timelike, with $c\Delta\tau = 4$ m, so $\Delta\tau = 4/c = 13.34$ ns. $(3, 5)$ gives $+16$ m^2, spacelike, with proper distance 4 m. $(5, 5)$ gives 0: null.

Worked Example 9.2 — Two lightning strikes

Two strikes are 1.000 km apart and 2.000 µs apart in the ground frame. Then $c\Delta t = 599.585$ m and

$$\Delta s^2 = -(599.585)^2 + (1000)^2 = -359\,502 + 1\,000\,000 = +640\,498 \text{ m}^2. \tag{9.3}$$

The separation is spacelike: neither strike could have caused the other. An observer moving at $v = c^2\Delta t/\Delta x = 0.59958c$ sees them as simultaneous, $\sqrt{640\,498} = 800.31$ m apart.

Worked Example 9.3 — Four dimensions

$\Delta x^\mu = (10, 2, 3, 4)$ m gives $\Delta s^2 = -100 + 4 + 9 + 16 = -71$ m^2: timelike, and a clock carried between the events at constant velocity reads $c\Delta\tau = \sqrt{71} = 8.43$ m, or 28.1 ns.

5. Connection to Einstein's Equations

Eq. (9.1) is Equation 1 of Part IV, and GR's line element $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$ (Section 12) is Eq. (9.2) with a position-dependent table; the sign of ds^2 still decides timelike, null and spacelike.

6. What I Should Remember

- An event has coordinates $x^\mu = (ct, x, y, z)$, with $\mu = 0$ for time measured in metres.
- The interval $\Delta s^2 = -(c\Delta t)^2 + \Delta x^2 + \Delta y^2 + \Delta z^2 = \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu$ is the same for all inertial observers: negative means timelike, zero null, positive spacelike.
- GR replaces $\eta_{\mu\nu}$ by a position-dependent metric $g_{\mu\nu}(x)$.

SECTION 10

Lorentz transformations

Difficulty: 5/10. Matching coefficients and $\cosh^2\phi - \sinh^2\phi = 1$; the hurdle is relativity of simultaneity.

1. What Is This Mathematics?

Lorentz transformations are the linear coordinate changes between inertial observers that keep the interval unchanged. Rotations keep $x^2 + y^2$ fixed, mixing x and y through cos and sin; **boosts** keep $-(ct)^2 + x^2$

fixed, mixing ct and x through the hyperbolic functions cosh and sinh: "rotations between space and time directions" (Carroll 1997, §1).

2. Why Does General Relativity Need It?

Locally inertial frames in GR are related by Lorentz transformations, which GR generalises to the position-dependent Jacobian $\partial x'^{\mu}/\partial x^{\nu}$ of an arbitrary coordinate change (Carroll 1997, before eq. 2.19).

3. How Does the Mathematics Work?

Look for linear formulas $ct' = A\,ct + B\,x$ and $x' = C\,ct + D\,x$ and demand $-(ct')^2 + x'^2 = -(ct)^2 + x^2$ for all ct and x. Expanding the left side and matching the coefficients of $(ct)^2$, $ct\,x$ and x^2:

$$-A^2 + C^2 = -1, \qquad -AB + CD = 0, \qquad -B^2 + D^2 = 1. \tag{10.1}$$

All three are satisfied by $A = D = \cosh\phi$ and $B = C = -\sinh\phi$, where the hyperbolic functions are $\cosh\phi = \tfrac{1}{2}(e^{\phi} + e^{-\phi})$, $\sinh\phi = \tfrac{1}{2}(e^{\phi} - e^{-\phi})$ and $\tanh\phi = \sinh\phi/\cosh\phi$; squaring and subtracting shows $\cosh^2\phi - \sinh^2\phi = 1$. The new observer's origin $x' = 0$ moves along $x/ct = \tanh\phi$, so $\tanh\phi = \beta \equiv v/c$; dividing $\cosh^2\phi - \sinh^2\phi = 1$ by $\cosh^2\phi$ then gives $\cosh\phi = \gamma$ and $\sinh\phi = \gamma\beta$. The number ϕ is the **rapidity**. As a matrix acting on (ct, x, y, z):

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}, \tag{10.2}$$

so $ct' = \gamma(ct - \beta x)$ and $x' = \gamma(x - \beta ct)$. The row label is the upper index μ and the column label the lower index ν, so $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$, with the repeated ν summed from 0 to 3 (the summation convention of Section 11), is matrix times column. Interval invariance, written as a matrix statement, is the defining property (compare rotations, $R^T R = I$):

$$\Lambda^T \eta\, \Lambda = \eta. \tag{10.3}$$

The matrix (10.2) does not depend on the signature; only η does. Successive boosts multiply as matrices and their rapidities add, the relativistic velocity-addition law: two $0.6c$ boosts make one of $\tanh(2\ln 2) = 15/17 = 0.882c$, not $1.2c$.

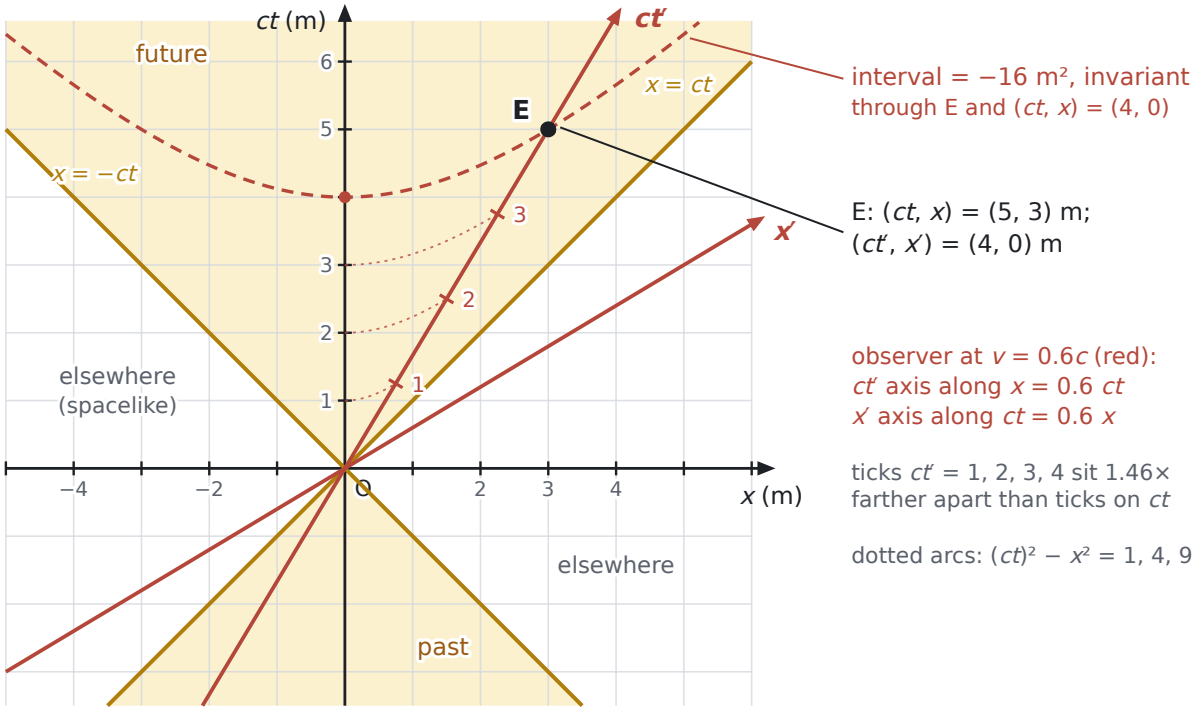


Figure 10.1. A Minkowski diagram with a boosted observer. The light cone (yellow) is the same for every observer. Axes of an observer moving at 0.6c (red) tilt toward the light lines. The event E lies on the moving clock's worldline, and the hyperbola through it joins all events with the same interval, −16 m².

4. Worked Examples

Worked Example 10.1 — The numbers at $v = 0.6c$

$\beta = 0.6$, $1 - \beta^2 = 0.64$, $\sqrt{0.64} = 0.8$, so $\gamma = 1.25$ and $\gamma\beta = 0.75$. The rapidity is $\phi = \operatorname{artanh} 0.6$, the inverse of tanh, which equals $\frac{1}{2}\ln\frac{1+\beta}{1-\beta}$: $\phi = \frac{1}{2}\ln\frac{1.6}{0.4} = \ln 2 = 0.693147$. Check: $\cosh(\ln 2) = (2 + \frac{1}{2})/2 = 1.25$ and $\sinh(\ln 2) = (2 - \frac{1}{2})/2 = 0.75$.

Worked Example 10.2 — Time dilation from the matrix

Transform the event $(ct, x, y, z) = (5, 3, 0, 0)$ m: $ct' = 1.25(5) - 0.75(3) = 4$ and $x' = -0.75(5) + 1.25(3) = 0$. The interval is $-25 + 9 = -16$ before and $-16 + 0 = -16$ after. Since $x' = 0$, the event sits on the moving clock's worldline: that clock reads 4 m of time (13.34 ns) while the ground frame says 5 m, and $5/4 = \gamma$.

Worked Example 10.3 — Checking $\Lambda^T \eta \Lambda = \eta$

On the t–x block, first $\eta\Lambda$, then Λ^T times that:

$$\begin{aligned} \eta\Lambda &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1.25 & -0.75 \\ -0.75 & 1.25 \end{pmatrix} = \begin{pmatrix} -1.25 & 0.75 \\ -0.75 & 1.25 \end{pmatrix}, \\ \Lambda^T(\eta\Lambda) &= \begin{pmatrix} -1.5625 + 0.5625 & 0.9375 - 0.9375 \\ 0.9375 - 0.9375 & -0.5625 + 1.5625 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \tag{10.4}$$

Also $\det\Lambda = \gamma^2 - \gamma^2\beta^2 = 1$: boosts preserve spacetime "area".

5. Connection to Einstein's Equations

Einstein's equations, being tensor equations, hold in every locally inertial frame and every coordinate system. The six parameters of Lorentz transformations (three rotations, three boosts) are what Section 15's

counting leaves free once a freely falling frame is chosen.

6. What I Should Remember

- Lorentz transformations are defined by preserving the interval: $\Lambda^T \eta \Lambda = \eta$.
- Boosts are hyperbolic rotations: $\gamma = \cosh\phi$, $\gamma\beta = \sinh\phi$, $\beta = \tanh\phi$; rapidities add, velocities do not.
- In GR the constant $\Lambda^\mu{}_\nu$ generalises to the Jacobian $\partial x'^\mu / \partial x^\nu$.

SECTION 11

Index notation: summation, upper and lower indices

Difficulty: 6/10. Not deep, but implied sums and index positions cause most beginners' errors (Moore 2006).

1. What Is This Mathematics?

Index notation is shorthand for sums over components. Its central rule, the **Einstein summation convention**, says that an index appearing once up and once down in a term is summed over all its values, with the $\sum$ sign left out (Carroll 1997, after eq. 1.9). Indices come in two heights because vectors have two natural kinds of components. **Contravariant** components V^i (upper index) say how many of each basis vector to add, like walking directions along streets; **covariant** components V_i (lower index) are the vector's dot products with each basis vector, like the shadows the trip casts on each street. They agree in a perpendicular unit-length basis but not on a skewed grid (Figure 11.1).

2. Why Does General Relativity Need It?

Every GR equation is written this way: one line such as $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ stands for sixteen equations, and index positions show at a glance what can be added, contracted or set equal.

3. How Does the Mathematics Work?

The rules (Carroll 1997, after eqs. 1.9 and 1.61; Moore 2006):

Rule	Example
Repeated index, once up and once down: sum it (a **dummy** index)	$a^\mu b_\mu = a^0 b_0 + a^1 b_1 + a^2 b_2 + a^3 b_3$
An index appearing once is **free**; it must appear at the same height in every term on both sides	$W^\mu = T^\mu{}_\nu V^\nu$ is four equations
No index may appear more than twice in a term	$a^\mu b_\mu c_\mu$ is meaningless
Dummy names can be changed freely	$a^\mu b_\mu = a^\nu b_\nu$
f free indices in n dimensions mean n^f equations	$G_{\mu\nu} = \ldots$ is $4^2 = 16$ equations

The line element with an implied double sum reads, in two dimensions,

$$\begin{aligned} g_{\mu\nu} dx^\mu dx^\nu &= g_{00} dx^0 dx^0 + g_{01} dx^0 dx^1 + g_{10} dx^1 dx^0 + g_{11} dx^1 dx^1 \\ &= g_{00} (dx^0)^2 + 2 g_{01} dx^0 dx^1 + g_{11} (dx^1)^2, \end{aligned} \tag{11.1}$$

using the symmetry $g_{01} = g_{10}$. For a skewed basis $\mathbf{e}_1$, $\mathbf{e}_2$, the two kinds of components and the metric are

$$\mathbf{V} = V^1 \mathbf{e}_1 + V^2 \mathbf{e}_2, \qquad V_i \equiv \mathbf{V} \cdot \mathbf{e}_i, \qquad g_{ij} \equiv \mathbf{e}_i \cdot \mathbf{e}_j, \tag{11.2}$$

and substituting the first into the second gives $V_i = g_{ij} V^j$: the metric converts one kind into the other.

Index ≠ exponent. In $x^\mu = (ct, x, y, z)$, x^2 means the coordinate y; the square of y is $(x^2)^2$. Carroll: "Don't start thinking of the superscripts as exponents" (Carroll 1997, §1). Some books write coordinate names as indices (u^t, u^r) to avoid this (Moore 2006).

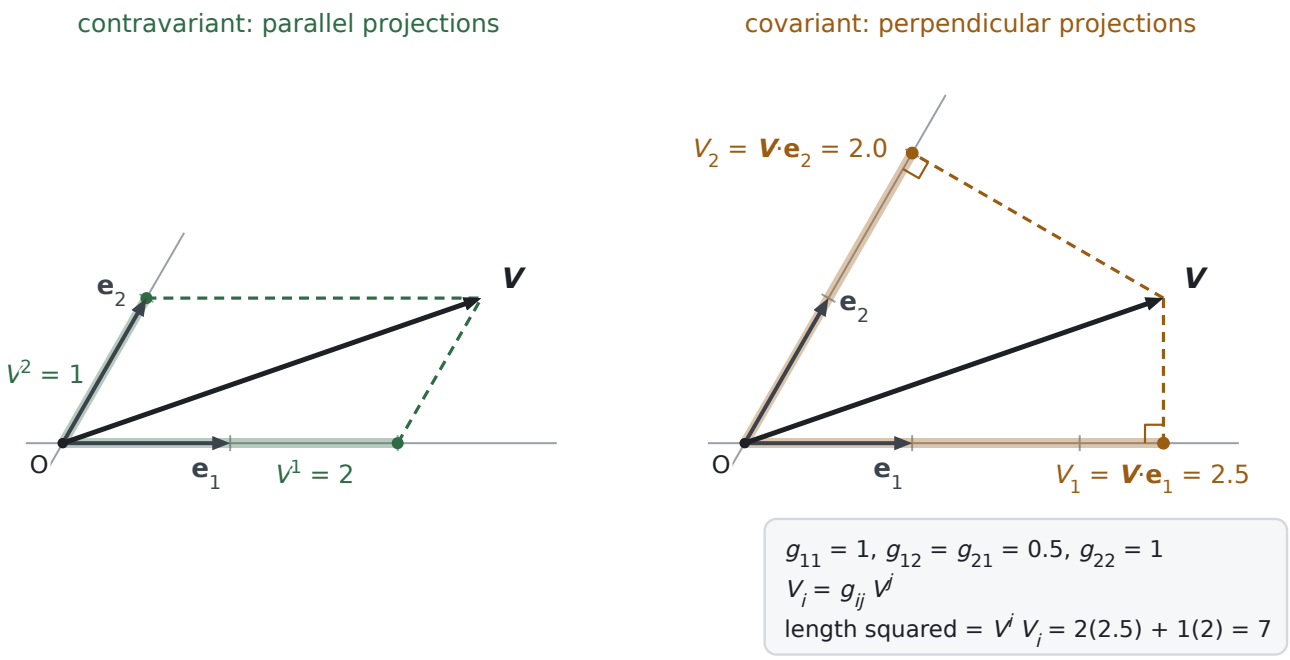


Figure 11.1. Contravariant versus covariant components on a 60° grid. The contravariant components (green) are the sides of the parallelogram built along the basis directions, (2, 1); the covariant components (orange) are perpendicular projections, (2.5, 2). They coincide only when the basis is perpendicular and of unit length.

4. Worked Examples

Worked Example 11.1 — A four-dimensional contraction

With $a^\mu = (3, 1, 2, -1)$ and $b_\mu = (-2, 4, 0, 5)$, the terms are $3(-2) = -6$, $1(4) = 4$, $2(0) = 0$ and $(-1)(5) = -5$, so $a^\mu b_\mu = -7$. Renaming the dummy, $a^\nu b_\nu = -7$ still.

Worked Example 11.2 — Components on a 60° grid

Let $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0.5, 0.8660)$, and $\mathbf{V} = 2\mathbf{e}_1 + 1\mathbf{e}_2 = (2.5, 0.8660)$, so $(V^1, V^2) = (2, 1)$. The covariant components are $V_1 = 2.5(1) + 0.866(0) = 2.5$ and $V_2 = 2.5(0.5) + 0.866(0.866) = 1.25 + 0.75 = 2.0$. The metric is $g_{11} = g_{22} = 1$, $g_{12} = g_{21} = 0.5$, and lowering checks: $V_1 = 1(2) + 0.5(1) = 2.5$ and $V_2 = 0.5(2) + 1(1) = 2$. The squared length is $V^i V_i = 2(2.5) + 1(2) = 7$, matching $2.5^2 + 0.866^2 = 7$. Double the basis vectors and the contravariant components halve to $(1, 0.5)$ while the covariant ones double to $(5, 4)$: they vary *contra* (against) and *co* (with) the basis, which is where the names come from.

Worked Example 11.3 — Spot the error

Each line is malformed (Moore 2006). $a^\mu b^\mu$: both indices up, so no sum is defined. $a^\mu = b_\nu$: free indices do not match. $\eta_{\mu\nu} A^\mu B^\nu = \eta_{\mu\mu} A^\mu B^\mu$: the right side uses μ four times and silently drops the off-diagonal terms. $\delta^\mu_\mu = 1$: written out, $\delta^0_0 + \delta^1_1 + \delta^2_2 + \delta^3_3 = 4$.

5. Connection to Einstein's Equations

In $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$, μ and ν are free: sixteen equations, ten distinct because every term is symmetric. In $R = g^{\mu\nu} R_{\mu\nu}$ both indices are dummies, giving a single number. In the geodesic equation, $\Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\tau} \frac{dx^\sigma}{d\tau}$ is a sixteen-term double sum for each value of μ.

6. What I Should Remember

- One index up and one down in a term means "sum over it"; free indices must match on both sides, no index may appear three times in a term, and dummy names are arbitrary.
- Upper (contravariant) components build the vector from basis arrows; lower (covariant) components are its dot products with them.
- A superscript is a label, not a power.

SECTION 12

The metric tensor, raising and lowering indices

Difficulty: 6/10. Matrices, indices and line elements combined; the risk is confusing $g_{\mu\nu}$ with its inverse $g^{\mu\nu}$.

1. What Is This Mathematics?

The **metric tensor** $g_{\mu\nu}$, a symmetric table of functions generalising the dot product, turns small coordinate steps into physical intervals of time or length. Lowering an index with $g_{\mu\nu}$ and raising it with the inverse $g^{\mu\nu}$ exchange the two kinds of components of Section 11.

2. Why Does General Relativity Need It?

In GR the metric *is* the gravitational field. In Carroll's list of its jobs, it "allows the computation of path length and proper time", "determines the 'shortest distance' between two points (and therefore the motion of test particles)" and "replaces the Newtonian gravitational field" (Carroll 1997, §2); its ten independent components are what Einstein's equations solve for.

3. How Does the Mathematics Work?

$$ds^2 = g_{\mu\nu}(x)\,dx^\mu dx^\nu, \qquad g_{\mu\nu} = g_{\nu\mu}, \qquad g^{\mu\lambda}g_{\lambda\nu} = \delta^\mu_\nu. \tag{12.1}$$

A symmetric 4×4 table has $4\cdot 5/2 = 10$ independent entries. To lower an index, contract with the metric; to raise one, contract with the inverse:

$$V_\mu = g_{\mu\nu}V^\nu, \qquad \omega^\mu = g^{\mu\nu}\omega_\nu. \tag{12.2}$$

Raising then lowering returns the original, because $g^{\mu\lambda}g_{\lambda\nu} = \delta^\mu_\nu$. The squared length of a vector is $V^\mu V_\mu = g_{\mu\nu}V^\mu V^\nu$. Some metrics you will meet:

Space and coordinates	Metric components	Note
Plane, (x, y)	$\mathrm{diag}(1, 1)$	Pythagoras
Plane, (r, θ)	$\mathrm{diag}(1, r^2)$	flat, yet position-dependent
Sphere, (θ, ϕ)	$\mathrm{diag}(a^2, a^2\sin^2\theta)$	curved
Minkowski, (ct, x, y, z)	$\mathrm{diag}(-1, 1, 1, 1)$	special relativity
Schwarzschild, (ct, r, θ, ϕ)	$\mathrm{diag}(-f, 1/f, r^2, r^2\sin^2\theta)$, $f = 1 - r_s/r$	outside a spherical mass

In Minkowski spacetime, lowering flips only the sign of the time component; in polar coordinates it multiplies the θ component by r^2.

Common confusion. A metric that changes from place to place does not mean curved space: polar coordinates on a flat plane are the counterexample. Only curvature (Part III), built from second derivatives of the metric, decides.

4. Worked Examples

Worked Example 12.1 — Lowering in Minkowski spacetime

For $V^\mu = (5, 3, 0, 0)$: $V_0 = \eta_{00} V^0 = -5$, $V_1 = 3$, so $V_\mu = (-5, 3, 0, 0)$, and $V^\mu V_\mu = 5(-5) + 3(3) = -16$, the interval of Worked Example 9.1. In a $(+,-,-,-)$ book you would get $V_\mu = (5, -3, 0, 0)$ and $+16$: same physics, opposite bookkeeping.

Worked Example 12.2 — Lowering in polar coordinates

At $r = 2$ take $V^\mu = (V^r, V^\theta) = (3, 0.5)$. Then $V_r = g_{rr} V^r = 3$ and $V_\theta = g_{\theta\theta} V^\theta = 4(0.5) = 2$. The squared length is $V^\mu V_\mu = 3(3) + 0.5(2) = 10$. Check with unit-vector components: the $\hat{\theta}$ component is $r V^\theta = 1$, and $3^2 + 1^2 = 10$. Raising back: $g^{\theta\theta} V_\theta = \frac{1}{4}(2) = 0.5$.

Worked Example 12.3 — Earth's surface

Earth's Schwarzschild radius is $r_s = 2GM/c^2 = 8.870$ mm (Equation 13 of Part IV). At the mean radius $r = 6.371 \times 10^6$ m, $r_s/r = 1.392 \times 10^{-9}$. A clock at rest has $dr = d\theta = d\phi = 0$, so $ds^2 = g_{00}(c\,dt)^2$, and with $ds^2 = -c^2 d\tau^2$ its rate is

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{r}} \approx 1 - \frac{r_s}{2r} = 1 - 6.96 \times 10^{-10}. \tag{12.3}$$

Over a day it loses $6.96 \times 10^{-10} \times 86\,400\ \text{s} = 60.1\ \mu\text{s}$ relative to a clock far away.

5. Connection to Einstein's Equations

The metric appears in every term: $g_{\mu\nu}$ inside $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$, $g^{\mu\nu}$ inside $R = g^{\mu\nu} R_{\mu\nu}$, and $g_{\mu\nu}$ beside the cosmological constant. The Christoffel symbols raise an index with $g^{\sigma\rho}$, and in weak gravity $g_{00} \approx -(1 + 2\Phi/c^2)$ carries Newton's potential.

6. What I Should Remember

- $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$; $g_{\mu\nu}$ is symmetric with 10 independent components in 4D.
- Lower with $g_{\mu\nu}$, raise with $g^{\mu\nu}$, and $g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu_\nu$; with $(-,+,+,+)$, lowering in Minkowski spacetime flips the sign of the time component only.
- The metric's appearance depends on coordinates; curvature, not variation, signals gravity.

SECTION 13

Tensor products, contractions and transformation laws

Difficulty: 6/10. Each rule is short; applying "one Jacobian per index" correctly takes practice.

1. What Is This Mathematics?

A tensor of **type** (k, l) has k upper and l lower indices, and n^{k+l} components in n dimensions (Carroll 1997, §1). The **tensor (outer) product** multiplies components with no sum, raising the rank: $T^{ij} = A^i B^j$. **Contraction** sums one upper index against one lower, removing both; the **trace** $M^i{}_i$ is the simplest case.

The **transformation law** says how components change between coordinate systems: one Jacobian factor per index.

2. Why Does General Relativity Need It?

The Ricci tensor and scalar are contractions of the Riemann tensor, and $T = T^{\mu}{}_{\mu}$ is the trace of the stress-energy tensor; the field equations are built from these. The transformation law guarantees that a tensor equation true in one coordinate system is true in all, the mathematical core of "general covariance".

3. How Does the Mathematics Work?

Contracting an outer product gives back the pairing of Section 6: if $M^{i}{}_{j} = A^{i}\omega_{j}$, then $M^{i}{}_{i} = A^{i}\omega_{i}$; with $A^{i} = (1, 2)$ and $\omega_{j} = (4, 1)$ both equal $4 + 2 = 6$. Symmetric and antisymmetric parts are written with round and square brackets:

$$T^{(ij)} = \tfrac{1}{2}\left(T^{ij} + T^{ji}\right), \qquad T^{[ij]} = \tfrac{1}{2}\left(T^{ij} - T^{ji}\right). \tag{13.1}$$

Under a change of coordinates $x \to x'$, upper indices pick up $\partial x'/\partial x$ and lower indices the inverse, $\partial x/\partial x'$ (Carroll 1997, eqs. 2.16–2.19):

$$\begin{aligned} V'^{\mu} &= \frac{\partial x'^{\mu}}{\partial x^{\nu}} V^{\nu}, \qquad \omega'_{\mu} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \omega_{\nu}, \\ g'_{\mu\nu} &= \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} g_{\alpha\beta}. \end{aligned} \tag{13.2}$$

Mnemonic: the primed coordinate sits where the index on the left sits. Why "true in one system, true in all": if $A_{\mu\nu} = B_{\mu\nu}$ in some coordinates, every component of $C_{\mu\nu} = A_{\mu\nu} - B_{\mu\nu}$ is zero there, and Eq. (13.2) multiplies zeros by Jacobian factors, so $C'_{\mu\nu} = 0$ in every system. This fails for non-tensors such as the partial derivative of a vector's components, which picks up an extra second-derivative term (Carroll 1997, eq. 2.26); Section 16 repairs that.

4. Worked Examples

Worked Example 13.1 — Outer product and its parts

With $A^{i} = (1, 2)$ and $B^{j} = (3, -1)$, $T^{ij} = A^{i}B^{j}$ has $T^{11} = 3$, $T^{12} = -1$, $T^{21} = 6$, $T^{22} = -2$. It is not symmetric. By Eq. (13.1) its symmetric part has entries 3, 2.5, 2.5, -2 and its antisymmetric part 0, -3.5, 3.5, 0; adding them returns T. A symmetric and an antisymmetric tensor always contract to zero, $S^{ij}A_{ij} = 0$; with Cartesian components, where lowering an index changes nothing: $3(0) + 2.5(-3.5) + 2.5(3.5) + (-2)(0) = 0$.

Worked Example 13.2 — An invariant contraction

At $(x, y) = (1, 1)$, so $r = \sqrt{2}$, $\theta = \pi/4$, take $V = \hat{\mathbf{x}}$, $(V^{x}, V^{y}) = (1, 0)$, and the dual vector $\omega = \nabla x$, $(\omega_{x}, \omega_{y}) = (1, 0)$. With $\partial r/\partial x = x/r = 0.7071$ and $\partial\theta/\partial x = -y/r^{2} = -0.5$, Eq. (13.2) gives $V^{r} = 0.7071$, $V^{\theta} = -0.5$. With $\partial x/\partial r = \cos\theta = 0.7071$ and $\partial x/\partial\theta = -r\sin\theta = -1$, it gives $\omega_{r} = 0.7071$, $\omega_{\theta} = -1$. The contraction is $1(1) + 0(0) = 1$ in Cartesian and $0.7071^{2} + (-1)(-0.5) = 0.5 + 0.5 = 1$ in polar: upper-with-lower contractions do not depend on coordinates.

Worked Example 13.3 — The trace of a perfect fluid

Looking ahead, a perfect fluid (Section 22) has $T^{\mu\nu} = (\rho + p/c^{2})u^{\mu}u^{\nu} + p\,g^{\mu\nu}$, where ρ is its mass density, p its pressure and u^{μ} its four-velocity (Section 14), a four-vector whose squared length is always $u_{\mu}u^{\mu} = -c^{2}$. Only that fact and $g_{\mu\nu}g^{\mu\nu} = \delta^{\mu}_{\mu} = 4$ are needed. The fluid's trace is

$$T = g_{\mu\nu}T^{\mu\nu} = (\rho + p/c^{2})(-c^{2}) + 4p = -\rho c^{2} + 3p. \tag{13.3}$$

For dust ($p = 0$), $T = -\rho c^2$; for radiation ($p = \rho c^2/3$), $T = 0$. (Carroll's notes, just after his eq. 4.41, say "$T > 0$ in matter"; with $(-,+,+,+)$ dust has $T < 0$, a sign slip that does not affect his argument.)

5. Connection to Einstein's Equations

$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$ and $R = g^{\mu\nu}R_{\mu\nu}$ are contractions; Einstein's equation is a tensor equation, so it holds in all coordinates; and its trace (Section 23) uses Eq. (13.3).

6. What I Should Remember

- Type (k, l) counts upper and lower indices; outer products raise the rank, and contractions, always of an upper with a lower index, lower it.
- Each index transforms with one Jacobian factor: $\partial x'/\partial x$ for upper, $\partial x/\partial x'$ for lower.
- A tensor that is zero in one coordinate system is zero in all, so tensor equations hold everywhere; partial derivatives of tensors break this.

SECTION 14

Proper time, four-velocity and four-acceleration

Difficulty: 5/10. Single-variable calculus applied to a worldline; the subtle step is a product-rule identity with indices.

1. What Is This Mathematics?

Proper time τ is what a clock carried along a worldline actually ticks: "τ actually measures the time elapsed on a physical clock carried along the path" (Carroll 1997, §1). Differentiating the worldline $x^\mu(\tau)$ gives the **four-velocity** $u^\mu = dx^\mu/d\tau$; times mass, the **four-momentum** $p^\mu = mu^\mu$; differentiating again, the **four-acceleration** $a^\mu = du^\mu/d\tau$, whose length an accelerometer on board reads.

2. Why Does General Relativity Need It?

GR's law of motion says a freely falling body has zero four-acceleration in the generalised sense of Section 18, and gravitational time dilation, GPS included, is proper time computed with $g_{\mu\nu}$ instead of $\eta_{\mu\nu}$.

3. How Does the Mathematics Work?

For a clock's worldline $ds^2 < 0$, and $d\tau^2 = -ds^2/c^2$. In flat spacetime this gives

$$d\tau = \sqrt{1 - \frac{v^2}{c^2}}\, dt, \qquad \tau = \int \sqrt{1 - \frac{v(t)^2}{c^2}}\, dt. \tag{14.1}$$

By the chain rule (1.2), $u^\mu = \frac{dt}{d\tau}\frac{dx^\mu}{dt} = \gamma(c, v_x, v_y, v_z)$, and its squared length is fixed:

$$u_\mu u^\mu = -\gamma^2 c^2 + \gamma^2 v^2 = -c^2, \qquad p_\mu p^\mu = -m^2 c^2. \tag{14.2}$$

With $p^\mu = (E/c,\, \mathbf{p})$, the second identity is $E^2 = (pc)^2 + (mc^2)^2$. Differentiate the constant $\eta_{\mu\nu}u^\mu u^\nu = -c^2$ with the product rule; the two terms are equal after renaming dummies and using $\eta_{\mu\nu} = \eta_{\nu\mu}$:

$$0 = \eta_{\mu\nu}a^\mu u^\nu + \eta_{\mu\nu}u^\mu a^\nu = 2\, u_\mu a^\mu. \tag{14.3}$$

So the four-acceleration is always perpendicular to the four-velocity: in the momentary rest frame $u^\mu = (c, 0, 0, 0)$ this forces $a^0 = 0$, and $a_\mu a^\mu$ is the squared acceleration the passenger feels. Moore lists this product-rule step among the commonest confusions (Moore 2006). In weak gravity, the metric

$g_{00} \approx -(1+2\Phi/c^2)$, $g_{ij} \approx (1-2\Phi/c^2)\delta_{ij}$ gives the GPS clock-rate formula APPROXIMATE (valid for $|\Phi|/c^2 \ll 1$ and $v \ll c$; derived in Equation 3 of Part IV):

$$\frac{d\tau}{dt} \approx 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2}. \tag{14.4}$$

4. Worked Examples

Worked Example 14.1 — Varying speed

A ship's speed rises steadily as $v(t) = 0.16c\,t$ (with t in years) from 0 to $0.8c$ over five years. By Eq. (14.1), $\tau = \int_0^5 \sqrt{1 - 0.0256t^2}\,dt$. Substitute $u = 0.16t$, $dt = 6.25\,du$, u from 0 to 0.8, and use the Calculus II result $\int \sqrt{1-u^2}\,du = \tfrac{1}{2}(u\sqrt{1-u^2} + \arcsin u)$:

$$\tau = 6.25 \cdot \tfrac{1}{2}\,(0.8 \times 0.6 + \arcsin 0.8) = 3.125 \times 1.407295 = 4.398 \text{ yr}. \tag{14.5}$$

The crew ages 4.40 years while 5 years pass on Earth.

Worked Example 14.2 — Energy and momentum at $0.6c$

For a 1000 kg ship at $0.6c$, $\gamma = 1.25$, so $u^\mu = (1.25c,\ 0.75c,\ 0,\ 0)$ and $u_\mu u^\mu = (-1.5625 + 0.5625)c^2 = -c^2$. The energy is $E = \gamma mc^2 = 1.1234 \times 10^{20}$ J and the kinetic energy $E - mc^2 = 2.2469 \times 10^{19}$ J; Newton's $\tfrac{1}{2}mv^2 = 1.6178 \times 10^{19}$ J is 28% low. Check: $-(E/c)^2 + p^2 = -1.40430 \times 10^{23} + 5.0555 \times 10^{22} = -8.9876 \times 10^{22} = -m^2c^2$ in kg² m² s⁻².

Worked Example 14.3 — A year at 1 g

With constant proper acceleration α the rapidity grows linearly, $\phi = \alpha\tau/c$, so $v = c\tanh\phi$, $ct = (c^2/\alpha)\sinh\phi$ and $x = (c^2/\alpha)(\cosh\phi - 1)$. For $\alpha = 9.80665$ m/s² (NIST) and $\tau = 1$ yr, $\phi = 1.0323$: $v = 0.7748c$, $\gamma = 1.582$, Earth time 1.187 yr, distance 0.564 light-years. The crew feels 1 g throughout; Earth sees the acceleration fall to $\alpha/\gamma^3 = 2.48$ m/s².

Worked Example 14.4 — The GPS clock budget

Use Eq. (14.4) for a satellite and a ground clock and subtract. The satellite's speed term is -8.3485×10^{-11} (Worked Example 1.3). The gravitational term compares the satellite's potential, $-GM/a$, with the ground clock's, fixed by the rotating geoid (the sea-level surface on which all ground clocks tick at the same rate) at $\Phi_0/c^2 = -6.969290 \times 10^{-10}$ (Ashby 2003):

$$\begin{aligned} \frac{-GM/a - \Phi_0}{c^2} &= -1.66970 \times 10^{-10} + 6.96929 \times 10^{-10} \\ &= +5.29959 \times 10^{-10} \quad (+45.8\ \mu\text{s/day}), \\ \text{net} &= 5.29959 \times 10^{-10} - 0.83485 \times 10^{-10} \\ &= 4.4647 \times 10^{-10} \quad (+38.6\ \mu\text{s/day}). \end{aligned} \tag{14.6}$$

The satellite clock gains 45.8 μs/day from weaker gravity, loses 7.2 μs/day from its speed, and nets **+38.6 μs/day**, the Beginner's Guide's figure, so engineers set the clocks to $10.23\,\text{MHz} \times (1 - 4.4647 \times 10^{-10}) = 10.22999999543$ MHz before launch (Ashby 2003). Some sources quote about 45.7 and 7.1 μs/day, booking the ground clock's rotation in the velocity term instead; the net is the same (Pascual-Sánchez, arXiv:gr-qc/0507121).

5. Connection to Einstein's Equations

The geodesic equation sets the four-acceleration to zero while accounting for curved coordinates, $d^2x^\mu/d\tau^2 + \Gamma^\mu_{\alpha\beta}u^\alpha u^\beta = 0$, with $g_{\mu\nu}u^\mu u^\nu = -c^2$; the simplest source in the field equations, dust, is $T^{\mu\nu} = \rho u^\mu u^\nu$.

6. What I Should Remember

- Proper time is wristwatch time, $d\tau^2 = -ds^2/c^2$, and in flat spacetime the unaccelerated worldline has the most of it.
- $u^\mu = dx^\mu/d\tau = \gamma(c, \mathbf{v})$ always has $u_\mu u^\mu = -c^2$ ($p^\mu = (E/c, \mathbf{p})$ has $p_\mu p^\mu = -m^2c^2$), and the four-acceleration, which an accelerometer reads, is perpendicular to it.
- In weak gravity $d\tau/dt \approx 1 + \Phi/c^2 - v^2/2c^2$: for GPS, +45.8 − 7.2 = +38.6 μs/day.

PART III

Curved Spacetime

Part III, the heart of the subject, asks how to do calculus when the basis vectors turn and space itself is curved, and ends with the field equations and the action they come from. It climbs the standard textbook ladder: flat polar coordinates (to see what curvature is *not*), then the two-sphere (small enough to compute by hand), then Schwarzschild spacetime (Carroll 1997; Tong GR notes).

Difficulty ratings climb to 8 and 9 here. Expect to read some sections twice; the goal is to recognise and interpret, not yet to compute everything yourself.

SECTION 15

Manifolds and tangent spaces

Difficulty: 6/10. Little calculation, but abstract: there is no surrounding space to lean on.

1. What Is This Mathematics?

A **manifold** is a space that looks like flat $\mathbb{R}^n$ (the set of lists of n real numbers) when you zoom in far enough: a sphere, a doughnut, curved spacetime. Like the Earth in an atlas, it is described by flat maps called **charts**, each covering part of it, with rules for translating where they overlap. Coordinates are labels, not rulers; distances come only from the metric. ANALOGY A city map looks flat because a small enough piece of any smooth curved surface is almost a plane (Einstein Online).

At each point there is a **tangent space**, like a sheet of paper touching a globe at one point; velocities, momenta and forces at that point live in it (Figure 15.1).

2. Why Does General Relativity Need It?

Spacetime in GR is a curved four-dimensional manifold that no single coordinate system need cover (Schwarzschild coordinates misbehave at the horizon). Local flatness, below, is the equivalence principle in mathematical form, and because each point has its own tangent space, comparing a vector at the Sun with one at the Earth needs a rule, the connection of Section 16.

3. How Does the Mathematics Work?

A **chart** assigns each point p in a patch its coordinates $x^\mu(p)$; the **dimension** is the number needed. Where charts overlap, the change of coordinates must be smooth, with the Jacobian of Section 5 as its derivative.

A curve $x^\mu(\lambda)$ through p has **tangent vector** components $V^\mu = dx^\mu/d\lambda$, and for any function f the chain rule gives

$$\frac{df}{d\lambda} = \frac{dx^\mu}{d\lambda}\,\partial_\mu f = V^\mu \partial_\mu f. \tag{15.1}$$

Since f is arbitrary, a vector can be identified with this directional derivative, $V = V^\mu \partial_\mu$: the partial derivatives ∂_μ act as the **coordinate basis** vectors (Carroll 1997, eqs. 2.9–2.10), and the metric gives their dot products, $g_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu$.

Local flatness. At any point p you can choose coordinates in which $g_{\mu\nu}(p) = \eta_{\mu\nu}$ and every first derivative $\partial_\sigma g_{\mu\nu}(p)$ vanishes, but in general not every second derivative (Carroll 1997, §2). Counting shows why. The metric at p has 10 components and the Jacobian offers 16 free numbers: enough, with 6 left over (the 3 rotations and 3 boosts of Section 10). The 40 first derivatives of the metric are matched by 40 free second derivatives of the coordinate change. But the 100 second derivatives of the metric face only 80 free third derivatives, leaving **20** that no coordinates can remove: curvature.

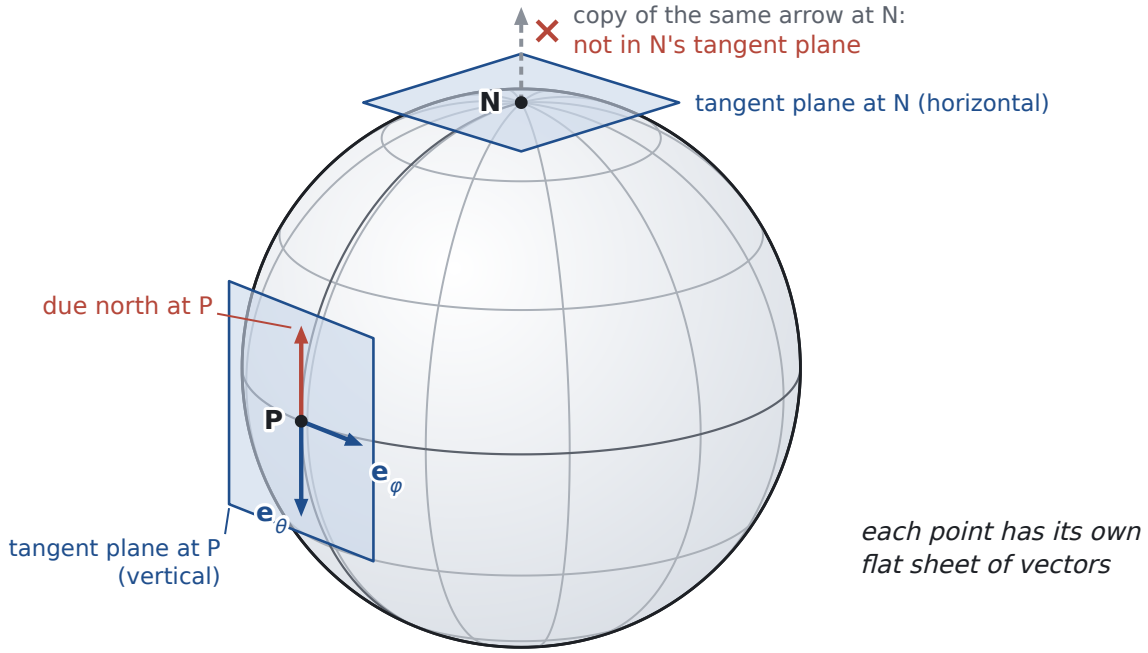


Figure 15.1. Every point of a curved space has its own tangent space. The tangent plane at the north pole is horizontal and the one at an equator point is vertical, so the vector "due north" at the equator does not even lie in the pole's tangent plane: comparing vectors at different points needs extra structure.

4. Worked Examples

Worked Example 15.1 — Why a sphere needs two charts

Latitude and longitude fail at the poles, where longitude is undefined and the metric determinant $a^4 \sin^2\theta$ vanishes, and along the seam where $\phi = 0$ meets $\phi = 2\pi$. Carroll uses two charts, stereographic projections from each pole (Carroll 1997, eqs. 2.4–2.6). The poles are ordinary points with bad coordinates, a lesson that returns at the Schwarzschild horizon.

Worked Example 15.2 — Local flatness on a sphere

Let $\rho = a\theta$ be the distance from the north pole along the surface. Eq. (8.2) becomes $ds^2 = d\rho^2 + a^2 \sin^2(\rho/a)\, d\phi^2$, and the Taylor series of sin gives

$$ds^2 = d\rho^2 + \rho^2\left(1 - \frac{\rho^2}{3a^2} + \cdots\right)d\phi^2. \tag{15.2}$$

The leading term is the flat polar metric; the first correction is second order in ρ and controlled by the curvature $1/a^2$, as the local-flatness theorem says.

5. Connection to Einstein's Equations

Einstein's equations are written in some chart, so they must be tensor equations to mean the same in every chart; the 20 numbers the counting leaves are the Riemann tensor's independent components (Section 19), half of which the field equations constrain.

Needs further study. Manifolds, charts and tangent spaces are made rigorous in a differential geometry course or the first chapters of graduate GR notes such as Carroll's or Tong's; Cambridge advises working through the differential geometry before graduate GR (Cambridge Part III preparation).

6. What I Should Remember

- A manifold looks like flat $\mathbb{R}^n$ up close; coordinates are labels, and some manifolds need several charts.
- A vector lives in the tangent space at one point; vectors at different points cannot be compared without a connection.
- At any point you can make the metric Minkowski with vanishing first derivatives, but 20 second-derivative numbers remain: curvature. This local flatness is the equivalence principle in mathematical form.

SECTION 16

The covariant derivative and Christoffel symbols

Difficulty: 8/10. The idea is one correction term, but the index bookkeeping is long and new.

1. What Is This Mathematics?

To ask whether a vector field is changing, you compare arrows at neighbouring points. In curved coordinates or on a curved surface the basis arrows themselves turn, so the ordinary derivative of the components mixes "the arrow really changed" with "my reference arrows turned". The **covariant derivative** ∇ removes the second effect. ANALOGY From a merry-go-round a fixed lamp post seems to swing around you; correcting for your own rotation shows it never moved. The correction coefficients, the turning rates of the coordinate basis, are the **Christoffel symbols** $\Gamma^{\lambda}_{\mu\nu}$ (capital gamma).

2. Why Does General Relativity Need It?

Physical laws must be tensor equations, and the partial derivative of a vector's components is not a tensor in curved coordinates; ∇ is. Most special-relativistic laws carry over by replacing ∂ with ∇, Carroll's "comma-goes-to-semicolon rule" (in the notation $V^{\nu}{}_{,\mu} \equiv \partial_\mu V^\nu$, $V^{\nu}{}_{;\mu} \equiv \nabla_\mu V^\nu$): energy conservation $\partial_\mu T^{\mu\nu} = 0$ becomes $\nabla_\mu T^{\mu\nu} = 0$ (Carroll 1997, eq. 4.23). The rule needs care where two derivatives act, because covariant derivatives do not commute (Section 19); in Carroll's words, "life is not always so easy".

3. How Does the Mathematics Work?

In flat space, define the Christoffel symbols by how the coordinate basis vectors change:

$$\partial_\mu \mathbf{e}_\nu = \Gamma^\lambda_{\mu\nu}\, \mathbf{e}_\lambda. \tag{16.1}$$

Differentiate a whole vector $\mathbf{V} = V^\nu \mathbf{e}_\nu$ with the product rule, then rename dummy indices ($\lambda \leftrightarrow \nu$) in the second term:

$$\partial_\mu \mathbf{V} = (\partial_\mu V^\nu)\mathbf{e}_\nu + V^\nu \Gamma^\lambda_{\mu\nu}\mathbf{e}_\lambda = \left(\partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda\right)\mathbf{e}_\nu. \tag{16.2}$$

The bracket is the covariant derivative. For a dual vector the correction has a minus sign, and a general tensor gets $+\Gamma$ for each upper index and $-\Gamma$ for each lower one (Carroll 1997, eqs. 3.1, 3.12, 3.13):

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda, \qquad \nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma^\lambda_{\mu\nu}\omega_\lambda. \tag{16.3}$$

On a curved manifold Eq. (16.1) has no meaning, because there is no surrounding space; instead GR fixes Γ by two demands. The connection is **torsion-free**, $\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}$, and **metric-compatible**, $\nabla_\rho g_{\mu\nu} = 0$, so the derivative preserves lengths and angles. Write $\nabla g = 0$ three times with the indices cycled, $\rho\mu\nu$, $\mu\nu\rho$ and $\nu\rho\mu$; add the second and third and subtract the first. Using the symmetry of Γ, four of the six Γ terms cancel in pairs and the other two are equal, leaving

$$\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu} = 2\,\Gamma^\lambda_{\mu\nu}\, g_{\lambda\rho}. \tag{16.4}$$

Multiply by $\frac{1}{2}g^{\sigma\rho}$ and use $g^{\sigma\rho}g_{\lambda\rho} = \delta^\sigma_\lambda$:

$$\Gamma^\sigma_{\mu\nu} = \tfrac{1}{2}\, g^{\sigma\rho}\left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right). \tag{16.5}$$

Carroll's advice: "commit it to memory" (Carroll 1997, eq. 3.21). Symmetric in its lower pair, Γ has $4 \times 10 = 40$ independent components in four dimensions.

> **Common confusion.** The Christoffel symbols are not a tensor. They vanish in Cartesian coordinates on a flat plane but not in polar coordinates, and a tensor that vanishes in one coordinate system vanishes in all. So $\Gamma \neq 0$ does not mean curvature, and $\Gamma = 0$ at a point does not mean flatness (Tong GR notes, ch. 1).

4. Worked Examples

Worked Example 16.1 — Polar coordinates from the formula

With $g_{rr} = 1$, $g_{\theta\theta} = r^2$ and inverse $g^{rr} = 1$, $g^{\theta\theta} = 1/r^2$, the only nonzero derivative is $\partial_r g_{\theta\theta} = 2r$. Eq. (16.5) gives

$$\begin{aligned} \Gamma^r_{\theta\theta} &= \tfrac{1}{2}g^{rr}\left(0 + 0 - \partial_r g_{\theta\theta}\right) = \tfrac{1}{2}(1)(-2r) = -r, \\ \Gamma^\theta_{r\theta} &= \Gamma^\theta_{\theta r} = \tfrac{1}{2}g^{\theta\theta}\,\partial_r g_{\theta\theta} = \tfrac{1}{2}\cdot\tfrac{1}{r^2}\cdot 2r = \tfrac{1}{r}, \end{aligned} \tag{16.6}$$

and all others vanish: the turning rates of the basis vectors found in Section 3, $\partial_\theta \mathbf{e}_\theta = -r\,\mathbf{e}_r$ and $\partial_\theta \mathbf{e}_r = \frac{1}{r}\mathbf{e}_\theta$.

Worked Example 16.2 — A constant field

The uniform field $\hat{\mathbf{x}}$ has polar components $V^r = \cos\theta$, $V^\theta = -\sin\theta/r$, whose partial derivatives are not zero. Add the corrections of Eq. (16.3):

$$\begin{aligned} \nabla_\theta V^r &= -\sin\theta + \Gamma^r_{\theta\theta}V^\theta = -\sin\theta + (-r)\left(-\tfrac{\sin\theta}{r}\right) = 0, \\ \nabla_r V^\theta &= \tfrac{\sin\theta}{r^2} + \Gamma^\theta_{r\theta}V^\theta = \tfrac{\sin\theta}{r^2} - \tfrac{\sin\theta}{r^2} = 0, \end{aligned} \tag{16.7}$$

and likewise $\nabla_r V^r = 0$ and $\nabla_\theta V^\theta = -\cos\theta/r + \cos\theta/r = 0$. A field that does not change has zero covariant derivative, even though its components change.

Worked Example 16.3 — The sphere

For $ds^2 = a^2(d\theta^2 + \sin^2\theta\, d\phi^2)$, the only nonzero derivative is $\partial_\theta g_{\phi\phi} = 2a^2 \sin\theta\cos\theta$, so $\Gamma^\theta_{\phi\phi} = \frac{1}{2} \cdot \frac{1}{a^2} \cdot (-2a^2 \sin\theta\cos\theta) = -\sin\theta\cos\theta$ and $\Gamma^\phi_{\theta\phi} = \frac{1}{2} \cdot \frac{1}{a^2\sin^2\theta} \cdot 2a^2 \sin\theta\cos\theta = \cot\theta$. The radius a cancels: Christoffel symbols cannot tell a big sphere from a small one, but curvature can (Section 19).

Worked Example 16.4 — The divergence formula

Contracting Eq. (16.5) (set $\sigma = \mu$ and sum: the first and third terms cancel after swapping the dummy names μ and ρ in one of them, and for a diagonal metric what is left is $\frac{1}{2}\sum_\mu \partial_\lambda g_{\mu\mu}/g_{\mu\mu} = \frac{1}{2}\partial_\lambda \ln\prod_\mu |g_{\mu\mu}|$, where the product is $|g|$) gives $\Gamma^\mu_{\mu\lambda} = \partial_\lambda \ln\sqrt{|g|}$, so $\nabla_\mu V^\mu = \frac{1}{\sqrt{|g|}}\partial_\mu(\sqrt{|g|}\, V^\mu)$ (Carroll 1997, eqs. 3.26–3.28). The field $x\hat{\mathbf{x}} + y\hat{\mathbf{y}}$ has Cartesian divergence 2 and polar components $V^r = r$, $V^\theta = 0$: the naive $\partial_r V^r = 1$ is wrong, while with $\sqrt{g} = r$, $\frac{1}{r}\partial_r(r \cdot r) = 2$ is right.

5. Connection to Einstein's Equations

Christoffel symbols enter the geodesic equation (Section 18), every covariant derivative including $\nabla_\mu T^{\mu\nu} = 0$, and the Riemann tensor (Section 19). In weak, static gravity $\Gamma^i_{00} \approx \partial_i \Phi/c^2$ APPROXIMATE (weak static field, $x^0 = ct$): the Christoffel symbol *is* Newton's gravitational field, which freely falling coordinates remove at any chosen point.

6. What I Should Remember

- $\nabla = \partial + \Gamma$-correction, $+\Gamma$ for each upper index and $-\Gamma$ for each lower; "comma goes to semicolon", $\partial \to \nabla$, carries most special-relativistic laws into GR.
- $\Gamma^\sigma_{\mu\nu} = \frac{1}{2}g^{\sigma\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$ follows from symmetry and $\nabla g = 0$.
- Christoffel symbols are not tensors: nonzero in flat polar coordinates, zero at any chosen point in curved space.

SECTION 17

Parallel transport

Difficulty: 7/10. A small system of first-order ODEs with Christoffel symbols as coefficients.

1. What Is This Mathematics?

Parallel transport carries an arrow along a path keeping it "as constant as possible", judged by the surface itself, not an outside space. On a flat sheet an arrow carried around any loop returns unchanged; on a sphere it returns rotated, and that rotation, the **holonomy**, is a fingerprint of curvature (Figure 17.1).

2. Why Does General Relativity Need It?

A freely falling gyroscope's spin axis is parallel-transported, so in curved spacetime it drifts: geodetic precession, measured by Gravity Probe B. Parallel transport also defines geodesics and, through its failure around small loops, the Riemann tensor.

3. How Does the Mathematics Work?

Along a path $x^\mu(\lambda)$, the covariant derivative in the direction of travel is $D/d\lambda = (dx^\sigma/d\lambda)\nabla_\sigma$. Parallel transport sets it to zero; by the chain rule $(dx^\sigma/d\lambda)\partial_\sigma V^\mu = dV^\mu/d\lambda$, so

$$\frac{DV^\mu}{d\lambda} = \frac{dV^\mu}{d\lambda} + \Gamma^\mu_{\nu\sigma}\frac{dx^\nu}{d\lambda}V^\sigma = 0 \tag{17.1}$$

(Carroll 1997, eq. 3.32). These first-order ODEs fix the vector along the whole path from its starting value, and metric compatibility preserves lengths and angles.

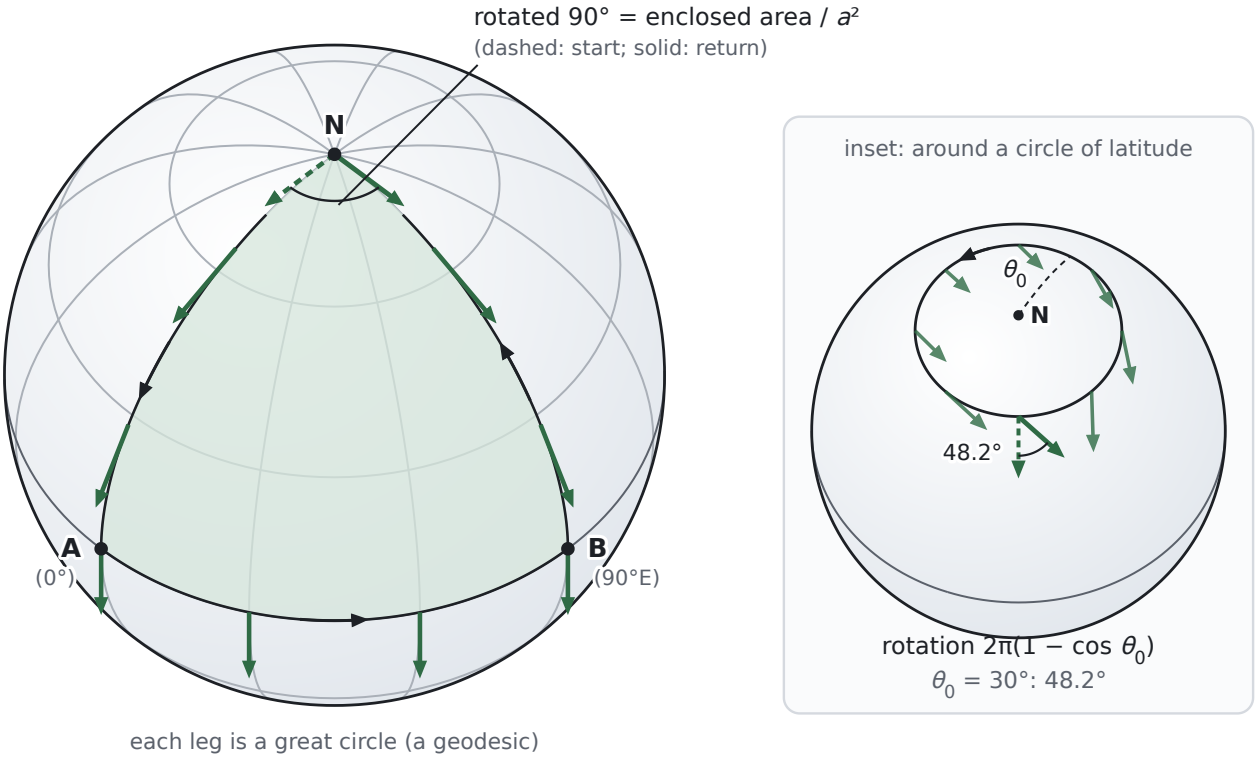


Figure 17.1. Parallel transport around a closed loop on a sphere. An arrow carried down a meridian, along the equator and back up another meridian, never turning relative to the surface, returns rotated by 90°, equal to the enclosed area divided by a². The inset shows transport around a circle of latitude.

4. Worked Examples

Worked Example 17.1 — No holonomy in a flat plane

Carry $\hat{\mathbf{x}}$ around the circle $r = R$, $\theta = \lambda$, so $dr/d\lambda = 0$ and $d\theta/d\lambda = 1$, with components $V^r = \cos\theta$, $V^\theta = -\sin\theta/R$. The r equation: $-\sin\theta + \Gamma^r_{\theta\theta}(1)V^\theta = -\sin\theta + (-R)(-\sin\theta/R) = 0$. The θ equation: $-\cos\theta/R + \Gamma^\theta_{\theta r}(1)V^r = -\cos\theta/R + \cos\theta/R = 0$. The components change, yet the arrow stays $\hat{\mathbf{x}}$.

Worked Example 17.2 — Around a circle of latitude

On the sphere take $\theta = \theta_0$ fixed and $\phi = \lambda$ from 0 to 2π. With $\Gamma^\theta_{\phi\phi} = -\sin\theta_0\cos\theta_0$ and $\Gamma^\phi_{\phi\theta} = \cot\theta_0$, Eq. (17.1) reads

$$\frac{dV^\theta}{d\lambda} - \sin\theta_0\cos\theta_0\,V^\phi = 0, \qquad \frac{dV^\phi}{d\lambda} + \cot\theta_0\,V^\theta = 0. \tag{17.2}$$

Differentiate the first and substitute the second: $d^2V^\theta/d\lambda^2 = -\cos^2\theta_0\,V^\theta$, simple harmonic motion with angular frequency $\cos\theta_0$. Starting due south ($V^\theta = 1$, $V^\phi = 0$), the solution is $V^\theta = \cos(\lambda\cos\theta_0)$, and the first equation gives V^ϕ. Along unit arrows pointing south and east, the arrow's components are proportional to $(\cos(\lambda\cos\theta_0), -\sin(\lambda\cos\theta_0))$: relative to these local directions, which return unchanged, it turns at the rate $\cos\theta_0$, by $2\pi\cos\theta_0$ per loop. A full turn changes nothing, so that equals a turn of $2\pi - 2\pi\cos\theta_0$ the other way: after one loop the arrow has rotated by

$$\Delta\alpha = 2\pi\,(1-\cos\theta_0)\,, \tag{17.3}$$

the enclosed polar-cap area $2\pi a^2(1-\cos\theta_0)$ divided by a^2: 48.2° at colatitude 30°, and on the equator, a geodesic, a full turn, the same as none. For the right-angled triangle of Figure 17.1 the arrow returns rotated by 90°: the angle excess $270^\circ - 180^\circ$, and also the area $\frac{1}{8}(4\pi a^2)$ divided by a^2, namely $\pi/2$. Holonomy, angle excess and enclosed curvature are one number.

Worked Example 17.3 — A gyroscope in orbit

Gravity Probe B's gyroscopes, in a 642 km polar orbit from 2004 to 2005, measured a geodetic drift, from parallel transport along an orbit that closes in space but not in spacetime (each lap happens later), of -6601.8 ± 18.3 milliarcseconds per year against GR's -6606.1 (Everitt et al. 2011): agreement to 0.07%, within the 0.28% uncertainty. A Foucault pendulum's turning is the same geometry on Earth's curved surface, but only an ANALOGY: Earth's rotation, not spacetime curvature, causes it.

5. Connection to Einstein's Equations

Around a tiny loop with sides $\delta a\,A^\nu$ and $\delta b\,B^\mu$, the change in a transported vector is $\delta V^\rho = (\delta a)(\delta b)\,A^\nu B^\mu R^\rho{}_{\sigma\mu\nu} V^\sigma$ (Carroll 1997, eq. 3.63). That is the Riemann tensor of Section 19, whose contraction stands on the left of Einstein's equations.

6. What I Should Remember

- Parallel transport solves $dV^\mu/d\lambda + \Gamma^\mu_{\nu\sigma}(dx^\nu/d\lambda)V^\sigma = 0$ and preserves lengths and angles.
- In flat space a transported vector returns unchanged; in curved space it returns rotated by the enclosed curvature, on a sphere area$/a^2$: $2\pi(1-\cos\theta_0)$ for a circle of latitude, 90° for the octant triangle.
- Gyroscopes in free fall are parallel-transported; Gravity Probe B measured the result.

SECTION 18

Geodesics and the geodesic equation

Difficulty: 7/10. One line, but a coupled nonlinear system of second-order ODEs.

1. What Is This Mathematics?

A **geodesic** is the straightest possible path in a curved space: on a sphere, a great circle. Carroll gives two equivalent definitions (Carroll 1997, §3): a curve that parallel-transports its own tangent vector ("never turns"), and a curve that makes its length, or for a clock its proper time, stationary.

2. Why Does General Relativity Need It?

It is GR's law of motion: planets, photons and astronauts in free fall follow geodesics, with gravity in the Christoffel symbols rather than a force term.

3. How Does the Mathematics Work?

Put $V^\mu = dx^\mu/d\lambda$ into the parallel-transport equation (17.1):

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\nu\sigma}\frac{dx^\nu}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0. \tag{18.1}$$

Carroll calls it "another one which you should memorize" (Carroll 1997, eq. 3.47). In Cartesian coordinates of flat space $\Gamma = 0$ and $d^2x^\mu/d\lambda^2 = 0$: straight lines. The form (18.1) holds only for an **affine** parameter, $\lambda = a\tau + b$. Massive particles use $\lambda = \tau$, with $g_{\mu\nu}u^\mu u^\nu = -c^2$; light has $d\tau = 0$, so it uses another affine parameter, with $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$. Timelike geodesics locally *maximise* proper time: the twin paradox in geometric form.

4. Worked Examples

Worked Example 18.1 — A straight line in polar coordinates

With $\Gamma^r_{\theta\theta} = -r$ and $\Gamma^\theta_{r\theta} = 1/r$, Eq. (18.1) gives $\ddot{r} - r\dot{\theta}^2 = 0$ and $\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} = 0$, the same equations as "zero acceleration" in Worked Example 3.2 (the factor 2 because $\Gamma^\theta_{r\theta}$ and $\Gamma^\theta_{\theta r}$ both appear). Test the line $x = b$, $y = s$, so $r = \sqrt{b^2 + s^2}$ and $\theta = \arctan(s/b)$:

$$\dot{r} = \frac{s}{r}, \quad \ddot{r} = \frac{b^2}{r^3}, \quad \dot{\theta} = \frac{b}{r^2}, \quad \ddot{\theta} = -\frac{2bs}{r^4}. \tag{18.2}$$

Then $\ddot{r} - r\dot{\theta}^2 = b^2/r^3 - b^2/r^3 = 0$ and $\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} = -2bs/r^4 + 2bs/r^4 = 0$. The Γ terms make a line that looks curved in polar language come out straight, and $r^2\dot{\theta} = b$ is a conserved angular momentum.

Worked Example 18.2 — Great circles, not parallels

On the sphere the geodesic equations are $\ddot{\theta} - \sin\theta\cos\theta\,\dot{\phi}^2 = 0$ and $\ddot{\phi} + 2\cot\theta\,\dot{\theta}\dot{\phi} = 0$ (Tong GR notes, ch. 3). A meridian (ϕ constant, $\theta = s/a$) satisfies both. So does the equator ($\theta = \pi/2$, where $\sin\theta\cos\theta = 0$). A circle of latitude with $\theta_0 \neq \pi/2$ has $\ddot{\theta} = 0$ but $\sin\theta_0\cos\theta_0\,\dot{\phi}^2 \neq 0$, so it fails: an aircraft flying due west along a parallel is not on the shortest route (Worked Example 8.2).

Worked Example 18.3 — Radial infall

In Schwarzschild spacetime, with $f = 1 - r_s/r$, the time component of the geodesic equation says $e \equiv f\,dt/d\tau$ is constant; $e = 1$ for a body dropped from rest far away. Put $dt/d\tau = e/f$ into the normalisation $-fc^2(dt/d\tau)^2 + f^{-1}(dr/d\tau)^2 = -c^2$ for radial motion and multiply by f:

$$\left(\frac{dr}{d\tau}\right)^2 = c^2(e^2 - f) = \frac{c^2 r_s}{r} = \frac{2GM}{r}. \tag{18.3}$$

Differentiating with respect to τ gives $d^2r/d\tau^2 = -GM/r^2$, Newton's law, EXACT in these variables. Caution: r is Schwarzschild's areal radius, not a ruler distance, and τ is the faller's clock; the Newtonian look is special to these variables.

5. Connection to Einstein's Equations

Einstein's equations determine the metric; the geodesic equation tells matter and light how to move in it. In the weak-field, slow-motion limit it becomes $\ddot{\mathbf{x}} = -\nabla\Phi$ (Equation 11 of Part IV); in Schwarzschild spacetime it gives circular orbits, the photon sphere at $3GM/c^2$ and the innermost stable orbit at $6GM/c^2$ (Equations 5 and 16).

6. What I Should Remember

- A geodesic parallel-transports its own tangent: $\ddot{x}^\mu + \Gamma^\mu_{\nu\sigma}\dot{x}^\nu\dot{x}^\sigma = 0$, with an affine parameter.
- Free particles follow timelike geodesics, which locally maximise proper time, and light follows null geodesics; there is no gravitational force term.
- In polar coordinates the Γ terms are "fictitious forces"; in curved spacetime they cannot be removed everywhere at once.

SECTION 19

Curvature: from two dimensions to the Riemann tensor

Difficulty: 9/10. One commutator to define, but four indices and long computations.

1. What Is This Mathematics?

Picture an ant that can never leave a surface. **Intrinsic** curvature is whatever it can detect from within (triangle angle sums, circle circumferences, the rotation of a carried arrow); **extrinsic** curvature, how the surface bends in the surrounding space, it cannot see. A cylinder unrolls flat without stretching, so it is intrinsically flat; a sphere cannot, which is why every world map distorts something (Carroll 1997, ch. 3). A surface needs one number per point, the **Gaussian curvature** K, which by Gauss's *Theorema Egregium* (1827) the metric alone determines.

In more dimensions every plane through a point can curve differently, and the full information is the **Riemann tensor** $R^{\rho}{}_{\sigma\mu\nu}$. Partial derivatives commute (Clairaut); covariant derivatives in curved space do not, and their failure to commute is the curvature.

2. Why Does General Relativity Need It?

The Christoffel symbols can be transformed away at a point; curvature cannot. Gravity's coordinate-independent content is tidal: neighbouring free-fall paths accelerate relative to one another (Section 20), and that relative acceleration is the Riemann tensor.

3. How Does the Mathematics Work?

The Riemann tensor is defined by the commutator of covariant derivatives acting on any vector field (Carroll 1997, eq. 3.66):

$$[\nabla_\mu, \nabla_\nu]V^\rho \equiv \nabla_\mu\nabla_\nu V^\rho - \nabla_\nu\nabla_\mu V^\rho = R^{\rho}{}_{\sigma\mu\nu}V^\sigma. \tag{19.1}$$

To compute it, expand $\nabla_\mu(\nabla_\nu V^\rho)$ with Eq. (16.3); $\nabla_\nu V^\rho$ has one upper and one lower index, so it gets a $+\Gamma$ and a $-\Gamma$ term. Subtract the same expression with $\mu \leftrightarrow \nu$. The $\partial_\mu\partial_\nu V^\rho$ terms cancel by Clairaut; the terms with one derivative of V are symmetric in $\mu\nu$ and cancel; the $\Gamma^\lambda_{\mu\nu}\nabla_\lambda V^\rho$ term cancels because Γ is symmetric. What survives multiplies V^σ with no derivative on it:

$$\begin{aligned} R^{\rho}{}_{\sigma\mu\nu} = {} & \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} \\ & + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \end{aligned} \tag{19.2}$$

Index roles: ρ is the component of the output, σ the slot for the input vector, and $\mu\nu$ the plane of the loop; reversing the loop reverses the change, so $R^{\rho}{}_{\sigma\mu\nu} = -R^{\rho}{}_{\sigma\nu\mu}$. With the first index lowered, $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^{\lambda}{}_{\sigma\mu\nu}$ obeys

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} = -R_{\rho\sigma\nu\mu} = R_{\mu\nu\rho\sigma}, \tag{19.3}$$

plus the cyclic identity of Section 23. These leave $n^2(n^2-1)/12$ independent components: 1 in two dimensions, 6 in three, 20 in four (Carroll 1997, eq. 3.85), as Section 15's counting found. In two dimensions $K = R_{1212}/\det g$.

4. Worked Examples

Worked Example 19.1 — The sphere, term by term

Set $\rho=\theta$, $\sigma=\phi$, $\mu=\theta$, $\nu=\phi$ in Eq. (19.2), with the sphere's $\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta$ and $\Gamma^\phi_{\theta\phi} = \cot\theta$:

$$\begin{aligned} \partial_\theta \Gamma^\theta_{\phi\phi} &= \sin^2\theta - \cos^2\theta, \qquad \partial_\phi \Gamma^\theta_{\theta\phi} = 0, \\ \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{\phi\phi} &= 0, \qquad \Gamma^\theta_{\phi\lambda}\Gamma^\lambda_{\theta\phi} = (-\sin\theta\cos\theta)(\cot\theta) = -\cos^2\theta, \\ R^\theta{}_{\phi\theta\phi} &= (\sin^2\theta - \cos^2\theta) - 0 + 0 - (-\cos^2\theta) = \sin^2\theta. \end{aligned} \tag{19.4}$$

Lowering, $R_{\theta\phi\theta\phi} = a^2\sin^2\theta$, and $K = R_{\theta\phi\theta\phi}/\det g = a^2\sin^2\theta/(a^4\sin^2\theta) = 1/a^2$, matching Worked Example 8.3 (Carroll 1997, eqs. 3.102–3.103).

Worked Example 19.2 — Flat after all

For the polar plane, $R^r{}_{\theta r\theta} = \partial_r\Gamma^r_{\theta\theta} - \partial_\theta\Gamma^r_{r\theta} + \Gamma^r_{r\lambda}\Gamma^\lambda_{\theta\theta} - \Gamma^r_{\theta\lambda}\Gamma^\lambda_{r\theta} = (-1) - 0 + 0 - (-r)(1/r) = 0$. Nonzero Christoffel symbols, zero curvature: Riemann is blind to the choice of coordinates. A cylinder, $ds^2 = dz^2 + a^2 d\phi^2$, has constant metric components, so every Γ and every curvature component vanishes.

5. Connection to Einstein's Equations

Only Riemann's contractions appear in Einstein's equations (Section 21): of its 20 components, the 10 in the Ricci tensor are fixed locally by matter, while the other 10, the Weyl tensor, carry tides and gravitational waves through empty space. Equation 6 of Part IV uses Riemann to show that the Schwarzschild horizon is not a place of infinite curvature.

Needs further study. Computing all components of Eq. (19.2) for a realistic metric takes hours by hand; relativists use computer algebra (Mathematica, Maple, SymPy) after doing a few by hand (Moore 2006). The full theory of curvature belongs to a differential geometry course.

6. What I Should Remember

- Intrinsic curvature is measurable from inside; a cylinder is flat, a sphere is not.
- $R^\rho{}_{\sigma\mu\nu}$, built from $\partial\Gamma$ and $\Gamma\Gamma$ (second derivatives of the metric), measures the failure of covariant derivatives to commute, equivalently the rotation of a vector carried around a small loop; it has 20 independent components in 4D.
- The sphere has $R^\theta{}_{\phi\theta\phi} = \sin^2\theta$ and $K = 1/a^2$; flat polar coordinates have zero curvature despite nonzero Γ.

SECTION 20

Geodesic deviation

Difficulty: 8/10. Second covariant derivatives, though the Newtonian version is a Taylor expansion you can do now.

1. What Is This Mathematics?

Two travellers start on the equator a short distance apart, both heading due north; neither ever turns, yet they meet at the North Pole (Figure 20.1). Freely falling objects do the same in spacetime: in a falling elevator, balls side by side drift together and balls one above the other drift apart. That relative acceleration is the **tidal** effect, computed from the Riemann tensor by the **geodesic deviation equation**.

2. Why Does General Relativity Need It?

The equivalence principle removes gravity at a point but not its variation across a region, and that variation is how curvature is physically measured: ocean tides, bodies stretched near black holes, every gravitational-wave detector (Carroll 1997, eq. 3.113; Tong GR notes, §5.2).

3. How Does the Mathematics Work?

Take a family of geodesics with four-velocity u^μ and let ξ^μ (xi) be the small separation vector from one to its neighbour. Using the geodesic equation, zero torsion and the commutator (19.1), Carroll derives (Carroll 1997, eq. 3.113; Tong GR notes, eq. 3.36)

$$\frac{D^2\xi^\mu}{d\tau^2} = R^\mu{}_{\nu\rho\sigma}\, u^\nu u^\rho \xi^\sigma, \tag{20.1}$$

where $D/d\tau = u^\alpha \nabla_\alpha$ is the covariant derivative along the reference geodesic. Since $R^\mu{}_{\nu\rho\sigma} = -R^\mu{}_{\nu\sigma\rho}$, some books write it with the last two indices swapped and a minus sign.

Now the Newtonian version, by Taylor expansion. Two particles at $\mathbf{x}$ and $\mathbf{x}+\boldsymbol{\xi}$ obey $\ddot{x}^i = -\partial_i\Phi(\mathbf{x})$ and $\ddot{x}^i + \ddot{\xi}^i = -\partial_i\Phi(\mathbf{x}+\boldsymbol{\xi})$. Expanding $\partial_i\Phi(\mathbf{x}+\boldsymbol{\xi}) \approx \partial_i\Phi(\mathbf{x}) + \xi^j\partial_j\partial_i\Phi$ and subtracting:

$$\frac{d^2\xi^i}{dt^2} = -\left(\partial_i\partial_j\Phi\right)\xi^j. \tag{20.2}$$

The matrix $\partial_i\partial_j\Phi$ is the **Newtonian tidal tensor**. Matching (20.1) with $u^\mu \approx (c,0,0,0)$ to (20.2) gives, in this document's convention,

$$R^i{}_{0j0} \approx \frac{\partial_i\partial_j\Phi}{c^2} \tag{20.3}$$

APPROXIMATE (weak field, slow motion). Its trace, $\nabla^2\Phi/c^2$, is the part that Poisson's equation ties to matter.

Common confusion. Books order the indices of Eq. (20.1) differently, so never mix lines from a source with another ordering into a calculation that uses Carroll's. A reliable check in any convention: meridians on a sphere must *converge*.

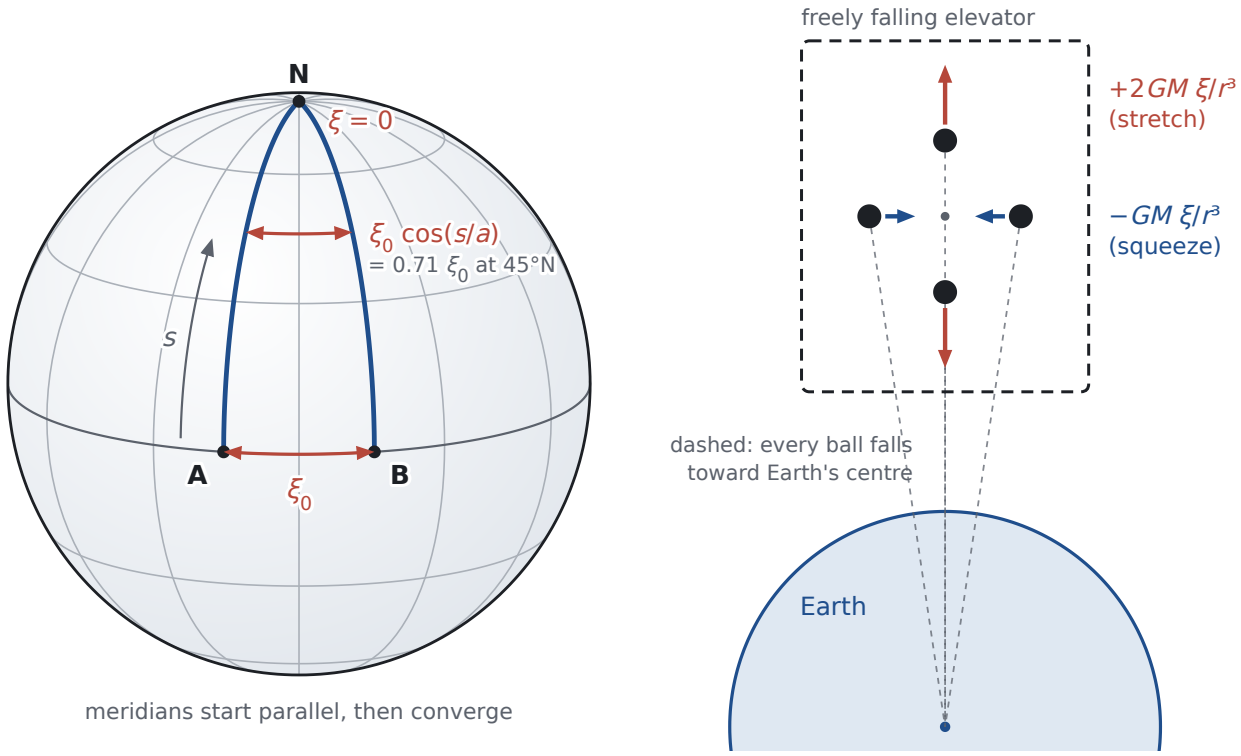


Figure 20.1. Geodesic deviation. Left: meridians that start parallel at the equator converge at the pole with no force acting. Right: in a freely falling elevator, test balls separate vertically and approach horizontally, the tidal pattern that is spacetime curvature.

4. Worked Examples

Worked Example 20.1 — Meridians on a sphere

Meridians at longitudes 0 and $\Delta\phi$ are separated at colatitude θ by $\xi = a\,\Delta\phi\sin\theta$. Measuring arc length s northward from the equator, $\theta = \pi/2 - s/a$, so $\xi(s) = \xi_0\cos(s/a)$ and $\xi'' = -\xi/a^2$. The deviation equation gives the same, with the orthonormal curvature component equal to $K = 1/a^2$ and the minus sign meaning convergence (Tong GR notes, eqs. 3.37–3.38). Initially parallel meridians meet after $s = \pi a/2$, about 10 008 km on Earth.

Worked Example 20.2 — The tidal tensor of a point mass

With $\Phi = -GM/r$, Worked Example 2.2 gave $\partial_i\Phi = GMx_i/r^3$; one more derivative gives

$$\partial_i\partial_j\Phi = \frac{GM}{r^3}\left(\delta_{ij} - \frac{3x_ix_j}{r^2}\right). \tag{20.4}$$

At $\mathbf{x} = (r,0,0)$ the radial entry is $\frac{GM}{r^3}(1-3) = -\frac{2GM}{r^3}$, so by Eq. (20.2) $\ddot{\xi}_{\text{radial}} = +\frac{2GM}{r^3}\xi$: stretching. Each transverse entry is $+\frac{GM}{r^3}$, so $\ddot{\xi}_\perp = -\frac{GM}{r^3}\xi$: squeezing. The trace, $-2+1+1=0$, is $\nabla^2\Phi = 0$ in empty space. At Earth's surface $2GM/R^3 = 3.08\times10^{-6}$ s^{-2}, so the head and feet of a 2 m person are pulled apart at 6.2×10^{-6} m/s^2, far too little to feel.

Worked Example 20.3 — The same tides in GR

For Schwarzschild spacetime, in the orthonormal frame of a static observer, the curvature components are $R^{\hat{r}}{}_{\hat{t}\hat{r}\hat{t}} = -r_s/r^3$ and $R^{\hat{\theta}}{}_{\hat{t}\hat{\theta}\hat{t}} = R^{\hat{\phi}}{}_{\hat{t}\hat{\phi}\hat{t}} = +r_s/2r^3$ (hats mark components along unit vectors). A particle released from rest beside the static observer has exactly $u = (c,0,0,0)$ in this frame, and Eq. (20.1) gives radial acceleration $+c^2r_s\xi/r^3 = +2GM\xi/r^3$ and transverse $-GM\xi/r^3$, EXACT for small separations between freely falling neighbours momentarily at rest or falling radially (a radial boost leaves these components

unchanged), with r the areal radius (Dray, *Geometry of General Relativity*). Newton's tidal formula survives unchanged. Two observers hovering at fixed r are not in free fall, so this is not their relative acceleration.

5. Connection to Einstein's Equations

Einstein's equations constrain the trace of the tidal operator, $R^{\mu}{}_{\nu\mu\sigma}$, the Ricci tensor (Section 21), as Poisson's equation constrains $\nabla^2\Phi$, the trace of the Newtonian tidal tensor. Gravitational-wave detectors are geodesic-deviation meters (Equation 18 of Part IV).

6. What I Should Remember

- $D^2\xi^\mu/d\tau^2 = R^\mu{}_{\nu\rho\sigma}u^\nu u^\rho \xi^\sigma$: relative acceleration of neighbours is curvature; on a sphere $\xi'' = -\xi/a^2$, so initially parallel meridians converge.
- Near a mass, free-fall neighbours are stretched radially ($+2GM\xi/r^3$) and squeezed sideways ($-GM\xi/r^3$), exactly as in Newton's tidal tensor.
- Tides are the part of gravity free fall cannot remove; check a book's sign convention with the sphere.

SECTION 21

The Ricci tensor, the Ricci scalar and the Einstein tensor

Difficulty: 8/10. The operations are contractions you already know; what is hard is their meaning.

1. What Is This Mathematics?

The **Ricci tensor** $R_{\mu\nu}$ is a trace of the Riemann tensor, a kind of average over directions. In Baez and Bunn's picture, a small ball of freely falling test particles, initially at rest relative to one another, has a volume V that starts to change as $\ddot{V}/V = -R_{tt}$ (local inertial coordinates, $c = 1$): "the Ricci tensor says how our ball of freely falling test particles starts changing in volume" (Baez & Bunn 2005). The rest of Riemann, the **Weyl tensor**, says how the ball changes shape (Figure 21.1). The **Ricci scalar** R is one more trace, and the **Einstein tensor** $G_{\mu\nu}$ is Ricci with its trace adjusted so that its divergence is automatically zero.

2. Why Does General Relativity Need It?

Ricci is the part of curvature that matter controls locally, which is what Einstein's equations say. In empty space Ricci vanishes but Weyl need not, which is how the Moon raises tides across a vacuum.

3. How Does the Mathematics Work?

Contract the first and third indices of Riemann, then contract again with the inverse metric (Carroll 1997, eqs. 3.90–3.92):

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}, \qquad R = g^{\mu\nu}R_{\mu\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu}. \tag{21.1}$$

Other contractions give nothing new (the first or last pair gives zero by antisymmetry). $R_{\mu\nu}$ is symmetric, so in four dimensions Riemann's 20 components split into 10 (Ricci) plus 10 (Weyl). Taking the trace of $G_{\mu\nu}$ with $g^{\mu\nu}g_{\mu\nu} = 4$ gives $G \equiv g^{\mu\nu}G_{\mu\nu} = R - 2R = -R$. So $G_{\mu\nu}$ is the **trace-reverse** of $R_{\mu\nu}$, and reversing again gets you back: $R_{\mu\nu} = G_{\mu\nu} - \tfrac{1}{2}G\,g_{\mu\nu}$.

In Baez and Bunn's units ($c = 8\pi G = 1$), Einstein's equation applied to the small ball reads $\ddot{V}/V = -\frac{1}{2}(\rho + P_x + P_y + P_z)$, where ρ is the energy density and P_x, P_y, P_z the pressures at the ball's centre (Baez & Bunn 2005). In SI units, for isotropic pressure p and mass density ρ, it becomes $\ddot{V}/V = -4\pi G(\rho + 3p/c^2)$ EXACT (for an infinitesimal ball initially at rest, in local inertial coordinates, with $\Lambda = 0$; a cosmological constant adds $+\Lambda c^2$ to the right side). Pressure gravitates.

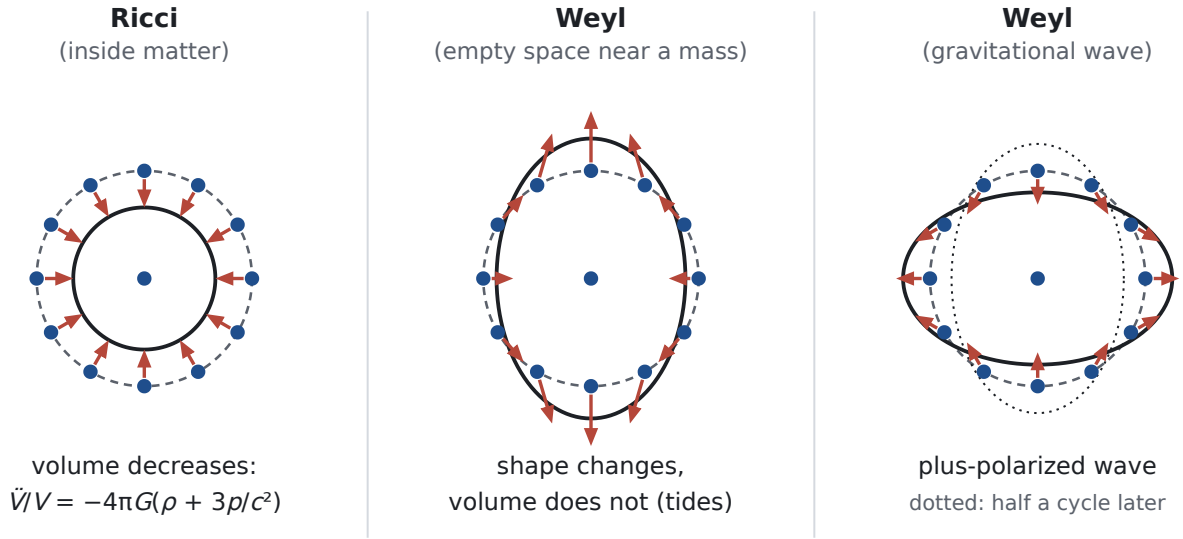


Figure 21.1. Ricci versus Weyl curvature. A small ball of freely falling particles shrinks in volume where there is matter (Ricci curvature), but in empty space only changes shape, stretching along one direction and squeezing across it (Weyl curvature: tides and gravitational waves).

4. Worked Examples

Worked Example 21.1 — Why two dimensions are dull

From Worked Example 19.1 and the symmetries, $R^{\phi}{}_{\theta\phi\theta} = 1$, so $R_{\theta\theta} = R^{\theta}{}_{\theta\theta\theta} + R^{\phi}{}_{\theta\phi\theta} = 0 + 1 = 1$, and similarly $R_{\phi\phi} = \sin^2\theta$, $R_{\theta\phi} = 0$. Then

$$R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{1}{a^2} + \frac{\sin^2\theta}{a^2\sin^2\theta} = \frac{2}{a^2}, \tag{21.2}$$

positive, so in this convention "spaces which everyone agrees to call positively curved actually have a positive Ricci scalar" (Carroll 1997, eq. 3.105). But $G_{\theta\theta} = 1 - \frac{1}{2}\cdot\frac{2}{a^2}\cdot a^2 = 0$ and likewise $G_{\phi\phi} = 0$. In two dimensions $G_{\mu\nu} = 0$ for every surface, so a two-dimensional "Einstein equation" would force $T_{\mu\nu} = 0$: no gravity at all.

Worked Example 21.2 — Empty but curved

For Schwarzschild spacetime, the orthonormal static-frame components (Worked Example 20.3) give

$$R_{\hat{t}\hat{t}} = R^{\hat{r}}{}_{\hat{t}\hat{r}\hat{t}} + R^{\hat{\theta}}{}_{\hat{t}\hat{\theta}\hat{t}} + R^{\hat{\phi}}{}_{\hat{t}\hat{\phi}\hat{t}} = -\frac{r_s}{r^3} + \frac{r_s}{2r^3} + \frac{r_s}{2r^3} = 0. \tag{21.3}$$

All ten components of $R_{\mu\nu}$ vanish outside the mass, yet Riemann does not: a ball of particles near Earth changes shape without changing volume.

5. Connection to Einstein's Equations

The left side of the field equations is $G_{\mu\nu} + \Lambda g_{\mu\nu}$. With $\Lambda = 0$ their trace gives $R = -(8\pi G/c^4)T$, and in vacuum they reduce to $R_{\mu\nu} = 0$, which Schwarzschild spacetime satisfies.

6. What I Should Remember

- $R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$ is a trace of Riemann; $R = g^{\mu\nu}R_{\mu\nu}$ is its trace; $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$.
- Ricci measures how a falling ball changes volume, Weyl how it changes shape, so Ricci $= 0$ does not mean flat: Schwarzschild spacetime is Ricci-flat but curved.
- In two dimensions $G_{\mu\nu} \equiv 0$; in four, $G_{\mu\nu}$ and $R_{\mu\nu}$ are trace-reverses of each other.

SECTION 22

The stress-energy tensor and covariant conservation

Difficulty: 7/10. The table itself reads quickly; "pressure is a flow of momentum" takes a while to absorb.

1. What Is This Mathematics?

The **stress-energy tensor** $T^{\mu\nu}$ is a symmetric 4×4 table of how much μ-momentum flows across a surface of constant x^ν, "0-momentum" meaning energy divided by c (Carroll 1997, §1): energy density in the top-left corner, energy flow along the top row, pressures on the space-block diagonal and shear stresses off it (Figure 22.1). Baez and Bunn: "It takes a while to figure out why pressure is really the flow of momentum, but it is eminently worth doing" (Baez & Bunn 2005).

2. Why Does General Relativity Need It?

It is the source of curvature, the right side of Einstein's equations: in GR energy, momentum flow and pressure gravitate, not only mass. Its conservation law, $\nabla_\mu T^{\mu\nu} = 0$, contains matter's equations of motion.

3. How Does the Mathematics Work?

With $x^0 = ct$, every component has units of J/m³ (equivalently pascals). The pieces are T^{00} (energy density), $T^{0i} = T^{i0}$ (energy flux divided by c, equal to c times momentum density) and T^{ij} (flux of i-momentum across a surface of constant x^j). For dust, pressureless matter of rest-frame mass density ρ, $T^{\mu\nu} = \rho u^\mu u^\nu$. A **perfect fluid** adds an isotropic pressure p:

$$T^{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) u^\mu u^\nu + p\, g^{\mu\nu}. \tag{22.1}$$

In special relativity, energy and momentum conservation is $\partial_\mu T^{\mu\nu} = 0$: whatever leaves a small box flows through its walls. In curved spacetime, by comma-goes-to-semicolon (Carroll 1997, eq. 4.23),

$$\nabla_\mu T^{\mu\nu} = 0. \tag{22.2}$$

This is local conservation. A conserved *total* energy needs a time symmetry (Section 24); an expanding universe has none, so photons lose energy to redshift with no global bookkeeping to restore it (Carroll 2010).

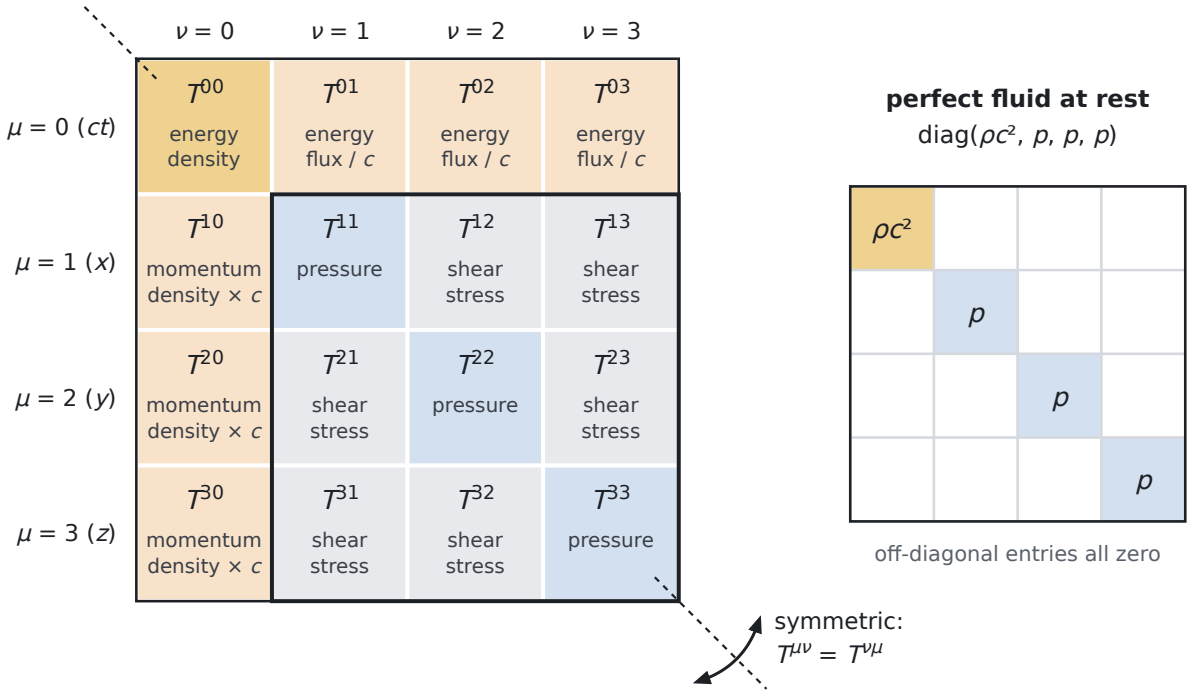


Figure 22.1. The layout of the stress-energy tensor. Energy density sits in the corner, energy flux and momentum density along the first row and column, and pressures and shear stresses in the 3 × 3 space block. A perfect fluid at rest is diagonal: ρc², p, p, p.

4. Worked Examples

Worked Example 22.1 — A perfect fluid at rest

In Minkowski spacetime the fluid's rest frame has $u^\mu = (c, 0, 0, 0)$, $g^{00} = -1$ and $g^{ii} = +1$. Eq. (22.1) gives

$$\begin{aligned} T^{00} &= \left(\rho + \frac{p}{c^2}\right) c^2 + p(-1) = \rho c^2, \\ T^{0i} &= 0, \qquad T^{ii} = 0 + p(+1) = p, \qquad T^{ij} = 0 \ (i \neq j). \end{aligned} \tag{22.3}$$

So $T^{\mu\nu} = \mathrm{diag}(\rho c^2, p, p, p)$. Pressure is usually a negligible source: for water at 1 atmosphere $p/\rho c^2 \approx 1.1 \times 10^{-15}$, and at the Sun's centre (density 1.622×10^5 kg/m³, pressure 2.477×10^{16} Pa; NASA Sun Fact Sheet) only 1.7×10^{-6}. In neutron stars pressure and energy density "contribute comparably" (Baez & Bunn 2005).

Worked Example 22.2 — The continuity equation

For slowly moving dust in flat spacetime, $T^{00} \approx \rho c^2$ and $T^{i0} \approx \rho c\, v^i$. The $\nu = 0$ component of $\partial_\mu T^{\mu\nu} = 0$ is $\partial_0 T^{00} + \partial_i T^{i0} = 0$; with $\partial_0 = \partial/\partial(ct)$ and a division by c,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{22.4}$$

the continuity equation: mass is neither created nor destroyed. In a weak gravitational field the $\nu = i$ components of Eq. (22.2) give Euler's fluid equation, with the gravitational force $-\rho \nabla \Phi$ supplied by the term $\Gamma^i_{00} T^{00}$.

Worked Example 22.3 — Cosmic dilution

For the expanding universe (Equation 19 of Part IV), with scale factor $a(t)$ and a fluid at rest in comoving coordinates, the $\nu = 0$ component of Eq. (22.2) works out to $\dot{\rho} + 3(\dot{a}/a)(\rho + p/c^2) = 0$ (Carroll 1997, §8).

With $p = w\rho c^2$, separate variables:

$$\frac{d\rho}{\rho} = -3(1+w)\frac{da}{a} \quad \Longrightarrow \quad \rho \propto a^{-3(1+w)}. \tag{22.5}$$

Matter ($w = 0$) dilutes as a^{-3}, with volume; radiation ($w = 1/3$) as a^{-4}, volume times each photon's redshift; a cosmological constant ($w = -1$) not at all.

5. Connection to Einstein's Equations

$T_{\mu\nu}$ is the entire right-hand side, multiplied by $8\pi G/c^4$. Because the left side's divergence vanishes identically (Section 23), the field equations *require* $\nabla_\mu T^{\mu\nu} = 0$: geometry will not accept a source that is not conserved.

6. What I Should Remember

- $T^{\mu\nu}$ is the flux of μ-momentum through surfaces of constant x^ν: energy density in the corner, stresses in the space block.
- A perfect fluid at rest is $\mathrm{diag}(\rho c^2, p, p, p)$, with trace $-\rho c^2 + 3p$; pressure gravitates, but matters only in extreme objects and cosmology.
- $\nabla_\mu T^{\mu\nu} = 0$ is local energy-momentum conservation and contains continuity and Euler's equation.

SECTION 23

The Bianchi identities and the field equations as differential equations

Difficulty: 9/10. Identities between derivatives of curvature, and the structure of a system of nonlinear PDEs.

1. What Is This Mathematics?

The **Bianchi identities** are statements about curvature that hold for every metric whatsoever, like the vector-calculus identity $\nabla \cdot (\nabla \times \mathbf{F}) = 0$: the divergence of a curl is always zero (the curl $\nabla \times \mathbf{F}$, met in Calculus III, measures how much a field circulates). MTW's slogan for them is "the boundary of a boundary is zero" (UW Physics 544 notes on MTW ch. 15). Their twice-contracted form says the Einstein tensor has zero divergence.

2. Why Does General Relativity Need It?

The contracted identity explains three things at once: why the field equations use $G_{\mu\nu}$ with exactly $\frac{1}{2}$, why they force energy-momentum conservation, and why their ten equations contain only six independent ones. It also rules out the first guess, $R_{\mu\nu} \propto T_{\mu\nu}$.

3. How Does the Mathematics Work?

The first Bianchi identity is algebraic and the second differential (Carroll 1997, eqs. 3.81 and 3.87):

$$\begin{aligned} &R^\rho{}_{\sigma\mu\nu} + R^\rho{}_{\mu\nu\sigma} + R^\rho{}_{\nu\sigma\mu} = 0, \\ &\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu} = 0. \end{aligned} \tag{23.1}$$

On the sphere the first identity reads $\sin^2\theta + 0 - \sin^2\theta = 0$ for $\rho = \theta$, $(\sigma\mu\nu) = (\phi\theta\phi)$. Now contract the second identity with $g^{\nu\sigma}$. Because $\nabla g = 0$, the metric slides through the derivatives, and the symmetries (19.3)

turn the first two Riemann terms into Ricci tensors (the third stays a Riemann tensor, now under a raised derivative ∇^ν); contracting again with $g^{\mu\lambda}$ turns every term into a Ricci tensor or the Ricci scalar:

$$\begin{aligned} &\nabla_\lambda R_{\rho\mu} - \nabla_\rho R_{\lambda\mu} + \nabla^\nu R_{\lambda\rho\mu\nu} = 0, \\ &\nabla^\mu R_{\rho\mu} - \nabla_\rho R + \nabla^\nu R_{\rho\nu} = 0. \end{aligned} \tag{23.2}$$

The first and third terms of the second line are equal, so $2\nabla^\mu R_{\rho\mu} = \nabla_\rho R$, and since $\nabla^\mu(R g_{\mu\nu}) = \nabla_\nu R$,

$$\nabla^\mu R_{\rho\mu} = \tfrac{1}{2}\nabla_\rho R \quad\Longrightarrow\quad \nabla^\mu G_{\mu\nu} = 0. \tag{23.3}$$

Why $\frac{1}{2}$. Try $X_{\mu\nu} = R_{\mu\nu} - \alpha R g_{\mu\nu}$. By Eq. (23.3), $\nabla^\mu X_{\mu\nu} = (\tfrac{1}{2} - \alpha)\nabla_\nu R$, which vanishes for every geometry only if $\alpha = \frac{1}{2}$. The coefficient is forced, not fitted.

Why not $R_{\mu\nu} = \kappa T_{\mu\nu}$, with κ a constant. Conservation would then force $\nabla^\mu R_{\mu\nu} = 0$, so $\nabla_\nu R = 0$ by Eq. (23.3); the trace $R = \kappa T$ would make T constant throughout spacetime, impossible when $T = 0$ in vacuum and $T = -\rho c^2$ in dust (Carroll 1997, eqs. 4.37–4.41).

History aside. Einstein's November 1915 papers to the Prussian Academy proposed, in effect, Ricci proportional to T (4 November), made it consistent by assuming $T = 0$ for all matter (11 November), and added the trace term, giving the final field equations (25 November) (Janssen & Renn 2015). Hilbert's action-based paper, presented on 20 November and published on 31 March 1916, survives in proofs that are not generally covariant and lack explicit field equations; most historians, though not all, conclude he did not derive them first (Corry, Renn & Stachel 1997; Janssen & Renn 2015).

The field equations as PDEs. $G_{\mu\nu}$ contains second derivatives of g (through $\partial\Gamma$) and products of first derivatives with inverse metrics (through $\Gamma\Gamma$), so Einstein's equations are second-order, nonlinear and coupled. Of the ten (both sides symmetric), the four identities (23.3) leave "only six truly independent equations", matching the four free functions in a choice of coordinates (Carroll 1997, after eq. 4.52). The four $G^{0\nu} = \kappa T^{0\nu}$ (here $\kappa = 8\pi G/c^4$ and $\Lambda = 0$) contain no second time derivatives: they are **constraints** on the initial data, while the six $G^{ij} = \kappa T^{ij}$ are **evolution** equations. Numerical relativity slices spacetime into spatial sheets, with a lapse function (proper time between sheets) and a shift vector (how coordinates slide), keeping the constraints satisfied as errors accumulate (Gourgoulhon 2007).

4. Worked Examples

Worked Example 23.1 — From PDEs to ODEs

For a static, spherically symmetric vacuum, try $ds^2 = -e^{2\alpha(r)}c^2dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2$, with α and β unknown functions of r only. The Ricci components are now ODEs (Carroll 1997, ch. 7). One combination of $R_{tt} = 0$ and $R_{rr} = 0$ gives $\frac{2}{r}(\alpha' + \beta') = 0$, so $\beta = -\alpha$. Writing $h = e^{2\alpha}$, the equation $R_{\theta\theta} = 0$ becomes

$$1 - e^{2\alpha}(2r\alpha' + 1) = 0 \quad\Longleftrightarrow\quad (rh)' = 1 \quad\Longrightarrow\quad h = 1 - \frac{r_s}{r}, \tag{23.4}$$

with r_s an integration constant. Matching $g_{tt} = -h\,c^2$ to the Newtonian $-(1 + 2\Phi/c^2)c^2$ with $\Phi = -GM/r$ fixes $r_s = 2GM/c^2$: the Schwarzschild solution, which Birkhoff's theorem says is the unique spherically symmetric vacuum solution, even outside a collapsing star.

Worked Example 23.2 — The trace-reversed form

Take the trace of $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ with $\kappa = 8\pi G/c^4$: $R - 2R + 4\Lambda = \kappa T$, so $R = 4\Lambda - \kappa T$. Substitute back into $R_{\mu\nu} = \kappa T_{\mu\nu} - \Lambda g_{\mu\nu} + \frac{1}{2}R g_{\mu\nu}$:

$$R_{\mu\nu} = \kappa\left(T_{\mu\nu} - \tfrac{1}{2}T\,g_{\mu\nu}\right) + \Lambda g_{\mu\nu}. \tag{23.5}$$

In vacuum with $\Lambda = 0$, this reduces to $R_{\mu\nu} = 0$.

5. Connection to Einstein's Equations

This section *is* their structure: $\nabla^\mu G_{\mu\nu} = 0$ makes $\nabla_\mu T^{\mu\nu} = 0$ a consequence of the field equations, not an extra assumption.

> **Needs further study.** Solving Einstein's equations in general needs PDE theory and numerical analysis; the first complete simulations of merging black holes came only in 2005 (Pretorius 2005). The route is courses in partial differential equations and numerical methods, then graduate GR.

6. What I Should Remember

- The Bianchi identities hold for every metric; contracted twice they give $\nabla^\mu G_{\mu\nu} = 0$, which forces the $\frac{1}{2}$ in $G_{\mu\nu}$ and forces $\nabla_\mu T^{\mu\nu} = 0$.
- Einstein's equations are ten coupled, nonlinear, second-order PDEs: six independent, four constraints.
- Symmetry reduces them to ODEs: spherical symmetry plus vacuum gives Schwarzschild.

SECTION 24

Symmetries, Killing vectors and conserved quantities

Difficulty: 7/10. Symmetry becomes a differential equation, with a conserved quantity you can use at once.

1. What Is This Mathematics?

A **Killing vector** field (after Wilhelm Killing) points in a direction along which you can slide the whole geometry without changing any distance: rotating a sphere, or shifting a plane. Each such symmetry gives a quantity constant along every geodesic, the geometric form of Noether's theorem: time-shift symmetry gives energy, rotational symmetry angular momentum.

2. Why Does General Relativity Need It?

Orbits around stars and black holes are solved with the energy and angular momentum conserved by Schwarzschild spacetime's symmetries. Whether energy is conserved at all depends on a time symmetry: an expanding universe has "no timelike Killing vector to give us a notion of conserved energy" (Carroll 1997, §8).

3. How Does the Mathematics Work?

A vector field ξ^μ is Killing when **Killing's equation** holds (Carroll 1997, eq. 5.42):

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0. \tag{24.1}$$

A shortcut: if the metric components do not depend on some coordinate $x^\star$, then $\partial_\star$ is a Killing vector. Along a geodesic with velocity u^μ,

$$u^\nu \nabla_\nu (\xi_\mu u^\mu) = u^\nu u^\mu \nabla_\nu \xi_\mu + \xi_\mu u^\nu \nabla_\nu u^\mu = 0. \tag{24.2}$$

The first term vanishes by Eq. (24.1), since $u^\nu u^\mu$ is symmetric; the second by the geodesic equation. So $\xi_\mu u^\mu$ is constant. An n-dimensional space has at most $n(n+1)/2$ Killing vectors: 3 in two dimensions, 10 in

four (Minkowski spacetime's 4 translations, 3 rotations and 3 boosts).

4. Worked Examples

Worked Example 24.1 — Which slides preserve a sphere?

The sphere's metric $a^2(d\theta^2 + \sin^2\theta\, d\phi^2)$ does not depend on ϕ, so ∂_ϕ, rotation about the polar axis, is Killing. ∂_θ is not: sliding points toward the equator lengthens the circles of latitude ($g_{\phi\phi}$ depends on θ). Rotations about the other two axes complete the maximum of 3.

Worked Example 24.2 — Orbital energy and angular momentum

Schwarzschild's metric does not depend on t or ϕ. In the equatorial plane, with dots meaning $d/d\tau$, the conserved quantities per unit mass are

$$E = \left(1 - \frac{r_s}{r}\right) c^2\, \dot{t}, \qquad L = r^2 \dot{\phi}. \tag{24.3}$$

Substitute $\dot{t}$ and $\dot{\phi}$ into the normalisation $-(1 - r_s/r)c^2\dot{t}^2 + (1 - r_s/r)^{-1}\dot{r}^2 + r^2\dot{\phi}^2 = -c^2$, multiply by $(1 - r_s/r)$, and rearrange using $r_s c^2/2 = GM$:

$$\begin{aligned} \tfrac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) &= \frac{E^2 - c^4}{2c^2}, \\ V_{\text{eff}} &= -\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2 r^3}. \end{aligned} \tag{24.4}$$

The first two terms are Newton's effective potential; the $1/r^3$ term is the entire GR correction (Carroll 1997, eqs. 7.47–7.48).

Worked Example 24.3 — Innermost stable orbit

Circular orbits sit where $dV_{\text{eff}}/dr = 0$. Multiplying that condition by r^4/GM gives the quadratic $r^2 - (L^2/GM)\, r + 3L^2/c^2 = 0$, whose two roots merge when its discriminant vanishes, $L^2 = 12G^2M^2/c^2$, at

$$r_{\text{ISCO}} = \frac{6GM}{c^2} = 3\, r_s. \tag{24.5}$$

Inside it no circular orbit is stable (Carroll 1997, eq. 7.55); for a 10-solar-mass black hole it is 88.6 km, and gas spiralling down to it radiates $1 - 2\sqrt{2}/3 \approx 5.7\%$ of its rest-mass energy.

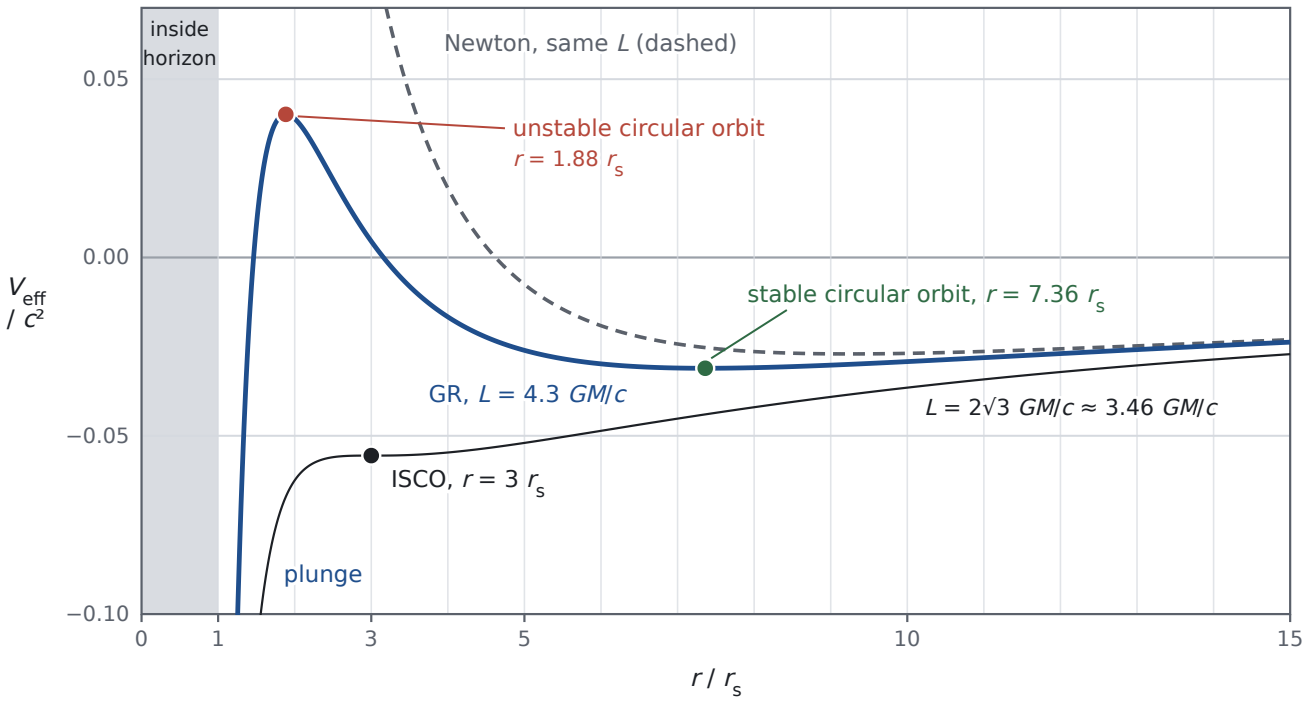


Figure 24.1. The Schwarzschild effective potential. The GR term $-GML^2/(c^2r^3)$ adds a peak and a plunge that Newtonian gravity (dashed) lacks; as angular momentum falls, the stable minimum and unstable maximum merge at the innermost stable circular orbit, 3 r_s.

5. Connection to Einstein's Equations

Symmetry makes the field equations solvable (Section 23) and their solutions usable: Equation 16 of Part IV uses Eq. (24.4) for Mercury's perihelion advance. With $\nabla_\mu T^{\mu\nu} = 0$, a Killing vector also yields a conserved total, $\nabla_\mu(T^{\mu\nu}\xi_\nu) = 0$.

6. What I Should Remember

- Killing vectors are directions of symmetry, $\nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu = 0$; if the metric does not depend on a coordinate, that coordinate's $\partial_\star$ is one.
- $\xi_\mu u^\mu$ is constant along geodesics; Schwarzschild gives conserved E and L.
- A globally conserved energy needs a timelike Killing vector, which an expanding universe lacks.

SECTION 25

Variational calculus and the Einstein–Hilbert action

Difficulty: 9/10. The Euler–Lagrange step is Calculus II integration by parts; varying a whole spacetime metric is graduate material.

1. What Is This Mathematics?

A **functional** eats a whole function or path and returns a number, such as a curve's length or travel time. Many laws of physics say the actual path makes some functional **stationary**, unchanged to first order when the path is nudged; the condition, the **Euler-Lagrange equation**, follows from integration by parts.

2. Why Does General Relativity Need It?

Geodesics are paths of stationary proper time, and Einstein's equations follow from making one number, the **Einstein–Hilbert action**, stationary under changes of the metric. The same calculation defines the stress-energy tensor of any matter field, and the action is the starting point of many approaches to quantum gravity (Carroll 1997, §4).

3. How Does the Mathematics Work?

Let $S[y] = \int_a^b L(x, y, y')\,dx$ with the endpoints fixed, and nudge the path, $y \to y + \epsilon\eta(x)$ with $\eta(a) = \eta(b) = 0$. By the chain rule, then integration by parts on the second term (its boundary term vanishes because η does):

$$\begin{aligned} \left.\frac{dS}{d\epsilon}\right|_0 &= \int_a^b \left(\frac{\partial L}{\partial y}\eta + \frac{\partial L}{\partial y'}\eta'\right) dx \\ &= \int_a^b \left[\frac{\partial L}{\partial y} - \frac{d}{dx}\frac{\partial L}{\partial y'}\right] \eta\, dx. \end{aligned} \tag{25.1}$$

For this to vanish for every bump η, the bracket must vanish everywhere:

$$\frac{\partial L}{\partial y} - \frac{d}{dx}\frac{\partial L}{\partial y'} = 0. \tag{25.2}$$

For geodesics, use $L = \frac{1}{2} g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu$ with an affine parameter (it gives the same paths as the proper-time integral and avoids a square root). Then $\partial L/\partial \dot{x}^\mu = g_{\mu\nu}\dot{x}^\nu$, its λ-derivative is $g_{\mu\nu}\ddot{x}^\nu + \partial_\alpha g_{\mu\nu}\dot{x}^\alpha \dot{x}^\nu$, and $\partial L/\partial x^\mu = \frac{1}{2}\partial_\mu g_{\alpha\beta}\dot{x}^\alpha \dot{x}^\beta$. Setting the difference to zero, symmetrising and multiplying by $g^{\sigma\mu}$ reproduces the geodesic equation (18.1) with exactly the Christoffel symbols (16.5). This is also the quickest way to *compute* Christoffel symbols: read them off the Euler–Lagrange equations, remembering that the coefficient of a cross term $\dot{x}^\nu \dot{x}^\sigma$ ($\nu \neq \sigma$) is 2Γ (Blau, *Lecture Notes on GR*, §3.1).

The **Einstein–Hilbert action**, with matter action S_M, is

$$S = \frac{c^4}{16\pi G}\int (R - 2\Lambda)\,\sqrt{-g}\; dt\, d^3x + S_M. \tag{25.3}$$

Units: R is in m^{-2}, $\sqrt{-g}$ is dimensionless when $x^0 = ct$, and $c^4/16\pi G = 2.408 \times 10^{42}$ N, so the integral over seconds and cubic metres gives J·s, the units of action. Why $\sqrt{-g}$: it makes $\sqrt{-g}\,d^4x$ the invariant volume element, the Jacobian of Section 5 in four dimensions (the minus sign because $\det g < 0$). Why R: any useful scalar must involve second derivatives of the metric, since first derivatives can be set to zero at a point, and R is the simplest.

Vary the inverse metric, $g^{\mu\nu} \to g^{\mu\nu} + \delta g^{\mu\nu}$, writing $R\sqrt{-g} = g^{\mu\nu}R_{\mu\nu}\sqrt{-g}$. Three pieces appear (Carroll 1997, eqs. 4.56–4.70; his notes, with $c = 1$, put the factor of 2 elsewhere, $S = \frac{1}{8\pi G}\int R\sqrt{-g}\,d^4x + S_M$ and $T_{\mu\nu} = -\frac{1}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}}$, which gives the same field equations): $R_{\mu\nu}\sqrt{-g}\,\delta g^{\mu\nu}$; a term $g^{\mu\nu}\delta R_{\mu\nu}$ that is a total divergence and integrates to a boundary term that vanishes; and $R\,\delta\sqrt{-g}$, using

$$\delta\sqrt{-g} = -\tfrac{1}{2}\sqrt{-g}\, g_{\mu\nu}\,\delta g^{\mu\nu}. \tag{25.4}$$

Collecting, with $T_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}}$ defining the stress-energy tensor (here $\delta S_M/\delta g^{\mu\nu}$, a **functional derivative**, means the coefficient of $\delta g^{\mu\nu}$ in the change of the matter action):

$$\delta S = \int \sqrt{-g}\left[\frac{c^4}{16\pi G}\left(G_{\mu\nu} + \Lambda g_{\mu\nu}\right) - \tfrac{1}{2}T_{\mu\nu}\right]\delta g^{\mu\nu}\, dt\, d^3x. \tag{25.5}$$

Demanding $\delta S = 0$ for every $\delta g^{\mu\nu}$ gives $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ EXACT.

4. Worked Examples

Worked Example 25.1 — The shortest path in a plane

Arc length is $S = \int \sqrt{1 + y'^2}\, dx$, so $L = \sqrt{1 + y'^2}$ with $\partial L/\partial y = 0$ and $\partial L/\partial y' = y'/\sqrt{1 + y'^2}$. Eq. (25.2) says $y'/\sqrt{1 + y'^2} = C$, a constant. Squaring, $y'^2(1 - C^2) = C^2$, so y' is constant and $y = mx + b$: a straight line.

Worked Example 25.2 — A thrown ball

A 1 kg ball is thrown up and caught after $T = 2$ s, with $g = 9.81$ m/s² and $L = \frac{1}{2}m\dot{y}^2 - mgy$. Eq. (25.2) gives $\ddot{y} = -g$, so the true path is $y_0 = \frac{g}{2}t(T - t)$, with action $S_0 = -mg^2T^3/24 = -32.0787$ J·s. Perturb it to $y_0 + \varepsilon \sin(\pi t/T)$. The term linear in ε vanishes after integration by parts, and the quadratic term is

$$S - S_0 = \frac{m}{2}\varepsilon^2 \int_0^T \frac{\pi^2}{T^2} \cos^2 \frac{\pi t}{T}\, dt = \frac{m\pi^2\varepsilon^2}{4T}, \tag{25.6}$$

which is $+0.012337$ J·s for $\varepsilon = \pm 0.1$ m, whichever sign: the true path is stationary.

Worked Example 25.3 — Sphere Christoffels from a Lagrangian

With $L = a^2\dot{\theta}^2 + a^2 \sin^2\theta\, \dot{\phi}^2$, the θ equation is $\frac{d}{d\lambda}(2a^2\dot{\theta}) - 2a^2 \sin\theta \cos\theta\, \dot{\phi}^2 = 0$, so $\ddot{\theta} - \sin\theta\cos\theta\, \dot{\phi}^2 = 0$ and $\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta$. The ϕ equation is $\frac{d}{d\lambda}(\sin^2\theta\, \dot{\phi}) = 0$, so $\ddot{\phi} + 2\cot\theta\, \dot{\theta}\dot{\phi} = 0$; the cross-term coefficient $2\cot\theta$ equals $2\Gamma^{\phi}_{\theta\phi}$, so $\Gamma^{\phi}_{\theta\phi} = \cot\theta$, matching Worked Example 16.3.

5. Connection to Einstein's Equations

Eq. (25.5) derives the field equations, cosmological constant included, from one scalar; the -2Λ becomes $+\Lambda g_{\mu\nu}$ in the equations and $+\Lambda c^2/3$ in the Friedmann equation, and because the action ignores the choice of coordinates, $\nabla_\mu T^{\mu\nu} = 0$ comes out automatically. On naming, Carroll says the equations "are rightly named after Einstein. The action, however, is rightly called the Hilbert action" (Carroll 1997, §4).

> **Needs further study.** The Euler–Lagrange equation is taught in an upper-level classical mechanics course (Lagrangian mechanics); varying actions of fields, including the step that discards the boundary term, belongs to classical field theory and graduate GR.

6. What I Should Remember

- A functional maps whole paths to numbers; making it stationary, via integration by parts, gives the Euler–Lagrange equation.
- Geodesics make proper time stationary, and $L = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ is the fastest way to find Christoffel symbols.
- $S = \frac{c^4}{16\pi G}\int (R - 2\Lambda)\sqrt{-g}\, dt\, d^3x + S_M$, with units of J·s; varying it with respect to $g^{\mu\nu}$ gives Einstein's equations and defines $T_{\mu\nu}$ as matter's response to the metric.

PART IV

Anatomy of the Great Equations

This Part dissects twenty equations, each equation needing only those before it (a few worked examples borrow a metric from later in this Part, and say so): the geometry of flat and curved spacetime, curvature, the field equations with their famous solutions and consequences, and finally the action from which everything can be derived. Each is taken apart in the same seven steps, A to G, with section numbers pointing back to where the mathematics was taught.

SECTION 26

Anatomy of the Great Equations of General Relativity

Conventions, once more: signature $(-,+,+,+)$; SI units with $x^0 = ct$ unless a line says otherwise; Greek indices run over 0–3, Latin over 1–3; an index repeated once up and once down is summed; superscripts on coordinates are labels, not powers. Every entry re-defines its symbols, so you can start anywhere.

Equation 1: The spacetime interval of special relativity

A. The equation

$$\Delta s^2 = -(c\,\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 = \eta_{\mu\nu}\,\Delta x^\mu \Delta x^\nu \tag{26.1}$$

EXACT in special relativity.

B. What it says in plain English

Inertial observers disagree about the time and the distance between two events but agree on this combination, whose sign says whether a clock, only light, or nothing at all can travel between them.

C. Every symbol

Symbol	Meaning	Units
Δs^2	the interval; a single name, which can be negative	m²
Δt; $\Delta x, \Delta y, \Delta z$	time and space separations in one inertial frame	s; m
c	speed of light, 299 792 458 (exact)	m/s
$(\,)^2$	a genuine square here, because it applies to a bracket, not an index	—
$-$ on the time term	the signature $(-,+,+,+)$: time enters opposite to space	—
$\eta_{\mu\nu}$	Minkowski metric, $\mathrm{diag}(-1,1,1,1)$	none
μ, ν	Greek indices, each 0 to 3; both repeated, so summed	—

Symbol	Meaning	Units
Δx^μ	$(c\Delta t, \Delta x, \Delta y, \Delta z)$; upper index marks vector components	m

D. The mathematical operations

Square four numbers and add them with one minus sign; in index form, a 16-term double sum with only the four diagonal terms nonzero (Sections 9, 11).

E. Physical interpretation

$\Delta s^2 < 0$: timelike; a clock present at both events reads the proper time $\Delta\tau = \sqrt{-\Delta s^2}/c$. $\Delta s^2 = 0$: null, connected by light. $\Delta s^2 > 0$: spacelike, causally disconnected. Because of the minus sign, the straight (unaccelerated) worldline between two events has the *longest* proper time: the twin paradox.

F. Worked example

Event B is $\Delta t = 1.000$ s and $\Delta x = 2.000 \times 10^8$ m from event A: $(c\Delta t)^2 = 8.98755 \times 10^{16}$ m² and $(\Delta x)^2 = 4.00000 \times 10^{16}$ m², so $\Delta s^2 = -4.98755 \times 10^{16}$ m², timelike, with $\Delta\tau = 2.23328 \times 10^8/c = 0.744943$ s. An observer moving at $0.6c$ ($\gamma = 1.25$) measures $\Delta t' = \gamma(\Delta t - v\Delta x/c^2) = 0.749654$ s and $\Delta x' = \gamma(\Delta x - v\Delta t) = 2.515566 \times 10^7$ m: different numbers but the same Δs^2, the invariance checked in Worked Example 10.2.

G. Connections to other equations

EXACT: Equation 2 with $g_{\mu\nu} = \eta_{\mu\nu}$; Equation 3 in flat spacetime gives $d\tau = dt/\gamma$. APPROXIMATE: holds in any small freely falling laboratory in curved spacetime, by local flatness (Section 15). ANALOGY: Pythagoras with one sign flipped.

Equation 2: The general metric line element

A. The equation

$$ds^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu = \sum_{\mu=0}^{3}\sum_{\nu=0}^{3} g_{\mu\nu}(x)\, dx^\mu dx^\nu \tag{26.2}$$

EXACT (a definition).

B. What it says in plain English

The metric is a position-dependent recipe that turns small steps in the coordinates into real distances and times.

C. Every symbol

Symbol	Meaning	Units
ds^2	squared interval between neighbouring events; $(ds)^2$, not $d(s^2)$	m²
$g_{\mu\nu}(x)$	metric components: a symmetric 4×4 table of functions of position x	none for (ct, x, y, z); m² for an angle coordinate
dx^μ	infinitesimal coordinate steps; upper index = vector components	coordinate units
μ, ν	indices 0–3, each summed (one up, one down)	—

D. The mathematical operations

A quadratic form, $dx^T g\, dx$: 16 products, 10 distinct coefficients since $g_{\mu\nu} = g_{\nu\mu}$ (Sections 11, 12). Lengths and times of whole paths integrate $\sqrt{|g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu|}\, d\lambda$.

E. Physical interpretation

The ten functions are the gravitational field: g_{00} sets clock rates, g_{ij} the geometry of space, and g_{0i} effects such as the dragging of frames by a rotating mass.

F. Worked example

In Schwarzschild spacetime (Equation 12) the radial ruler distance between r_1 and r_2 is $L = \int dr/\sqrt{1 - r_s/r}$, with antiderivative (check by differentiating) $F(r) = \sqrt{r(r - r_s)} + r_s \ln(\sqrt{r} + \sqrt{r - r_s})$. For a 10-solar-mass black hole ($r_s = 29.53$ km), from r_s to $2r_s$:

$$L = F(2r_s) - F(r_s) = \left(\sqrt{2} + \ln(1 + \sqrt{2})\right) r_s = 2.2956\, r_s = 67.79 \text{ km}, \tag{26.3}$$

against a coordinate difference of 29.53 km. Near the Sun the effect is tiny: from its surface to twice its radius, the ruler distance exceeds Δr by $\frac{r_s}{2} \ln 2 = 1.02$ km out of 696 000 km.

G. Connections to other equations

EXACT: Equation 1 is the flat case, Equation 3 this along a clock's path, and Equation 4 builds the Christoffel symbols from $g_{\mu\nu}$. APPROXIMATE: for weak gravity, $g_{00} \approx -(1 + 2\Phi/c^2)$ (Equation 11).

Equation 3: Proper time

A. The equation

$$\tau_{AB} = \frac{1}{c} \int_A^B \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}\, d\lambda \tag{26.4}$$

EXACT. Two special cases:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \text{ (flat)}, \qquad \frac{d\tau}{dt} \approx 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2} \text{ (weak field)}. \tag{26.5}$$

The first is EXACT; the second is APPROXIMATE (valid for $|\Phi|/c^2 \ll 1$ and $v \ll c$).

B. What it says in plain English

The time a clock carried along a path actually ticks depends on how fast it moves and how deep in a gravitational potential it sits.

C. Every symbol

Symbol	Meaning	Units
τ	proper time, read by the travelling clock	s
$1/c$	converts metres of ds into seconds	s/m
$-$ inside the root	makes the radicand positive for timelike paths in $(-,+,+,+)$	—
λ	any parameter labelling points of the path	any
$dx^\mu/d\lambda$	tangent vector of the path	m per unit λ
Φ	Newtonian potential, negative near masses	J/kg = m²/s²
v	speed relative to the coordinates	m/s

D. The mathematical operations

Evaluate the line element along the path, take a square root, integrate (Sections 1, 14); the weak-field form needs one Taylor step.

E. Physical interpretation

Proper time is what clocks measure; coordinate time is a label. Between two events not too far apart, a freely falling clock records more proper time than any clock on a nearby path. The maximum is only local: a satellite completing one orbit and a clock launched straight up from the same event that falls back to

meet it are both in free fall, yet record different proper times (Section 18). The GPS budget, $+45.8 - 7.2 = +38.6$ μs/day, is Eq. (26.5) applied twice (Worked Example 14.4).

F. Worked example

Derive the weak-field form. With $ds^2 = -(1+2\Phi/c^2)c^2dt^2 + (1-2\Phi/c^2)|d\mathbf{x}|^2$ and $v = |d\mathbf{x}|/dt$:

$$\left(\frac{d\tau}{dt}\right)^2 = 1 + \frac{2\Phi}{c^2} - \left(1 - \frac{2\Phi}{c^2}\right)\frac{v^2}{c^2} \approx 1 + \frac{2\Phi}{c^2} - \frac{v^2}{c^2}, \tag{26.6}$$

dropping the tiny product $\Phi v^2/c^4$; then $\sqrt{1+\epsilon} \approx 1 + \epsilon/2$. Near the ground $\Delta\Phi = gh$: a clock 3000 m up gains $gh/c^2 = 3.27 \times 10^{-13}$, or 28.3 ns per day. In 2010 optical clocks detected the shift for a 33 cm height change, about 4×10^{-17} against a predicted 3.6×10^{-17} (Physics Today 2010).

G. Connections to other equations

EXACT: making τ stationary gives the geodesic equation, Equation 5 (Section 25). APPROXIMATE ($|\Phi|/c^2 \ll 1$ and $v \ll c$): the weak-field form uses Equation 11's metric. ANALOGY: a car's odometer, recording the length of road actually driven.

Equation 4: The Christoffel symbols

A. The equation

$$\Gamma^{\sigma}_{\mu\nu} = \tfrac{1}{2}\, g^{\sigma\rho}\left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right) \tag{26.7}$$

EXACT for the torsion-free, metric-compatible connection GR uses.

B. What it says in plain English

Correction terms, built from the slopes of the metric, that describe how the coordinate directions twist and stretch from point to point.

C. Every symbol

Symbol	Meaning	Units ($x^0 = ct$)
Γ (capital gamma)	connection coefficient; **not** a tensor	m^{-1}
σ (upper)	which component of the correction	—
μ, ν (lower)	the two input directions; Γ is symmetric in them	—
$\frac{1}{2}$	undoes the doubling from adding two permuted equations	—
$g^{\sigma\rho}$	inverse metric; ρ is summed	none
$\partial_\mu g_{\nu\rho}$	partial derivative of a metric component with respect to x^μ	m^{-1}
three terms, last negative	the cyclic combination left by $\nabla g = 0$	—

D. The mathematical operations

Differentiate the metric, invert it, multiply and sum over ρ (Section 16): 40 independent symbols in four dimensions, most of them zero for diagonal metrics.

E. Physical interpretation

In the geodesic equation $-\Gamma^i_{00}c^2$ is the Newtonian gravitational acceleration; because free fall makes Γ zero at any point, gravity's "force" is a coordinate effect: the equivalence principle.

F. Worked example

Schwarzschild spacetime with t in seconds has $g_{tt} = -(1 - r_s/r)c^2$ and $g^{rr} = 1 - r_s/r$; only $\rho = r$ contributes and $\partial_t g = 0$, so

$$\Gamma^r_{tt} = \tfrac{1}{2} g^{rr} \left(-\partial_r g_{tt}\right) = \tfrac{1}{2}\left(1 - \tfrac{r_s}{r}\right) \frac{r_s c^2}{r^2} = \frac{GM}{r^2}\left(1 - \frac{r_s}{r}\right). \tag{26.8}$$

At Earth's surface ($GM/R^2 = 9.820$ m/s², $r_s/R = 1.39 \times 10^{-9}$), $\Gamma^r_{tt} = 9.820$ m/s², equal to GM/R^2 to about one part in a billion: the familiar g.

G. Connections to other equations

EXACT: feeds Equations 5 and 6. APPROXIMATE: $\Gamma^i_{00} \approx \partial_i \Phi / c^2$ in weak static fields (Equation 11). ANALOGY: the centrifugal and Coriolis terms of rotating or curved coordinates.

Equation 5: The geodesic equation

A. The equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 \tag{26.9}$$

EXACT for freely falling test bodies. Light uses an affine parameter λ in place of τ.

B. What it says in plain English

A freely falling body moves as straight as spacetime allows.

C. Every symbol

Symbol	Meaning	Units
$x^\mu(\tau)$	the worldline: coordinates as functions of proper time	m
$d^2x^\mu/d\tau^2$	coordinate four-acceleration	m/s²
$dx^\alpha/d\tau = u^\alpha$	four-velocity, with $g_{\mu\nu}u^\mu u^\nu = -c^2$	m/s
$\Gamma^\mu_{\alpha\beta}$	Christoffel symbols (Equation 4)	m⁻¹
μ	free index: four equations	—
α, β	summed: 16 terms in each equation	—

D. The mathematical operations

Four coupled, second-order, nonlinear ODEs in τ, with a correction quadratic in the velocity (Sections 7, 18).

E. Physical interpretation

There is no gravitational force: free fall *is* inertial motion, which is why astronauts float. Orbits, light bending and the shapes of black-hole images all come from this equation.

F. Worked example

For a circular equatorial orbit in Schwarzschild spacetime ($\dot r = \ddot r = 0$), the r component of Eq. (26.9) keeps only $\Gamma^r_{tt}\dot t^2 + \Gamma^r_{\phi\phi}\dot\phi^2 = 0$. With the time coordinate t in seconds (as in Equation 4's worked example), $\Gamma^r_{tt} = c^2 r_s(r - r_s)/2r^3$ and, at $\theta = \pi/2$, $\Gamma^r_{\phi\phi} = -(r - r_s)$,

$$\left(\frac{d\phi}{dt}\right)^2 = -\frac{\Gamma^r_{tt}}{\Gamma^r_{\phi\phi}} = \frac{c^2 r_s}{2r^3} = \frac{GM}{r^3}, \tag{26.10}$$

Kepler's third law, exact in Schwarzschild coordinate time. Light circles where its own angular speed from $ds^2 = 0$, $(d\phi/dt)^2 = (1 - r_s/r)c^2/r^2$, matches this: $c^2(1 - 2GM/c^2 r) = GM/r$, so $r = 3GM/c^2 = 1.5\,r_s$, the **photon sphere** (44.3 km for 10 solar masses).

G. Connections to other equations

EXACT: paths of stationary proper time (Equation 3; Section 25), and neighbouring geodesics give Equation 17. APPROXIMATE (weak, static field; speeds much less than c): reduces to $d^2\mathbf{x}/dt^2 = -\nabla\Phi$ (Equation 11). ANALOGY: Newton's first law.

Equation 6: The Riemann curvature tensor

A. The equation

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma} \tag{26.11}$$

EXACT, defined by $[\nabla_\mu, \nabla_\nu]V^\rho = R^{\rho}{}_{\sigma\mu\nu}V^\sigma$ (Carroll/MTW sign convention).

B. What it says in plain English

Carry a vector around a tiny closed loop and it returns rotated; the Riemann tensor says by how much, or equivalently how neighbouring free-fall paths accelerate toward or away from each other.

C. Every symbol

Symbol	Meaning	Units
$R^{\rho}{}_{\sigma\mu\nu}$	Riemann tensor; 256 components, 20 independent in 4D	m^{-2}
ρ	component of the change in the vector	—
σ	slot for the vector being carried	—
μ, ν	the two directions spanning the loop; antisymmetric	—
$\partial_\mu\Gamma$ terms	derivatives of Christoffel symbols: second derivatives of g	m^{-2}
$\Gamma\Gamma$ terms	products, with λ summed	m^{-2}

D. The mathematical operations

Differentiate the Christoffel symbols, multiply pairs of them, antisymmetrise in $\mu\nu$ (Section 19).

E. Physical interpretation

Curvature is tidal acceleration, the part of gravity no frame removes. Invariants built from it decide whether a singularity is real or an artefact of coordinates.

F. Worked example

In a static observer's orthonormal frame in Schwarzschild spacetime there are exactly six independent nonzero components: $R_{\hat{t}\hat{r}\hat{t}\hat{r}} = -r_s/r^3$, $R_{\hat{t}\hat{\theta}\hat{t}\hat{\theta}} = R_{\hat{t}\hat{\phi}\hat{t}\hat{\phi}} = r_s/2r^3$, $R_{\hat{r}\hat{\theta}\hat{r}\hat{\theta}} = R_{\hat{r}\hat{\phi}\hat{r}\hat{\phi}} = -r_s/2r^3$ and $R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}} = r_s/r^3$. Each appears four times, squared, in the **Kretschmann scalar** (every term is positive, since raising a $\hat{t}$ index flips a sign and each component has two or none; the two $\hat{t}$–angle and the two $\hat{r}$–angle components are grouped):

$$\begin{aligned} R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} &= 4\left[\frac{r_s^2}{r^6} + 2\cdot\frac{r_s^2}{4r^6} + 2\cdot\frac{r_s^2}{4r^6} + \frac{r_s^2}{r^6}\right] \\ &= \frac{12\,r_s^2}{r^6} = \frac{48\,G^2M^2}{c^4 r^6}. \end{aligned} \tag{26.12}$$

At the horizon it is $12/r_s^4$, finite (1.58×10^{-13} m^{-4} for one solar mass), so the horizon is only a coordinate singularity; as $r \to 0$ it diverges, a true curvature singularity.

G. Connections to other equations

EXACT: its contractions give Equation 7, and it drives geodesic deviation, Equation 17. APPROXIMATE: $R^{i}{}_{0j0} \approx \partial_i \partial_j \Phi / c^2$ in weak fields. ANALOGY: a surface's Gaussian curvature; the rotation of a carried arrow (holonomy).

Equation 7: The Ricci tensor and the Ricci scalar

A. The equation

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}, \qquad R = g^{\mu\nu} R_{\mu\nu} \tag{26.13}$$

EXACT. Written out with Christoffel symbols:

$$R_{\mu\nu} = \partial_\lambda \Gamma^{\lambda}_{\mu\nu} - \partial_\nu \Gamma^{\lambda}_{\mu\lambda} + \Gamma^{\lambda}_{\lambda\sigma} \Gamma^{\sigma}_{\mu\nu} - \Gamma^{\lambda}_{\nu\sigma} \Gamma^{\sigma}_{\mu\lambda}. \tag{26.14}$$

B. What it says in plain English

The Ricci tensor averages Riemann over directions and tells how a small ball of freely falling particles starts to change volume; the Ricci scalar is one number for the overall curvature at a point.

C. Every symbol

Symbol	Meaning	Units
$R_{\mu\nu}$	Ricci tensor: symmetric, 10 components	m^{-2}
λ	contracted (1st and 3rd slots of Riemann summed)	—
R	Ricci scalar; not the radius r and not a constant	m^{-2}
$g^{\mu\nu}$	inverse metric, used for the second trace	none

D. The mathematical operations

A contraction, then a trace with the inverse metric (Sections 13, 21).

E. Physical interpretation

In empty space $R_{\mu\nu} = 0$ though Riemann is not zero: falling balls change shape, not volume. Inside matter, Einstein's equations make $R_{\mu\nu}$ follow the local energy and pressure.

F. Worked example

For a uniform expanding universe (Equation 19, t in seconds), $R = \frac{6}{c^2}\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right) + \frac{6k}{a^2}$ (Carroll 1997, eq. 8.14, with c restored); take flat space, $k = 0$. Exponential growth, $a = e^{Ht}$: $\dot{a}/a = H$, $\ddot{a}/a = H^2$, so $R = 12H^2/c^2$. Radiation, $a \propto t^{1/2}$: $\dot{a}/a = 1/2t$ and $\ddot{a}/a = -1/4t^2$, so the bracket is $-1/4t^2 + 1/4t^2 = 0$ and $R = 0$, matching radiation's zero trace. Matter, $a \propto t^{2/3}$: $-2/9t^2 + 4/9t^2 = 2/9t^2$, so $R = 4/3c^2t^2$.

G. Connections to other equations

EXACT: builds the Einstein tensor (Equation 8); tracing the field equations gives $R = 4\Lambda - (8\pi G/c^4)T$. APPROXIMATE: $R_{00} \approx \nabla^2 \Phi / c^2$ in weak static fields. ANALOGY: the Laplacian as "average minus centre value".

Equation 8: The Einstein tensor, fully expanded

A. The equation

Three nested layers, each a definition:

$$
\begin{aligned}
G_{\mu\nu} &= R_{\mu\nu} - \tfrac{1}{2}\, g_{\mu\nu}\, g^{\alpha\beta} R_{\alpha\beta}, \\
R_{\mu\nu} &= \partial_\lambda \Gamma^\lambda_{\mu\nu} - \partial_\nu \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{\mu\nu} - \Gamma^\lambda_{\nu\sigma}\Gamma^\sigma_{\mu\lambda}, \\
\Gamma^\lambda_{\mu\nu} &= \tfrac{1}{2} g^{\lambda\rho}\left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right).
\end{aligned}
\tag{26.15}
$$

EXACT. Schematically, $G_{\mu\nu} \sim g^{-1}\partial^2 g + g^{-1}g^{-1}(\partial g)(\partial g)$.

B. What it says in plain English

Written out in the metric, each of the ten field equations is a long sum of products of the metric, its inverse, and its first and second derivatives.

C. Every symbol

Symbol	Meaning	Units
$G_{\mu\nu}$	Einstein tensor	m⁻²
$\frac{1}{2}$	the trace-reversal coefficient forced by the Bianchi identity	—
$g^{\alpha\beta}R_{\alpha\beta}$	the Ricci scalar R	m⁻²
$\alpha, \beta, \lambda, \rho, \sigma$	dummy (summed) indices	—
μ, ν	free indices	—
$\partial^2 g$, $(\partial g)^2$	second derivatives; products of first derivatives	m⁻²

D. The mathematical operations

Matrix inversion, first and second partial derivatives, the product rule, contractions and a trace. A computer-algebra count for a general metric (inverse metric kept as separate symbols) gives 7 or 8 terms in a single Christoffel symbol (7 in Γ^0_{00}), 417 in R_{00}, 2262 in R and **2679** in G_{00}.

E. Physical interpretation

Linear in the second derivatives, which lets the equations be posed as an evolution problem, but nonlinear in g and its first derivatives: gravitational fields gravitate, and solutions cannot be added.

F. Worked example

For the weak static metric $g = \mathrm{diag}(-(1+2\Phi/c^2),\, 1-2\Phi/c^2,\, 1-2\Phi/c^2,\, 1-2\Phi/c^2)$ to first order in Φ, every product of Christoffel symbols is second order and drops out, and the 2679 terms of G_{00} collapse to one:

$$
G_{00} \approx \frac{2}{c^2}\nabla^2\Phi, \qquad R_{00} \approx \frac{1}{c^2}\nabla^2\Phi.
\tag{26.16}
$$

G. Connections to other equations

EXACT: Equations 4, 6 and 7 chained. **APPROXIMATE**: linearised, it becomes a wave operator (Equation 18) or the Laplacian (Equation 11). **ANALOGY**: the left side, $\nabla^2\Phi$, of Poisson's equation.

Equation 9: The stress-energy tensor and the perfect fluid

A. The equation

$$
T^{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) u^\mu u^\nu + p\, g^{\mu\nu}, \qquad T^{\mu\nu}_{\text{rest}} = \mathrm{diag}(\rho c^2, p, p, p)
\tag{26.17}
$$

EXACT (the definition of a perfect fluid; the second form in its rest frame in flat spacetime). Its trace is $T = -\rho c^2 + 3p$.

B. What it says in plain English

One symmetric 4×4 table holds everything that sources gravity: energy density, energy flow, momentum, pressure and shear stress.

C. Every symbol

Symbol	Meaning	Units
$T^{\mu\nu}$	flux of μ-momentum across surfaces of constant x^ν	J/m³ = Pa
ρ	rest-frame mass density (energy density ρc^2)	kg/m³
p; p/c^2	pressure; pressure expressed as a mass density	Pa; kg/m³
u^μ	the fluid's four-velocity, $u_\mu u^\mu = -c^2$	m/s
$g^{\mu\nu}$	inverse metric	none
T^{00}, T^{0i}, T^{ij}	energy density; energy flux/c; stresses	J/m³

D. The mathematical operations

An outer product $u^\mu u^\nu$ plus a multiple of the inverse metric (Sections 13, 22).

E. Physical interpretation

Dust has $p = 0$; radiation $p = \rho c^2/3$, so $T = 0$; vacuum energy $p = -\rho c^2$, exactly like a cosmological constant. For everyday matter $p \ll \rho c^2$, so only T^{00} matters in the Newtonian limit.

F. Worked example

Dust moving at $0.6c$ along x has $u^\mu = \gamma(c, v, 0, 0)$ with $\gamma^2 = 1.5625$: $T^{00} = \gamma^2\rho c^2 = 1.5625\,\rho c^2$, $T^{0x} = \gamma^2\rho c v = 0.9375\,\rho c^2$ and $T^{xx} = \gamma^2\rho v^2 = 0.5625\,\rho c^2$. The energy density gains two factors of γ, one from each particle's energy and one from length contraction packing particles closer, while the trace, $\eta_{\mu\nu}T^{\mu\nu} = (-1.5625 + 0.5625)\rho c^2 = -\rho c^2$, is the same as at rest: a scalar.

G. Connections to other equations

EXACT: the right side of Equation 10; $\nabla_\mu T^{\mu\nu} = 0$ (Section 22); Λ equals a vacuum fluid. APPROXIMATE: $T^{00} \approx \rho c^2$ alone when $v \ll c$ and $p \ll \rho c^2$. ANALOGY: the engineering stress tensor, extended by a time row and column.

Equation 10: Einstein's field equations

A. The equation

With and without the cosmological constant, and in trace-reversed form:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\,T_{\mu\nu}, \qquad R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = \frac{8\pi G}{c^4}\,T_{\mu\nu} \tag{26.18}$$

$$R_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mu\nu} - \tfrac{1}{2}T\,g_{\mu\nu}\right) + \Lambda g_{\mu\nu} \tag{26.19}$$

EXACT. In geometric units ($G = c = 1$): $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$.

B. What it says in plain English

The curvature of spacetime at a point, in the averaged combination $G_{\mu\nu}$, equals a universal constant times the density and flow of energy and momentum there. Equivalently, a small ball of freely falling particles, initially at rest, starts to shrink at a rate proportional to the energy density plus the three pressures at its centre (Baez & Bunn 2005).

C. Every symbol

Symbol	Meaning	Units
μ, ν	free indices 0–3: 16 equations, 10 distinct, 6 independent	—
$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$	Einstein tensor; divergence-free	m^{-2}
$R_{\mu\nu}$, R	Ricci tensor and scalar	m^{-2}
$g_{\mu\nu}$	the metric: the unknown being solved for	none
Λ	cosmological constant, about 1.1×10^{-52}	m^{-2}
G	Newton's constant, 6.67430×10^{-11}	$\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$
c^4	converts energy density into curvature	$\mathrm{m}^4/\mathrm{s}^4$
8π	4π from Gauss's law times 2 from trace reversal (Equation 11)	—
$8\pi G/c^4$	Einstein's constant $\kappa = 2.0766 \times 10^{-43}$	N^{-1}
$T_{\mu\nu}$, T	stress-energy tensor and its trace	$\mathrm{J/m}^3$

Why G: the equations must reproduce Newton's gravity. Why c^4: $T_{\mu\nu}$ is in $\mathrm{J/m}^3$ while curvature is in m^{-2}, and $(\mathrm{N}^{-1})(\mathrm{J\,m}^{-3}) = \mathrm{m}^{-2}$. Why 8π: derived in Equation 11.

D. The mathematical operations

Compute the curvature of a trial metric (Equations 4, 6–8) and $T_{\mu\nu}$, and require equality component by component. "Solving" means finding the ten functions $g_{\mu\nu}(x)$: ten coupled, nonlinear, second-order PDEs, four automatically consistent by the Bianchi identity (Section 23).

E. Physical interpretation

Spacetime is stiff: $1/\kappa = c^4/8\pi G = 4.8 \times 10^{42}$ N, so noticeable curvature needs enormous energy densities. Moved to the right side, Λ becomes a vacuum fluid with energy density $\Lambda c^4/8\pi G$ and equal negative pressure.

F. Worked example

With Planck 2018 values of the Hubble constant (today's expansion rate, Equation 19), $H_0 = 67.4$ km/s/Mpc, where 1 Mpc (megaparsec) $= 3.086 \times 10^{22}$ m, and of $\Omega_\Lambda = 0.6847$, the fraction of the flat universe's energy density carried by Λ (Planck 2018), flat-universe cosmology gives $\Lambda = 3\Omega_\Lambda H_0^2/c^2 = 1.09 \times 10^{-52}\ \mathrm{m}^{-2}$, and a vacuum energy density

$$u_\Lambda = \frac{\Lambda}{\kappa} = \frac{1.09 \times 10^{-52}\ \mathrm{m}^{-2}}{2.0766 \times 10^{-43}\ \mathrm{N}^{-1}} = 5.25 \times 10^{-10}\ \mathrm{J/m}^3, \tag{26.20}$$

equivalent to 5.8×10^{-27} kg/m³, about 3.5 proton masses per cubic metre, with pressure -5.25×10^{-10} Pa. Water's energy density, by contrast, gives curvature $\kappa\rho c^2 = 1.87 \times 10^{-23}\ \mathrm{m}^{-2}$, a radius of curvature of roughly 1.5 astronomical units, the Beginner's Guide's illustration of stiffness.

G. Connections to other equations

EXACT: the trace gives $R = 4\Lambda - \kappa T$ and hence Eq. (26.19); the Bianchi identity forces $\nabla_\mu T^{\mu\nu} = 0$; in vacuum with $\Lambda = 0$, $R_{\mu\nu} = 0$, solved by Equation 12. APPROXIMATE: weak, slow, static fields give Poisson's equation (Equation 11). ANALOGY: Hooke's law, strain proportional to stress.

Equation 11: The Newtonian limit

A. The equation

$$g_{00} \approx -\left(1 + \frac{2\Phi}{c^2}\right), \qquad \frac{d^2x^i}{dt^2} = -\partial_i \Phi, \qquad \nabla^2 \Phi = 4\pi G\rho \tag{26.21}$$

APPROXIMATE: valid for speeds small compared with c, a weak ($|\Phi|/c^2 \ll 1$), static field, pressure small compared with ρc^2, and Λ negligible (Carroll 1997, §4.1).

B. What it says in plain English

For weak gravity and slow motion GR becomes Newton's theory, with the Newtonian potential in the time-time component of the metric: everyday gravity is mostly the warping of time.

C. Every symbol

Symbol	Meaning	Units
g_{00}	time-time metric component, with $x^0 = ct$	none
$h_{00} = -2\Phi/c^2$	small deviation from flat, in $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$	none
Φ	Newtonian potential	m²/s²
x^i, ∂_i	spatial coordinates and derivatives, $i = 1, 2, 3$	m, m⁻¹
$\nabla^2 = \delta^{ij}\partial_i\partial_j$	Laplacian (δ^{ij} is 1 when $i = j$)	m⁻²
4π, G, ρ	solid angle of a sphere; Newton's constant; mass density	—; SI; kg/m³

D. The mathematical operations

Motion. Slow motion keeps only the Γ^μ_{00} term of Equation 5; static and weak fields give $\Gamma^i_{00} = -\frac{1}{2}\partial_i h_{00} = +\partial_i\Phi/c^2$; and $\Gamma^0_{00} = 0$ makes $dt/d\tau$ constant, so

$$\frac{d^2x^i}{dt^2} = -c^2\,\Gamma^i_{00} = -\partial_i\Phi. \tag{26.22}$$

Field. With $R_{00} \approx \partial_i\Gamma^i_{00} = \nabla^2\Phi/c^2$ (Equation 8) and, for dust, $T_{00} = \rho c^2$, $T = -\rho c^2$, the trace-reversed equation (26.19) with unknown constant κ reads

$$\frac{\nabla^2\Phi}{c^2} = \kappa\left(\rho c^2 - \tfrac{1}{2}\rho c^2\right) = \tfrac{1}{2}\kappa\rho c^2. \tag{26.23}$$

Matching Poisson's $\nabla^2\Phi = 4\pi G\rho$ requires $\frac{1}{2}\kappa c^4 = 4\pi G$, so $\kappa = 8\pi G/c^4$: the 4π is Gauss's, the 2 undoes trace reversal's $\frac{1}{2}$.

E. Physical interpretation

At Earth's surface $|\Phi|/c^2$ is about 7×10^{-10}: Newtonian gravity is a part-per-billion distortion of the rate of time, and slow particles fall because their geodesics follow it.

F. Worked example

Where the limit holds: GM/rc^2 is 1.67×10^{-10} at GPS altitude, 6.95×10^{-10} at Earth's surface (equatorial radius) and 2.12×10^{-6} at the Sun's surface. Where it fails: 0.17 at the surface of a 1.4-solar-mass, 12 km neutron star; and for light, which also feels the spatial part $g_{ij} \approx (1 - 2\Phi/c^2)\delta_{ij}$, so starlight grazing the Sun bends by $4GM_\odot/(c^2R_\odot) = 1.75''$ (arcseconds; $1'' = 1/3600$ of a degree), twice the value from g_{00} alone (Will 2014).

G. Connections to other equations

APPROXIMATE: the weak-field limits of Equations 5 and 10. EXACT: a clock at rest in a static field ticks at $d\tau/dt = \sqrt{-g_{00}}$ (Equation 3). APPROXIMATE (first order in $|\Phi|/c^2 \ll 1$): $d\tau/dt \approx 1 + \Phi/c^2$. ANALOGY: electrostatics, $\nabla^2 V = -\rho_q/\varepsilon_0 = -4\pi k \rho_q$, with Coulomb's constant $k = 1/(4\pi\varepsilon_0)$; its 4π, hidden inside ε_0 in SI units, also comes from Gauss's law.

Equation 12: The Schwarzschild metric

A. The equation

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right), \qquad r_s = \frac{2GM}{c^2}. \tag{26.24}$$

EXACT vacuum solution outside any spherical, non-rotating, uncharged mass.

B. What it says in plain English

The recipe for spacetime outside a star or black hole: near the mass, clocks tick slowly (the dt^2 coefficient shrinks) and radial distances stretch (the dr^2 coefficient grows).

C. Every symbol

Symbol	Meaning	Units
t	time kept by static clocks far away	s
r	areal radius: the sphere labelled r has area $4\pi r^2$; **not** distance from the centre	m
θ, ϕ	colatitude and longitude	rad
$d\theta^2 + \sin^2\theta\, d\phi^2$	unit-sphere metric, often written $d\Omega^2$	—
M	mass measured from distant orbits	kg
$r_s = 2GM/c^2$	Schwarzschild radius (Equation 13)	m
$1 - r_s/r$	dimensionless factor; $\sqrt{1 - r_s/r}$ is a static clock's rate	—
$(\,)^{-1}$	reciprocal	—

D. The mathematical operations

A diagonal quadratic form (Equation 2): a radial ruler gives $\int \sqrt{g_{rr}}\, dr$ and a static clock $\int \sqrt{-g_{tt}}\, dt/c$ (Section 12).

E. Physical interpretation

At $r = r_s$, g_{rr} blows up but curvature stays finite (Equation 6): a horizon, not a wall. Inside, g_{tt} and g_{rr} swap sign, so decreasing r becomes as unavoidable as the future. At $r = 0$ curvature is infinite.

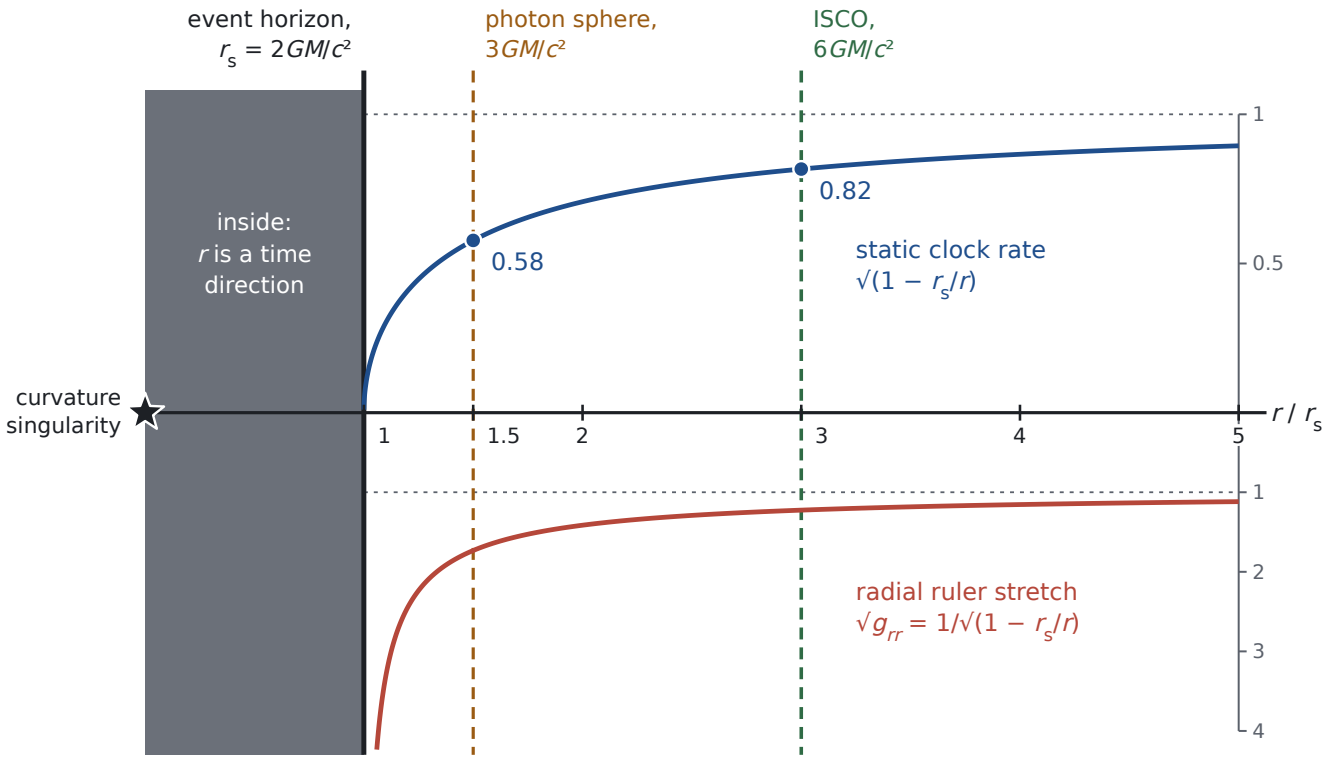


Figure 26.1. Landmarks of Schwarzschild spacetime. Clocks held at rest slow (blue) and radial rulers stretch (red) as the horizon is approached; the photon sphere and the innermost stable circular orbit lie at 1.5 and 3 Schwarzschild radii.

F. Worked example

At the surface of an illustrative 1.4-solar-mass neutron star of radius 12 km, $r_s = 4.135$ km and $r_s/R = 0.3445$, so $\sqrt{-g_{tt}}/c = \sqrt{0.6555} = 0.8096$ (surface clocks run at 81% of the far-away rate) and $\sqrt{g_{rr}} = 1/0.8096 = 1.235$ (radial rulers stretched by 23.5%). The weak-field estimates $1 - r_s/2r = 0.828$ and $1 + r_s/2r = 1.172$ are off by 2% and 5%; on Earth, where $r_s/r = 1.4 \times 10^{-9}$, they agree with the exact values to within 10^{-18}.

G. Connections to other equations

EXACT: the unique spherically symmetric solution of $R_{\mu\nu} = 0$ (Birkhoff's theorem; derived in Worked Example 23.1), the non-rotating case of the Kerr metric (the exact solution for a spinning black hole), with Equations 13–16 (bar the approximate perihelion formula) and the Schwarzschild tidal values of Equation 17 as exact consequences. APPROXIMATE: for $r \gg r_s$, $g_{rr} \approx 1 + r_s/r = 1 - 2\Phi/c^2$ with $\Phi = -GM/r$, giving Equation 11's weak field once r is read as Newton's distance; $g_{tt} = -(1 + 2\Phi/c^2)c^2$ holds exactly.

Equation 13: The Schwarzschild radius

A. The equation

$$r_s = \frac{2GM}{c^2} \qquad \text{(geometric units: } r_s = 2M\text{)} \qquad (26.25)$$

EXACT as the horizon radius of a non-rotating black hole.

B. What it says in plain English

If a mass M lies entirely inside the radius r_s, nothing, not even light, can get back out from within it.

C. Every symbol

Symbol	Meaning	Units
r_s	Schwarzschild radius (an areal radius)	m
2	fixed by matching the solution to Newtonian orbits	—
G, M, c^2	Newton's constant, mass, speed of light squared	SI

Units check: $(\mathrm{m^3\,kg^{-1}\,s^{-2}})(\mathrm{kg})/(\mathrm{m^2\,s^{-2}}) = \mathrm{m}$.

D. The mathematical operations

Multiplication and division; since r_s grows linearly with M, the mean density inside it falls as $1/M^2$.

E. Physical interpretation

At $r = r_s$, $g_{tt} = 0$: a clock held there would not tick, so no observer can stay at rest at or inside the horizon.

F. Worked example

Using IAU GM values (IAU 2015 Resolution B3): the Sun, $r_s = 2953$ m; Earth, 8.870 mm; a 10-solar-mass black hole, 29.53 km; Sagittarius A*, the Milky Way's central black hole, at 4.30×10^6 solar masses (GRAVITY 2022), 1.27×10^{10} m or 0.085 astronomical units; M87*, at 6.5×10^9 solar masses (EHT 2019), 1.92×10^{13} m or 128 astronomical units, with a mean density inside of only 0.44 kg/m³, less than air.

G. Connections to other equations

EXACT: the root of $g_{tt} = 0$ in Equation 12; the photon sphere and innermost stable orbit sit at $1.5r_s$ and $3r_s$ (Equations 5, 16). ANALOGY: setting the Newtonian escape speed $\sqrt{2GM/r}$ equal to c gives the same formula, but by coincidence: Newtonian light below escape speed would rise and fall back, Newton's r is not the areal radius, and a horizon is a property of the causal structure of spacetime.

Equation 14: Gravitational time dilation for a static clock

A. The equation

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{r}} = \sqrt{1 - \frac{2GM}{rc^2}} \approx 1 - \frac{GM}{rc^2} = 1 + \frac{\Phi}{c^2} \tag{26.26}$$

The square root is EXACT for a static clock outside r_s; the last form is APPROXIMATE (valid for $GM/rc^2 \ll 1$).

B. What it says in plain English

A clock held at rest at radius r ticks more slowly than an identical clock far away, the more so the deeper in the potential well.

C. Every symbol

Symbol	Meaning	Units
τ	proper time read by the static clock	s
t	coordinate time = proper time of a static clock at infinity	s
$d\tau/dt$	rate ratio	—
r, r_s	areal radius; Schwarzschild radius	m
$\Phi = -GM/r$	Newtonian potential	m²/s²

"Static" means held in place by a rocket or the ground, not freely falling.

D. The mathematical operations

Set $dr = d\theta = d\phi = 0$ in Equation 12, use $ds^2 = -c^2 d\tau^2$, take a square root, then Taylor-expand (Section 1).

E. Physical interpretation

Lower clocks run slow, and both observers agree, because light signals between them carry the effect (Equation 15); with the speed term, it gives the GPS budget (Worked Example 14.4).

F. Worked example

Clock	r_s/r	$d\tau/dt$	Daily loss vs. infinity
Earth's surface, $r = 6.371 \times 10^6$ m	1.392×10^{-9}	$1 - 6.96 \times 10^{-10}$	60.1 μs
GPS orbit, $r = 2.656 \times 10^7$ m	3.34×10^{-10}	$1 - 1.67 \times 10^{-10}$	14.4 μs
Sun's surface	4.245×10^{-6}	$1 - 2.12 \times 10^{-6}$	0.183 s (67 s per year)
Neutron star, 1.4 solar masses, 12 km	0.3445	0.8096	4.6 hours
$r = 1.01\, r_s$	0.990	0.0995	10 s pass far away per 1 s here

At Earth the exact and approximate forms agree to about 10^{-19}; at the neutron star the approximation is 2% off (Equation 12).

G. Connections to other equations

EXACT: a special case of Equation 12, whose ratio at two radii gives Equation 15; a clock on a circular geodesic orbit, which also moves, ticks at $\sqrt{1 - 3GM/rc^2}$ (EXACT), the origin of the $3GM/2ac^2$ term in GPS bookkeeping. APPROXIMATE ($GM/rc^2 \ll 1$): $1 + \Phi/c^2$. ANALOGY: in an accelerating rocket, a clock at the nose runs fast by gh/c^2.

Equation 15: Gravitational redshift

A. The equation

$$1 + z \equiv \frac{\lambda_R}{\lambda_E} = \frac{\nu_E}{\nu_R} = \sqrt{\frac{1 - r_s/r_R}{1 - r_s/r_E}} \approx 1 + \frac{\Phi_R - \Phi_E}{c^2} \tag{26.27}$$

EXACT for a static emitter and receiver in Schwarzschild spacetime; the last form is APPROXIMATE (weak field), and over a small height h near a planet it becomes $z \approx gh/c^2$.

B. What it says in plain English

Light climbing out of a gravitational well arrives redder (lower in frequency) than it left, because the emitter's clock runs slow compared with the receiver's; light falling in is blueshifted.

C. Every symbol

Symbol	Meaning	Units
z	redshift; $z > 0$ means redder	—
λ, ν	wavelength, frequency	m, Hz
subscripts E, R	emitter, receiver	—
r_E, r_R	areal radii of emitter and receiver	m
Φ, g, h	potential; local gravitational acceleration; height difference	m²/s², m/s², m

D. The mathematical operations

A ratio of square roots and a first-order Taylor expansion. For tiny effects compute the weak-field form: in double precision the exact form subtracts numbers equal to 1 in their first 15 digits, and for a 22.5 m

tower comes out 0.6% wrong.

E. Physical interpretation

Redshift tests the equivalence principle (every clock runs at a rate set by g_{tt}), not the full field equations (Will 2014). It is distinct from the cosmological redshift, $1+z=1/a$, which comes from expansion.

F. Worked example

Pound and Rebka's Harvard tower: $gh/c^2 = 9.81 \times 22.5/8.98755 \times 10^{16} = 2.46 \times 10^{-15}$. The Pound–Rebka–Snider experiments of 1960–65 reached 1% accuracy using the Mössbauer effect (Will 2014). For sunlight, $cz = GM_\odot/(R_\odot c) = 636$ m/s for a receiver far away, or 633.1 m/s after correcting for the Sun's potential at Earth's orbit and Earth's own potential; laser-comb spectra of sunlight reflected by the Moon measured 638 ± 6 m/s (González Hernández et al. 2020).

G. Connections to other equations

EXACT: the ratio of Equation 14 at two radii, via the photon energy conserved because Schwarzschild spacetime has the time symmetry of Section 24. APPROXIMATE (weak field; small h for the second): $\Delta\Phi/c^2$ and gh/c^2. ANALOGY: the Doppler shift of a signal sent up an accelerating rocket.

Equation 16: Orbits in Schwarzschild spacetime

A. The equation

$$\frac{1}{2}\left(\frac{dr}{d\tau}\right)^2 + V_{\text{eff}}(r) = \frac{E^2 - c^4}{2c^2}, \qquad V_{\text{eff}} = -\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2 r^3} \tag{26.28}$$

EXACT for equatorial geodesics. The resulting perihelion advance per orbit is

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}, \tag{26.29}$$

APPROXIMATE (first order in GM/c^2a).

B. What it says in plain English

The radial part of an orbit behaves like a ball rolling in a one-dimensional landscape, to which GR adds one attractive term that makes orbits precess and, close in, lets them plunge.

C. Every symbol

Symbol	Meaning	Units
r, τ	areal radius; the orbiting body's proper time	m, s
$E = (1-r_s/r)c^2\,dt/d\tau$	conserved energy per unit mass; c^2 for a body at rest far away	J/kg
$L = r^2\,d\phi/d\tau$	conserved angular momentum per unit mass	m²/s
V_{eff}	effective potential per unit mass	J/kg
$-GML^2/c^2r^3$	the entire GR correction	J/kg
a, e	semi-major axis, eccentricity; $a(1-e^2)$ is the semi-latus rectum	m, —

D. The mathematical operations

Substitute the two Killing-vector constants into $g_{\mu\nu}u^\mu u^\nu = -c^2$ (Worked Example 24.2); Eq. (26.29) treats the $1/r^3$ term as a small perturbation of a Newtonian ellipse.

E. Physical interpretation

Stable circular orbits end at $6GM/c^2$ (the innermost stable circular orbit, the inner edge of accretion discs); light circles at $3GM/c^2$.

F. Worked example

Mercury: $a = 57.909 \times 10^9$ m, $e = 0.2056$, period 87.969 days (NASA Mercury Fact Sheet). Then $1 - e^2 = 0.957729$ and $a(1-e^2) = 5.54611 \times 10^{10}$ m, so

$$\Delta\phi = \frac{6\pi \times 1.3271244 \times 10^{20}}{8.98755 \times 10^{16} \times 5.54611 \times 10^{10}} = 5.0186 \times 10^{-7}\ \text{rad} = 0.10352''. \tag{26.30}$$

With $36\,525/87.969 = 415.20$ orbits per century, that is **42.98″ per century**, the observed anomaly (Will 2014).

G. Connections to other equations

EXACT: from Equations 5 and 12 with Killing vectors (Section 24). APPROXIMATE (first order in GM/c^2a): the precession formula. The same physics gives 4.23° per year of periastron advance in the Hulse–Taylor binary pulsar (a pulsar orbiting another neutron star, discovered in 1974) and shapes the orbit of the star S2 around Sagittarius A* (GRAVITY 2022).

Equation 17: The geodesic deviation equation

A. The equation

$$\frac{D^2\xi^\mu}{d\tau^2} = R^\mu{}_{\nu\rho\sigma}\, u^\nu u^\rho \xi^\sigma, \qquad \frac{d^2\xi^i}{dt^2} \approx -\left(\partial_i\partial_j\Phi\right)\xi^j \tag{26.31}$$

The first is EXACT for an infinitesimal separation (Carroll/MTW convention, Carroll 1997, eq. 3.113); being linear in ξ^μ, for two particles a finite distance apart it is the leading term, accurate while ξ is small compared with the distance over which the curvature changes. The second, its Newtonian form with $R^i{}_{0j0} \approx \partial_i\partial_j\Phi/c^2$, is APPROXIMATE (weak field, slow motion).

B. What it says in plain English

Two freely falling particles that start side by side, at rest relative to each other, do not stay so: their separation accelerates in proportion to curvature times distance. Curvature shows up physically as tides.

C. Every symbol

Symbol	Meaning	Units
ξ^μ (xi)	separation vector to a neighbouring geodesic	m
τ	proper time along the reference geodesic	s
$D/d\tau = u^\alpha\nabla_\alpha$	covariant derivative along the curve; applied twice, relative acceleration	— ; m/s²
u^ν	four-velocity of the reference observer	m/s
$R^\mu{}_{\nu\rho\sigma}$	Riemann tensor: μ output, ν and ρ fed u, σ fed ξ	m⁻²
repeated indices	64 terms for each μ	—
$\partial_i\partial_j\Phi$	Newtonian tidal tensor	s⁻²

Units check: $(\text{m}^{-2})(\text{m/s})^2(\text{m}) = \text{m/s}^2$.

D. The mathematical operations

Two covariant derivatives along a curve, and a rank-4 tensor contracted with three vectors: a linear map from separation to acceleration (Section 20).

E. Physical interpretation

Radial stretching ($+2GM\xi/r^3$) and sideways squeezing ($-GM\xi/r^3$) near a mass, spaghettification near small black holes, and every gravitational-wave detector. Sign check for any book: meridians on a sphere must converge.

F. Worked example

The radial stretch across a 2 m body is $2GM(2\,\mathrm{m})/r^3$: standing on Earth, 6.2×10^{-6} m/s², about $6\times10^{-7}g$ (Worked Example 20.2); at the horizon of a 10-solar-mass black hole ($r = 29.5$ km), 2.1×10^{8} m/s², about $2\times10^{7}g$, lethal long before the horizon; at the horizon of Sagittarius A* (4.3×10^{6} solar masses), 1.1×10^{-3} m/s², about $10^{-4}g$, unnoticeable. Since $r_s \propto M$, the tide at the horizon scales as $M/r_s^3 \propto 1/M^2$: big black holes are gentle at the edge.

G. Connections to other equations

EXACT: follows from Equations 5 and 6; in Schwarzschild spacetime $2GM\xi/r^3$ is exact (for small ξ) for freely falling neighbours momentarily at rest or falling radially (Worked Example 20.3); and the detectors of Equation 18 respond through it. APPROXIMATE (weak field, slow motion): the Newtonian tidal law. ANALOGY: ocean tides.

Equation 18: Gravitational waves: the linearised field equations

A. The equation

Write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$, define the trace-reversed perturbation $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$, and choose coordinates with $\partial^\mu \bar{h}_{\mu\nu} = 0$ (the Lorenz gauge). Then

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \qquad \Box \equiv -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2, \tag{26.32}$$

and in vacuum $\Box \bar{h}_{\mu\nu} = 0$ (Carroll 1997, eqs. 6.20–6.23, with c restored). A plane wave travelling along z has, in the transverse-traceless (TT) gauge,

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cos\left[\omega\left(t - \frac{z}{c}\right)\right]. \tag{26.33}$$

APPROXIMATE: first order in h; within that linear theory the wave equation is exact.

B. What it says in plain English

Far from their sources, small metric ripples obey the same wave equation as light and travel at c, stretching space along one axis while squeezing it along the perpendicular one, then reversing.

C. Every symbol

Symbol	Meaning	Units
$\eta_{\mu\nu}$	flat Minkowski metric	none
$h_{\mu\nu}$; $h = \eta^{\mu\nu}h_{\mu\nu}$	small perturbation (strain); its trace	none
$\bar{h}_{\mu\nu}$	trace-reversed perturbation	none
$\partial^\mu = \eta^{\mu\nu}\partial_\nu$	derivative with index raised	m^{-1}
$\Box$	d'Alembertian (wave operator)	m^{-2}
$16\pi G/c^4$	$2\kappa = 4.15\times10^{-43}$	N^{-1}
h_+, $h_\times$	amplitudes of the two polarisations, "plus" and "cross"	none

Symbol	Meaning	Units
ω; $t - z/c$	angular frequency; retarded time	rad/s; s

D. The mathematical operations

Linearisation (keep first order in h), a gauge choice (a use of coordinate freedom), and second partial derivatives forming the wave operator. With sources, far away and for slow sources, the solution is the quadrupole formula $h_{ij} \approx \frac{2G}{c^4 r}\ddot{I}_{ij}(t - r/c)$, where $I_{ij} = \int \rho\, x^i x^j\, d^3x$ and the dots mean d/dt.

E. Physical interpretation

Only h_+ and $h_\times$ are physical; the rest is coordinate choice. There is no dipole radiation, because an isolated system's centre of mass cannot oscillate, and the waves are weak because G/c^4 is tiny: the Earth–Sun system radiates about 200 W (Figure 26.2).

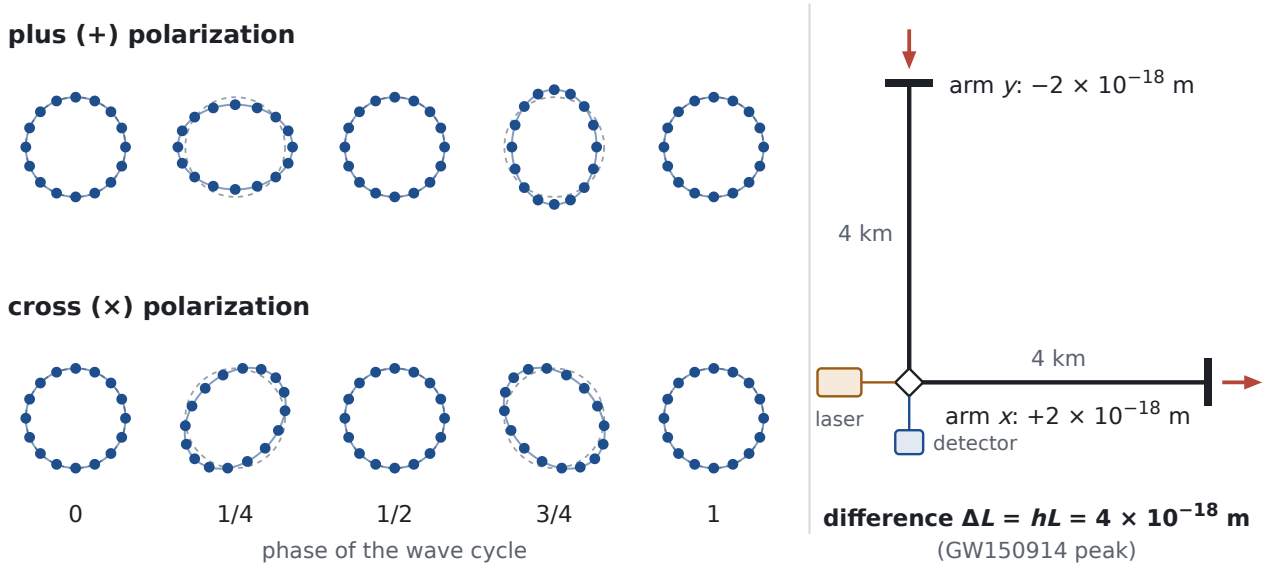


Figure 26.2. The two polarisations of a gravitational wave acting on a ring of free particles (distortion hugely exaggerated). An L-shaped detector measures the difference between the changes in its two arm lengths.

F. Worked example

GW150914 had a peak strain of 1.0×10^{-21} on LIGO's 4 km arms (Abbott et al. 2016). For an optimally oriented "plus" wave each arm changes by $\pm\frac{1}{2}hL = \pm 2 \times 10^{-18}$ m, in opposite directions, so the measured difference between the arm lengths is

$$\Delta L = \delta L_x - \delta L_y = hL = 1.0 \times 10^{-21} \times 4000\ \text{m} = 4 \times 10^{-18}\ \text{m}, \tag{26.34}$$

a few hundred times smaller than a proton (LIGO). The Hulse–Taylor binary pulsar's orbit shrinks at 0.9983 ± 0.0016 times the rate this radiation predicts (Weisberg & Huang 2016).

G. Connections to other equations

APPROXIMATE: Equation 10 to first order in h. EXACT (within linear theory): the vacuum wave equation, which detectors register through geodesic deviation (Equation 17). ANALOGY: electromagnetic waves, which however start at dipole order.

Equation 19: The Friedmann equations

A. The equation

For the uniform, expanding (FLRW) universe,

$$ds^2 = -c^2 dt^2 + a(t)^2 \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right], \tag{26.35}$$

$$H^2 \equiv \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \qquad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{\Lambda c^2}{3}. \tag{26.36}$$

EXACT for an exactly homogeneous, isotropic universe; APPROXIMATE for ours, which is uniform only on the largest scales.

B. What it says in plain English

The square of the expansion rate is a budget: matter and energy, minus a curvature term, plus dark energy. Matter and pressure slow the expansion; a positive cosmological constant speeds it up.

C. Every symbol

Symbol	Meaning	Units
t	cosmic time, kept by observers at rest in the cosmic fluid	s
$a(t)$	scale factor, 1 today	none
$\dot{a}$, $\ddot{a}$	first and second time derivatives (here dots mean d/dt)	s^{-1}, s^{-2}
$H = \dot{a}/a$	Hubble parameter; H_0 today	s^{-1}
r, θ, ϕ	comoving coordinates; galaxies keep fixed values	m, rad
k	spatial curvature: positive closed, zero flat, negative open	m^{-2}
ρ; p	total mass density (all forms of energy $/c^2$); pressure	kg/m^3; Pa
Λ	cosmological constant	m^{-2}
$8\pi G/3$, $4\pi G/3$	descendants of $8\pi G/c^4$	$m^3\ kg^{-1}\ s^{-2}$
$\rho + 3p/c^2$	the combination that gravitates: pressure counts three times	kg/m^3

D. The mathematical operations

Insert the FLRW metric into the field equations: symmetry turns ten PDEs into ODEs in t. For the first equation, with $c = 1$ for this one step, $G_{00} = 3(\dot{a}^2 + k)/a^2$, $\Lambda g_{00} = -\Lambda$ and $8\pi G T_{00} = 8\pi G \rho$, so $3(\dot{a}^2 + k)/a^2 - \Lambda = 8\pi G \rho$; dividing by 3 gives the first line of Eq. (26.36) (Carroll 1997, §8).

E. Physical interpretation

They describe the Big Bang, the dilution of matter as a^{-3} and radiation as a^{-4} (Worked Example 22.3), and today's accelerating expansion.

F. Worked example

With $H_0 = 67.4$ km/s/Mpc $= 2.1843 \times 10^{-18}$ s^{-1} (Planck 2018), the critical density for a flat universe is $\rho_c = 3H_0^2/8\pi G = 8.53 \times 10^{-27}$ kg/m^3, about five hydrogen atoms per cubic metre. A flat universe with matter fraction $\Omega_m = 0.315$ and $\Omega_\Lambda = 0.685$ has $H = H_0 \sqrt{\Omega_m a^{-3} + \Omega_\Lambda}$; since $dt = da/(aH)$, substituting $u = a^{3/2}$ turns its age into a Calculus II integral:

$$t_0 = \int_0^1 \frac{a^{1/2}\,da}{H_0\sqrt{\Omega_m + \Omega_\Lambda a^3}} = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\,\operatorname{arsinh}\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}. \tag{26.37}$$

Here arsinh, the inverse of sinh, equals $\ln(x + \sqrt{x^2+1})$, and the integral is the standard form $\int du/\sqrt{A+Bu^2} = B^{-1/2}\operatorname{arsinh}(u\sqrt{B/A})$. Numbers: $\sqrt{0.685/0.315} = 1.47465$, $\operatorname{arsinh}(1.47465) = \ln(3.25640) = 1.18062$, $2/(3\sqrt{0.685}) = 0.80550$, so $H_0 t_0 = 0.95099$. With $1/H_0 = 14.507$ billion years, $t_0 = 13.80$ billion years, matching Planck's 13.797 ± 0.023.

G. Connections to other equations

EXACT: the time-time and space-space components of Equation 10; the fluid equation $\dot\rho + 3H(\rho + p/c^2) = 0$ is $\nabla_\mu T^{\mu 0} = 0$ and follows from the two Friedmann equations, the Bianchi identity at work; the $\Lambda c^2/3$ terms descend from the -2Λ in Equation 20. ANALOGY: Newton's energy equation for a uniform expanding ball gives the first equation's form with $\Lambda = 0$, but treats k as an energy and misses the pressure term.

Equation 20: The Einstein–Hilbert action

A. The equation

$$S = \frac{c^4}{16\pi G}\int (R - 2\Lambda)\,\sqrt{-g}\;dt\,d^3x + S_M = \frac{c^3}{16\pi G}\int (R - 2\Lambda)\,\sqrt{-g}\;d^4x + S_M \tag{26.38}$$

with $d^4x = d(ct)\,d^3x$ in the second form, and $T_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}}$. EXACT: $\delta S = 0$ for all variations of the metric gives Equation 10.

B. What it says in plain English

Among all possible spacetime geometries, the real one makes this single number stationary: unchanged, to first order, by any small change of the metric.

C. Every symbol

Symbol	Meaning	Units
S	action: one number for a whole spacetime history	J·s
$c^4/16\pi G$	prefactor, 2.408×10^{42}	N
16π	$2 \times 8\pi$; the 2 balances the -2 in the definition of $T_{\mu\nu}$	—
$\int \dots dt\,d^3x$	integral over a four-dimensional region	s·m³
g	determinant of $g_{\mu\nu}$; negative, so $-g > 0$	none
$\sqrt{-g}$	turns coordinate volume into invariant volume	none
R	Ricci scalar	m⁻²
-2Λ	becomes $+\Lambda g_{\mu\nu}$ in the field equations	m⁻²
S_M	action of all matter and radiation	J·s
δ	a small variation of a whole function	—

D. The mathematical operations

Vary $g^{\mu\nu}$, expand to first order, discard a total divergence that integrates to the boundary, use $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$, and demand that the coefficient of $\delta g^{\mu\nu}$ vanish everywhere (Section 25).

E. Physical interpretation

All of GR is packaged in one scalar, whose coordinate independence makes energy-momentum conservation automatic. Its units are those of Planck's constant (J·s), which matters for quantum approaches to gravity.

F. Worked example

Check the key variation formula on the two-dimensional metric $g = \mathrm{diag}(-A, B)$, for which $\sqrt{-g} = \sqrt{AB}$. Changing B to $B(1+\varepsilon)$ gives $\delta\sqrt{-g} = \tfrac{1}{2}\varepsilon\sqrt{AB}$ to first order, while $g^{11} = 1/B$ changes by $\delta g^{11} = -\varepsilon/B$, and the formula gives

$$-\tfrac{1}{2}\sqrt{-g}\, g_{11}\, \delta g^{11} = -\tfrac{1}{2}\sqrt{AB}\cdot B\cdot\left(-\frac{\varepsilon}{B}\right) = +\tfrac{1}{2}\varepsilon\sqrt{AB}. \tag{26.39}$$

They agree. Units: $(2.408\times10^{42}\ \mathrm{N})(\mathrm{m}^{-2})(\mathrm{s}\cdot\mathrm{m}^{3}) = \mathrm{N\,m\,s} = \mathrm{J\,s}$.

G. Connections to other equations

EXACT: $\delta S/\delta g^{\mu\nu} = 0$ gives Equation 10, and the -2Λ gives the $\Lambda c^2/3$ of Equation 19. APPROXIMATE: expanded to second order in $h_{\mu\nu}$, it gives the linearised theory of Equation 18. ANALOGY: Hamilton's principle of stationary action in mechanics (Worked Example 25.2). Hilbert's action-based paper of November 1915 gives the action its name (Section 23).

PART V

Maps, Notation and Further Study

Looking back, the whole apparatus of GR is one chain: a metric, its first derivatives (Christoffel symbols), their derivatives and products (curvature), two traces (Ricci tensor and scalar), one combination forced by an identity (the Einstein tensor), and an equation setting it proportional to energy and momentum. Only three links bring genuinely new ideas: index positions, the covariant derivative and the Bianchi identity. The rest is careful bookkeeping, which is why the next courses supply practice more than insight.

SECTION 27

The equation dependency map

Figure 27.1 shows how the great equations feed one another: read it top to bottom for the logic of the field equations, and along the side branches for the predictions.

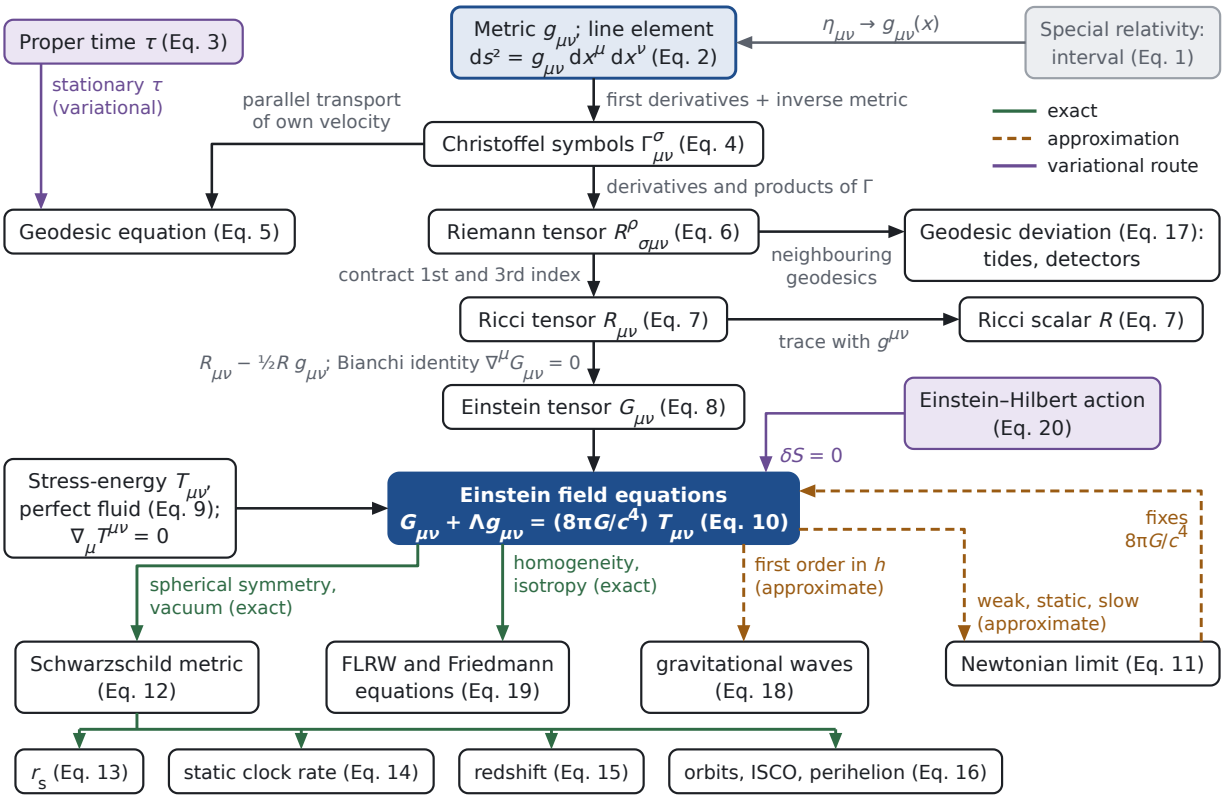


Figure 27.1. The equation dependency map. The spine runs from the metric through the Christoffel symbols, the Riemann and Ricci tensors and the Einstein tensor to the field equations; solutions under symmetry (green) and approximations (orange) branch off below, and the variational routes (purple) give the geodesic and field equations a second derivation.

From	Operation	To	Type
Interval (Eq. 1)	replace $\eta_{\mu\nu}$ by $g_{\mu\nu}(x)$	Line element (Eq. 2)	generalisation
Line element	evaluate along a clock's path	Proper time (Eq. 3)	EXACT
Metric	first derivatives, inverse metric	Christoffel symbols (Eq. 4)	EXACT
Christoffel symbols	parallel-transport the velocity	Geodesic equation (Eq. 5)	EXACT
Proper time	make it stationary	Geodesic equation	EXACT
Christoffel symbols	derivatives and products	Riemann (Eq. 6)	EXACT
Riemann	contract, then trace	Ricci tensor, scalar (Eq. 7)	EXACT
Ricci	subtract $\frac{1}{2}Rg_{\mu\nu}$	Einstein tensor (Eq. 8)	EXACT
Einstein tensor, stress-energy (Eq. 9)	set proportional	Field equations (Eq. 10)	EXACT
Field equations	weak, static, slow	Newtonian limit (Eq. 11)	APPROXIMATE
Field equations	spherical symmetry, vacuum	Schwarzschild (Eqs. 12–16)	EXACT
Riemann, geodesics	neighbouring paths	Geodesic deviation (Eq. 17)	EXACT (to first order in the separation)
Field equations	first order in h	Gravitational waves (Eq. 18)	APPROXIMATE

From	Operation	To	Type
Field equations	homogeneity, isotropy	Friedmann (Eq. 19)	EXACT
Einstein–Hilbert action (Eq. 20)	vary the metric	Field equations	EXACT

SECTION 28

Notation reference

Signature $(-,+,+,+)$; SI units with $x^0 = ct$; Carroll/MTW curvature signs.

Notation	Meaning	Section
x^μ, $\mu = 0,1,2,3$	spacetime coordinates (ct, x, y, z); superscripts are labels	9
x^i, $i = 1,2,3$	spatial coordinates (Latin indices)	9
V^μ, ω_μ	vector (contravariant) and dual-vector (covariant) components	6, 11
repeated up–down index	summed over all values (Einstein convention)	11
δ^μ_ν	Kronecker delta: 1 if $\mu = \nu$, else 0; $\delta^\mu_\mu = 4$	4, 11
∂_μ, $A_{,\mu}$	partial derivative $\partial/\partial x^\mu$	2
∇_μ, $A_{;\mu}$	covariant derivative	16
$D/d\tau$	covariant derivative along a curve, $u^\alpha \nabla_\alpha$	17, 20
$\dot{x}$	$dx/d\tau$ or $dx/d\lambda$; dx/dt in Newtonian formulas and cosmology	1
∇, ∇^2, $\Box$	gradient; Laplacian; d'Alembertian $-c^{-2}\partial_t^2 + \nabla^2$	7; 7; 26
$T_{(\mu\nu)}$, $T_{[\mu\nu]}$	symmetric and antisymmetric parts	13
$\eta_{\mu\nu}$	Minkowski metric $\mathrm{diag}(-1,1,1,1)$	9
$g_{\mu\nu}$, $g^{\mu\nu}$	metric and inverse metric; $g^{\mu\lambda}g_{\lambda\nu} = \delta^\mu_\nu$	12
g, $\sqrt{-g}$	determinant of $g_{\mu\nu}$ (negative); invariant volume factor	4, 25
$\Lambda^\mu{}_\nu$	Lorentz transformation matrix	10
$\Gamma^\lambda_{\mu\nu}$	Christoffel symbols (not a tensor)	16
$R^\rho{}_{\sigma\mu\nu}$	Riemann tensor	19
$R_{\mu\nu}$, R	Ricci tensor $R^\lambda{}_{\mu\lambda\nu}$; Ricci scalar $g^{\mu\nu}R_{\mu\nu}$	21
$G_{\mu\nu}$	Einstein tensor $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$	21
$T^{\mu\nu}$, T	stress-energy tensor; its trace	22
κ	Einstein's constant $8\pi G/c^4 = 2.0766 \times 10^{-43}$ N^{-1}	23, 26
Λ	cosmological constant, about 1.1×10^{-52} m^{-2}	26
τ, u^μ, p^μ, a^μ	proper time; four-velocity, -momentum, -acceleration	14
ξ^μ	separation vector (Section 20) or Killing vector (Section 24)	20, 24
$h_{\mu\nu}$, $\bar{h}_{\mu\nu}$	metric perturbation; its trace-reverse	26
r_s	Schwarzschild radius $2GM/c^2$	26
$a(t)$, H	cosmic scale factor; Hubble parameter $\dot{a}/a$	26

Some symbols carry more than one meaning; the text says which at each use.

Symbol	Meanings
ρ	an index (Greek rho) in tensor formulas; mass density in physics formulas; distance from the pole along a sphere (Worked Example 15.2)

Symbol	Meanings
Λ	cosmological constant; with indices, $\Lambda^{\mu}{}_{\nu}$, a Lorentz matrix
G	Newton's constant; with indices, $G_{\mu\nu}$, the Einstein tensor
R vs r	Ricci scalar vs radial coordinate
ϕ	longitude (azimuthal angle); rapidity in Section 10
θ	colatitude (angle from the pole), not latitude
δ	Kronecker delta δ^{μ}_{ν}; "small variation of" in δS
T	trace of $T_{\mu\nu}$; also a time interval in worked examples
a	sphere radius (Sections 8, 19); GPS orbit radius (Sections 1, 14); scale factor $a(t)$ (Section 22, Eq. 19); semi-major axis (Eq. 16); a^{μ}, four-acceleration (Section 14)
g	metric $g_{\mu\nu}$; its determinant (in $\sqrt{-g}$); local gravitational acceleration, 9.81 m/s² (Worked Example 25.2, Eqs. 3, 15)
L	Lagrangian (Section 25); angular momentum per unit mass (Section 24, Eq. 16); a length (Section 8, Eqs. 2, 18)
η	Minkowski metric $\eta_{\mu\nu}$; the bump $\eta(x)$ added to a path (Section 25)
λ, ν	indices; λ also a path parameter (Sections 1, 17, 18); wavelength and frequency in Eq. 15
e, E	e: eccentricity (Eq. 16) or energy per unit rest energy (Worked Example 18.3); $\mathbf{e}_{\mu}$: basis vectors; E: energy (Section 14) or energy per unit mass (Section 24, Eq. 16)

Geometric units ($G = c = 1$) turn masses and times into lengths. To convert a geometric-units formula to SI, replace every mass M by GM/c^2, every time t by ct, and $8\pi T_{\mu\nu}$ by $(8\pi G/c^4)T_{\mu\nu}$.

Quantity	Geometric equivalent
1 second	299 792 458 m
1 kilogram	$GM/c^2 = 7.426 \times 10^{-28}$ m
1 solar mass	1.477 km, or 4.925 μs (from the IAU value of $GM_{\odot}$)
1 Earth mass	4.435 mm
1 joule	8.263×10^{-45} m

SECTION 29

Further study

Courses, in the order you would usually meet them. GR courses list the same prerequisites (multivariable calculus, linear algebra, differential equations, Lagrangian mechanics and special relativity), with electromagnetism added at graduate level (MIT course catalog; UChicago catalog; UC Santa Cruz PHYS 171); Hartle argues that a "physics-first" undergraduate course needs only intermediate mechanics as its essential physics prerequisite (Hartle 2006). A typical US sequence from your position:

Stage	Mathematics	Physics	What it unlocks here
Now	Calculus II (series, integration)	Physics I (mechanics)	Sections 1, 8, 14; Taylor expansions; proper-time integrals
Year 2	Calculus III (multivariable); linear algebra; differential equations	Physics II (electromagnetism); modern physics with special relativity	Sections 2–7, 9–14
Year 3	Mathematical methods (tensors, calculus of variations, PDEs)	Classical mechanics (Lagrangians); upper-level electromagnetism	Sections 13, 18, 25; wave equations
Year 3–4	Optional differential geometry (mathematics department)	Undergraduate GR	Part III in full

Stage	Mathematics	Physics	What it unlocks here
Graduate	Differential geometry and topology; numerical methods	Graduate GR	manifolds, actions, solving the field equations

At MIT "Calculus II" means multivariable calculus, which most US universities call Calculus III (MIT mathematics catalog); Cambridge's graduate course calls finite-dimensional vector spaces and the Euler–Lagrange equations "essential" (Cambridge Part III GR guide).

Books, gentlest first. Taylor and Wheeler's *Spacetime Physics* (special relativity, free online); Fleisch's *A Student's Guide to Vectors and Tensors* (with Calculus III); Hartle's *Gravity* or Moore's *A General Relativity Workbook* (undergraduate GR); Schutz's *A First Course in General Relativity* (advanced undergraduate); Carroll's *Spacetime and Geometry* (first-year graduate); then Misner, Thorne and Wheeler's *Gravitation* and Wald's *General Relativity*. Boas's *Mathematical Methods in the Physical Sciences* covers Parts I and II, including Jacobians, the calculus of variations and tensors (Boas, table of contents).

Free notes and lectures. Carroll's 1997 notes, the source of most equation numbers here (Carroll 1997); Tong's Cambridge notes, followable "for students who haven't had a first course in the subject" (Tong); Blau's thousand-page Bern notes (Blau); Scott Hughes's 23 recorded MIT 8.962 lectures, taught from Carroll's book (MIT OCW 8.962); and eigenchris's "Tensors for Beginners" videos for index notation (eigenchris).

Questions to ask a GR professor, drawn from real choices that courses make:

Question	Why it is a good one
Does your course go physics-first (Hartle), math-first (Schutz, Carroll) or interleave the two (Moore), and why?	Instructors genuinely disagree (Hartle 2006; Moore 2006)
Which linear algebra matters most for GR: change of basis, dual spaces, determinants, eigenvalues?	Tells you what to emphasise
Should I take differential geometry from the mathematics department before or after GR?	Graduate programmes advise doing it first
Which signature and curvature sign conventions do you use?	Avoids sign errors across books (Conventions, front matter)
Will we compute Christoffel symbols and curvature by hand or with computer algebra?	Both are needed; the balance varies
Do you use abstract index notation, as in Wald, or components?	Matters for reading graduate texts (Wald 2005)
Why is the Einstein–Hilbert action R and not some other scalar, and what changes if it is not?	Leads to modified theories of gravity
How do numerical relativists keep the four constraint equations satisfied?	Leads to the 3+1 formulation (Section 23)

SECTION 30

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